

How Much Can Gaussian Dependence Inflate the Benjamini–Hochberg Procedure’s FDR?

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Abstract

We study the worst-case false discovery rate (FDR) of the Benjamini–Hochberg procedure for both one- and two-sided Gaussian tests when the correlation matrix is otherwise unrestricted. In each setting we construct finite Gaussian models for which the ratio of their FDR to q diverges as $q \downarrow 0$, disproving any universal multiplicative FDR bound. For two-sided tests, the supremum over the number of hypotheses, mean vector, and correlation matrix is at least an explicit $\ell_=(q) > q$ satisfying

$$\ell_=(q) = \frac{q\sqrt{\log(1/q)}}{2\sqrt{\pi}} + c_\ell q + o(q), \quad c_\ell = 0.6492828\dots$$

For the one-sided hypotheses $H_i : \theta_i \leq 0$, a sign-reversed one-common-factor construction gives the stronger explicit lower bound $\ell_\leq(q) > q$, with

$$\ell_\leq(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + o(q).$$

The nulls are positively correlated, whereas null–nonnull correlations are negative. Finally, we prove an $O\{q\sqrt{\log(1/q)}\}$ upper bound for the two-sided one-common-factor class and the matching one-sided upper bound $q\sqrt{\log(1/q)}/\sqrt{\pi} + O(q)$.

1 Introduction

We observe a Gaussian vector $T = (T_1, \dots, T_N)$ with mean θ and covariance matrix Σ , where $\Sigma_{ii} = 1$ for $i = 1, \dots, N$. We consider two testing problems. For two-sided tests, $H_i : \theta_i = 0$ and

$$P_i = p(|T_i|), \quad p(x) = 2\bar{\Phi}(x) = 2\{1 - \Phi(x)\}.$$

For one-sided tests, $H_i : \theta_i \leq 0$ and

$$P_i = p_+(T_i), \quad p_+(x) = \bar{\Phi}(x) = 1 - \Phi(x).$$

Here Φ and $\phi = \Phi'$ are the standard normal CDF and density. The normal hazard is

$$\lambda(x) = \frac{\phi(x)}{\bar{\Phi}(x)}. \tag{1}$$

We use the conventions $p^{-1}(0) = p_+^{-1}(0) = \infty$. The Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] at level q orders the p -values as

$$P_{(1)} \leq \dots \leq P_{(N)}$$

and rejects the hypotheses corresponding to

$$P_{(1)}, \dots, P_{(R)}, \quad R = \max\{1 \leq k \leq N : P_{(k)} \leq qk/N\},$$

with $R = 0$ if the set is empty. If V is the number of rejected true nulls, then the false discovery proportion (FDP) and false discovery rate (FDR) are

$$\text{FDP} = \frac{V}{R \vee 1}, \quad \text{FDR} = \mathbb{E}(\text{FDP}).$$

A long-standing conjecture for correlated two-sided Gaussian z -tests is that BH controls the FDR at its nominal level q , regardless of the correlation matrix [Reiner-Benaim, 2007, Benjamini, 2010, Roux, 2018, Sarkar, 2023, Sarkar and Zhang, 2025]. Benjamini [2010, p. 407] summarized the evidence as follows:

“The modification to general dependence is often not needed: convincing simutheoretical evidence indicates that the same holds for two-sided z -tests with any correlation structure.”

Benjamini then noted that a complete theoretical proof was still missing. These works stated or supported this conjecture. However, recently, Dobriban [2026] found a counterexample where $\text{FDR} > 0.0104$ when $q = 0.01$.

In this paper, we ask a more refined question in each testing problem: how large can the FDR be as a function of q ? For two-sided tests, we seek a lower bound $\ell_{=}(q)$ such that

$$\sup_{\substack{N \in \mathbb{N}, \theta \in \mathbb{R}^N, \Sigma \in \mathbb{S}_+^N \\ \Sigma_{ii}=1, 1 \leq i \leq N}} \text{FDR}_{\theta, \Sigma}(\text{BH}_q) \geq \ell_{=}(q). \quad (2)$$

Here $\mathbb{S}_+^N$ denotes the cone of $N \times N$ symmetric positive semidefinite matrices. A relaxed folklore conjecture is that the worst-case FDR in (2) is bounded above by Cq for some universal constant $C > 0$. However, Theorem 2.1 and Proposition 2.2 together show that the worst-case FDR is at least a quantity satisfying, as $q \downarrow 0$,

$$\ell_{=}(q) = \frac{q\sqrt{\log(1/q)}}{2\sqrt{\pi}} + c_\ell q + o(q), \quad c_\ell = 0.6492828\dots$$

In particular, the ratio of the supremum in (2) to q diverges as $q \downarrow 0$.

Moreover, we derive an exact formula for $\ell_{=}(q)$. We prove that $\ell_{=}(q) > q$ for every $q \in (0, 1)$. Instead of seeking a counterexample for a given nominal level q , Theorem 2.1 and Proposition 2.3 together disprove the conjecture that BH controls the FDR for correlated two-sided Gaussian tests. Figure 1 illustrates the resulting two-sided FDR inflation.

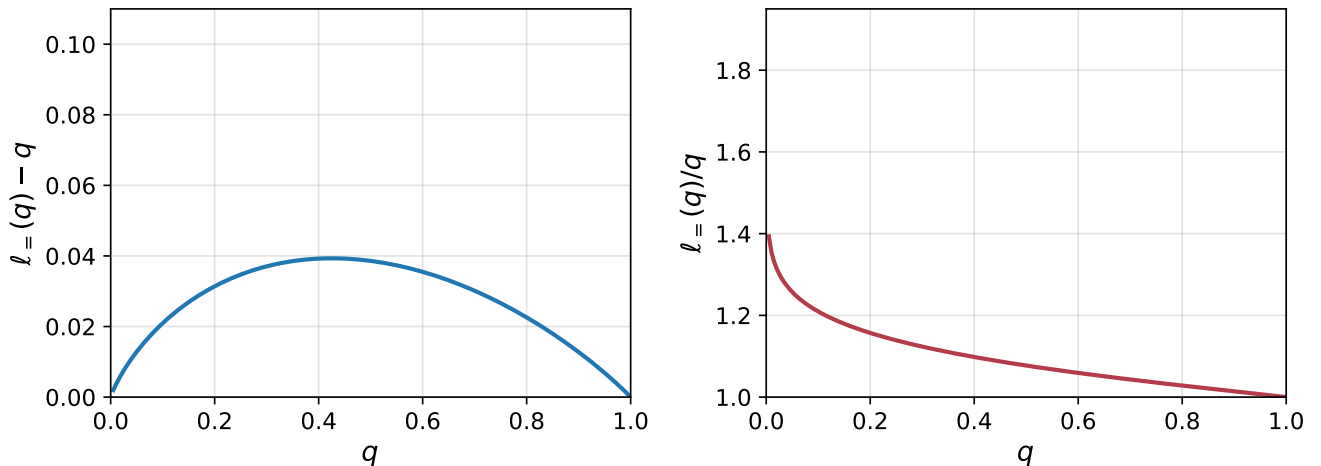


Figure 1: Numerical evaluation of $\ell_{=}(q) - q$ and $\ell_{=}(q)/q$ over the grid $q = 0.005k$, $k = 1, \dots, 199$.

q	0.01	0.05	0.10	0.20
$\ell_{\leq}(q) - q$	0.0036	0.0129	0.0209	0.0314
$\ell_{\leq}(q)/q$	1.3553	1.2571	1.2094	1.1570

Table 1: The absolute inflation $\ell_{\leq}(q) - q$ and relative inflation $\ell_{\leq}(q)/q$ at commonly used nominal levels.

For one-sided Gaussian tests, the classical negatively correlated examples of Hochberg and Rom [1995] and Samuel-Cahn [1996] show that the Simes inequality, and hence BH under the global null, can be anticonservative. Chi et al. [2025] prove useful upper bounds under several notions of negative dependence, but the adversarial construction used in their sharpness comparison, due to Su [2018], is not jointly Gaussian. Our one-sided construction instead retains positively correlated nulls and reverses the loading of the nonnulls. Theorem 3.1 and Proposition 3.3 give, for every $q \in (0, 1)$,

$$\ell_{\leq}(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + \bar{\Phi}\left(\sqrt{2\log(1/q)}\right) > q,$$

and Proposition 3.2 gives

$$\ell_{\leq}(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + o(q).$$

This is a larger leading constant than in the two-sided construction and again implies that the worst-case FDR ratio diverges. Figure 2 illustrates the resulting one-sided FDR inflation.

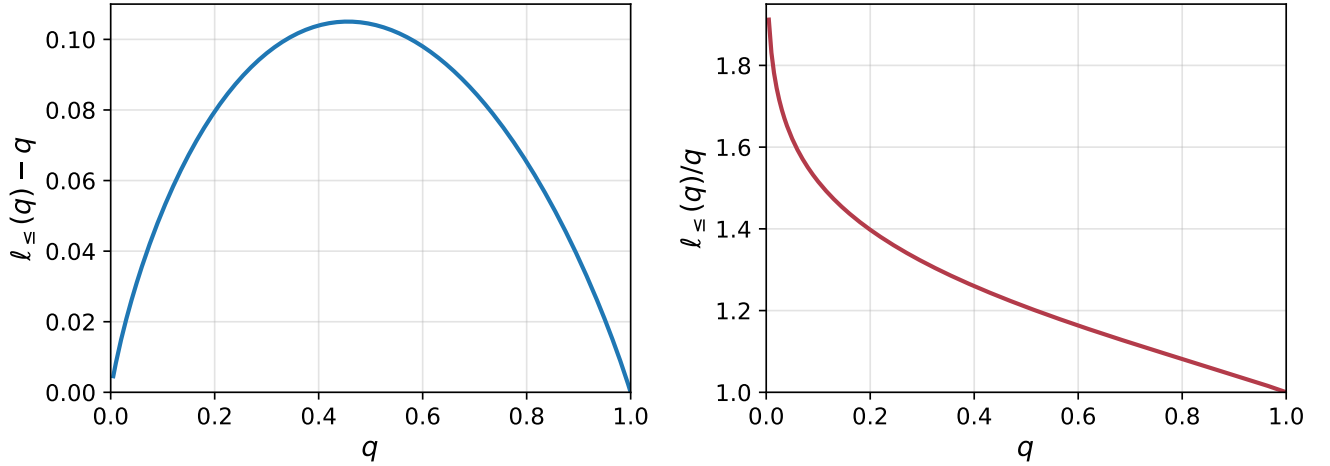


Figure 2: Numerical evaluation of $\ell_{\leq}(q) - q$ and $\ell_{\leq}(q)/q$ over the grid $q = 0.005k$, $k = 1, \dots, 199$.

Table 2: The explicit one-sided lower bound and its inflation factor.

q	0.01	0.05	0.10	0.20
$\ell_{\leq}(q)$	0.01831	0.08101	0.15155	0.27955
$\ell_{\leq}(q)/q$	1.8311	1.6203	1.5155	1.3977

The dependence mechanisms differ sharply between the two constructions. Every covariance in the two-sided construction is positive, but the map $T_i \mapsto 2\bar{\Phi}(|T_i|)$ is not monotone and need not make the full vector of two-sided p -values positively regression dependent on the subset of true nulls (PRDS). For one-sided Gaussian p -values, by contrast, nonnegative correlations between every true null and every

coordinate imply that the full p -value vector is PRDS on the subset of true nulls and hence FDR control [Benjamini and Yekutieli, 2001]. Negative dependence touching a true null is therefore essential in the one-sided construction: null–null correlations are positive, while null–nonnull correlations are negative. Thus, the full vector of one-sided p -values in our construction is not PRDS; Section 3 verifies this directly.

The lower-bound rate $q\sqrt{\log(1/q)}$ should be compared with the FDR-linking theorem of Su [2018]. Positive regression dependence within nulls (PRDN) relaxes PRDS by imposing the relevant positive dependence assumption only on the null p -values. Under PRDN, the FDR-linking argument gives an $O(q\log(1/q))$ bound. Section 4 proves sharper Gaussian bounds. For two-sided one-common-factor models,

$$\text{FDR}(\text{BH}_q) = O(q\sqrt{\log(1/q)}),$$

uniformly over the number of hypotheses, means, and loadings. Thus, in this one-common-factor Gaussian class, the upper-bound rate improves on the general PRDN linking bound and matches the asymptotic order of $\ell_=(q)$ as $q \downarrow 0$. Moreover, as explained in Remark 4.3, the two-sided null p -values in this class satisfy the PRDN property by the absolute-value Gaussian MTP₂ criterion of Karlin and Rinott [1981]. For one-sided one-common-factor models, Theorem 4.4 gives an upper bound with leading term $q\sqrt{\log(1/q)}/\sqrt{\pi}$, and Corollary 4.5 shows that this matches the leading constant in $\ell_=(q)$. Both bounds are uniform over the number of hypotheses, means, and scalar loadings.

2 Lower bound for two-sided tests

2.1 Main result

$$M(y) = \sup_{a \geq 0} e^{-a^2/2} \cosh(ay). \quad (3)$$

Appendix A identifies the likelihood-ratio envelope: $M = 1$ on $[0, 1]$, and M is continuous, strictly increasing, and unbounded on $(1, \infty)$. Hence there is a unique $y_q > 1$ such that

$$qM(y_q) = 1. \quad (4)$$

Set

$$\ell_=(q) = q\Phi(1) + q \int_1^{y_q} M(y)\phi(y) dy + \bar{\Phi}(y_q). \quad (5)$$

The next theorem shows that the lower bound can be approached arbitrarily closely by finite-dimensional Gaussian models with positive definite correlation matrices.

Theorem 2.1. *For $T \sim N(\theta, \Sigma)$, let $\text{FDR}_{\theta, \Sigma}(\text{BH}_q)$ denote the FDR of BH_q applied to T . For every $q \in (0, 1)$ and $\delta > 0$, there are $N \in \mathbb{N}$, $\theta \in \mathbb{R}^N$, and a positive definite correlation matrix Σ such that*

$$\text{FDR}_{\theta, \Sigma}(\text{BH}_q) > \ell_=(q) - \delta. \quad (6)$$

Consequently,

$$\sup_{\substack{N \in \mathbb{N}, \theta \in \mathbb{R}^N, \\ \Sigma \succ 0, \text{diag}(\Sigma) = \mathbf{1}_N}} \text{FDR}_{\theta, \Sigma}(\text{BH}_q) \geq \ell_=(q). \quad (7)$$

The next proposition gives the small- q expansion of the lower bound.

Proposition 2.2. *As $q \downarrow 0$,*

$$\ell_=(q) = \frac{q\sqrt{\log(1/q)}}{2\sqrt{\pi}} + c_\ell q + o(q),$$

where

$$\begin{aligned} c_\ell &= \Phi(1) - \frac{1}{2\sqrt{2\pi}} + \int_1^\infty \left\{ M(y)\phi(y) - \frac{1}{2\sqrt{2\pi}} \right\} dy \\ &= 0.6492828 \dots \end{aligned}$$

Proposition 2.3 further shows that $\ell_=(q) > q$ for every $q \in (0, 1)$. Together with Theorem 2.1, this implies that BH fails to control the FDR at every nontrivial nominal level in the correlated two-sided Gaussian setting.

Proposition 2.3. *For every $q \in (0, 1)$,*

$$\ell_=(q) > q.$$

Figure 1 in the Introduction displays the numerical behavior of the two-sided lower bound. Table 1 gives the absolute inflation $\ell_=(q) - q$ and relative inflation $\ell_=(q)/q$ for commonly used nominal levels.

2.2 Roadmap

Fix $q \in (0, 1)$ and a design law ν_q . Let $\Omega_N = ((\Theta_1, B_1), \dots, (\Theta_N, B_N))$, where the pairs are i.i.d. draws from ν_q , independently of $Z, \varepsilon_1, \dots, \varepsilon_N \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$, and set

$$T_i = \Theta_i + B_i Z + \sqrt{1 - B_i^2} \varepsilon_i, \quad i = 1, \dots, N.$$

Conditional on a realization of Ω_N , this is a finite deterministic Gaussian design. Since BH is a step-up procedure, its rejection set is $\{i : |T_i| \geq \widehat{C}_N\}$. We will show in Subsection 2.4 that, conditional on $Z = z$, $\widehat{C}_N \rightarrow C(z)$ and $\text{FDP}_N \rightarrow D(z)$. Consequently,

$$\mathbb{E}_{\Omega_N}[\text{FDR}_{\Omega_N}(\text{BH}_q)] = \mathbb{E}_{\Omega_N, Z, \varepsilon}[\text{FDP}_N] \longrightarrow \mathbb{E}[D(Z)].$$

Thus, for any $\delta > 0$ and all sufficiently large N , at least one deterministic realization ω_N of Ω_N satisfies

$$\text{FDR}_{\omega_N}(\text{BH}_q) \geq \mathbb{E}[D(Z)] - \delta.$$

We next construct a family of laws $\nu_{q,r}$. Write D_r for the conditional FDP limit associated with $\nu_{q,r}$. For every $z \notin \{-1, -y_q\}$, we prove that $D_r(z) \rightarrow D_*(z)$ as $r \downarrow 0$, where

$$D_*(z) = \begin{cases} q, & z \geq -1, \\ qM(-z), & -y_q < z < -1, \\ 1, & z \leq -y_q. \end{cases}$$

Because $0 \leq D_r, D_* \leq 1$, dominated convergence then gives

$$\mathbb{E}[D_r(Z)] \longrightarrow \mathbb{E}[D_*(Z)] = \ell_=(q).$$

The argument therefore takes the iterated limit $\lim_{r \downarrow 0} \lim_{N \rightarrow \infty}$: first $N \rightarrow \infty$ for each fixed r , and then $r \downarrow 0$. To obtain a single finite example, we choose r sufficiently small and then N sufficiently large. A small independent Gaussian perturbation finally makes its covariance matrix positive definite.

2.3 Population quantities

Put $u_q = p^{-1}(q)$. Let ν_q be a probability measure on $\mathbb{R} \times [-1, 1]$. Write

$$\pi_0 = \nu_q(\{0\} \times [-1, 1]), \tag{8}$$

and assume $\pi_0 > 0$.

For $(\theta, b) \in \mathbb{R} \times [-1, 1]$, $z \in \mathbb{R}$, and $s \in (0, 1]$, let $u_s = p^{-1}(s)$ and define

$$F_{\theta, b, z}(s) = \mathbb{P}\{p(|T_i|) \leq s \mid \Theta_i = \theta, B_i = b, Z = z\}.$$

If $|b| < 1$, then

$$F_{\theta,b,z}(s) = \bar{\Phi}\left(\frac{u_s - \theta - bz}{\sqrt{1-b^2}}\right) + \bar{\Phi}\left(\frac{u_s + \theta + bz}{\sqrt{1-b^2}}\right). \quad (9)$$

If $|b| = 1$, then

$$F_{\theta,b,z}(s) = \mathbf{1}\{p(|\theta + bz|) \leq s\}. \quad (10)$$

Set $F_{\theta,b,z}(0) = 0$.

Conditional on $Z = z$, the CDFs of the mixture p -value and the null p -value are

$$R_z(s) = \int F_{\theta,b,z}(s) \nu_q(d\theta, db), \quad (11)$$

$$R_{0,z}(s) = \frac{1}{\pi_0} \int_{\{0\} \times [-1,1]} F_{0,b,z}(s) \nu_q(d\theta, db). \quad (12)$$

For $c \geq u_q$, define the normalized quantities

$$\bar{R}_z(c) = \frac{qR_z\{p(c)\}}{p(c)}, \quad (13)$$

$$\bar{R}_{0,z}(c) = \frac{q\pi_0 R_{0,z}\{p(c)\}}{p(c)}. \quad (14)$$

Then the limiting cutoff can be written as

$$C(z) = \inf\{c \geq u_q : \bar{R}_z(c) \geq 1\}. \quad (15)$$

Here and below, $\inf \emptyset = \infty$. The limiting conditional FDP is defined by

$$D(z) = \begin{cases} \bar{R}_{0,z}\{C(z)\}, & C(z) < \infty, \\ 0, & C(z) = \infty. \end{cases} \quad (16)$$

We call a finite cutoff a regular first crossing when it satisfies the following requirements.

Definition 2.4. A finite cutoff $C(z) > u_q$ is regular if

$$\sup_{u_q \leq c \leq C(z) - \rho} \bar{R}_z(c) < 1 \quad \text{for every } 0 < \rho < C(z) - u_q, \quad (17)$$

$$\sup_{C(z) < c < C(z) + \rho} \bar{R}_z(c) > 1 \quad \text{for every } \rho > 0, \quad (18)$$

and $c \mapsto \bar{R}_{0,z}(c)$ is continuous at $C(z)$.

The following proposition reduces regularity to a local monotonicity check.

Proposition 2.5. Let $C(z) > u_q$ be finite. Suppose there exist $u_q < a < C(z) < b$ such that the map $c \mapsto \bar{R}_z(c)$ is differentiable on (a, b) and satisfies

$$\bar{R}'_z(c) > 0, \quad a < c < b. \quad (19)$$

Suppose also that $c \mapsto \bar{R}_{0,z}(c)$ is continuous at $C(z)$. Then $C(z)$ is regular.

Proof. The map $c \mapsto R_z\{p(c)\}$ is a survival function and hence is upper semicontinuous. Since p is positive and continuous, $c \mapsto \bar{R}_z(c)$ is also upper semicontinuous. By the definition of $C(z)$, $\bar{R}_z(c) < 1$ for every $c < C(z)$. Therefore, for every $0 < \rho < C(z) - u_q$, upper semicontinuity and compactness give

$$\sup_{u_q \leq c \leq C(z) - \rho} \bar{R}_z(c) < 1,$$

which proves (17).

There is a sequence $c_n \geq C(z)$ decreasing to $C(z)$ such that $\bar{R}_z(c_n) \geq 1$. Continuity on (a, b) therefore gives $\bar{R}_z\{C(z)\} = 1$. The mean value theorem and (19) imply strict monotonicity on (a, b) , which yields

$$\bar{R}_z(c) > 1 \quad \text{for } C(z) < c < b.$$

This gives (18). Continuity of the null contribution at $C(z)$ completes Definition 2.4. $\square$

2.4 Limiting cutoff and FDR

Let

$$\Omega_N = ((\Theta_1, B_1), \dots, (\Theta_N, B_N)), \quad (\Theta_i, B_i) \stackrel{\text{i.i.d.}}{\sim} \nu_q. \quad (20)$$

Since the distribution of $(T_1, \dots, T_N)$ is fully determined by Ω_N , we write $\text{FDR}_{\theta, \Sigma}(\text{BH}_q)$ as $\text{FDR}_{\Omega_N}(\text{BH}_q)$.

Define $P_i = p(|T_i|)$.

$$\hat{R}_N(s) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{P_i \leq s\}, \quad (21)$$

$$\hat{R}_{0,N}(s) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\Theta_i = 0, P_i \leq s\}. \quad (22)$$

The following lemma gives a uniform conditional empirical-CDF bound. For a function f on $[0, 1]$, write $\|f\|_{\infty, [0, 1]} = \sup_{0 \leq s \leq 1} |f(s)|$.

Lemma 2.6. *For every $z \in \mathbb{R}$ and $t > 0$,*

$$\mathbb{P}\left(\max\left\{\|\hat{R}_N - R_z\|_{\infty, [0, 1]}, \|\hat{R}_{0,N} - \pi_0 R_{0,z}\|_{\infty, [0, 1]}\right\} > t \mid Z = z\right) \leq 4e^{-2Nt^2}. \quad (23)$$

Proof. Conditional on $Z = z$, the P_i are i.i.d. with CDF R_z . Also define

$$W_i = P_i \mathbf{1}\{\Theta_i = 0\} + 2\mathbf{1}\{\Theta_i \neq 0\}.$$

For $0 \leq s \leq 1$,

$$\mathbf{1}\{W_i \leq s\} = \mathbf{1}\{\Theta_i = 0, P_i \leq s\}, \quad \mathbb{P}(W_i \leq s \mid Z = z) = \pi_0 R_{0,z}(s).$$

Apply the Dvoretzky–Kiefer–Wolfowitz inequality twice and take a union bound. $\square$

The next lemma gives an exact characterization of the BH cutoff and FDP through the empirical CDFs.

Lemma 2.7. *Let R_N be the number of BH rejections and put*

$$\hat{\tau}_N = \frac{qR_N}{N}. \quad (24)$$

If $R_N \geq 1$, then

$$\hat{R}_N(\hat{\tau}_N) = \frac{R_N}{N}, \quad \text{FDP}_N = \frac{q\hat{R}_{0,N}(\hat{\tau}_N)}{\hat{\tau}_N}. \quad (25)$$

With $\hat{C}_N = p^{-1}(\hat{\tau}_N)$,

$$\hat{C}_N = \inf \left\{ c \geq u_q : \frac{q\hat{R}_N\{p(c)\}}{p(c)} \geq 1 \right\}. \quad (26)$$

Then the BH rejection set is $\{i : |T_i| \geq \hat{C}_N\}$. If $R_N = 0$, the set in (26) is empty and $\hat{C}_N = \infty$.

Proof. Let $P_{(1)} \leq \dots \leq P_{(N)}$. If more than R_N values were at most qR_N/N , then

$$P_{(R_N+1)} \leq qR_N/N < q(R_N+1)/N,$$

contradicting the maximality of R_N . Hence $N\hat{R}_N(\hat{\tau}_N) = R_N$, and (25) follows. Since p is decreasing, the BH step-up definition is equivalent to (26). $\square$

The next theorem establishes the limiting cutoff conditional on the common factor Z .

Theorem 2.8. Assume that $C(z)$ is regular for almost every z . Then, for every z at which $C(z)$ is regular, conditional on $Z = z$,

$$\hat{C}_N \longrightarrow C(z), \quad \text{FDP}_N \longrightarrow D(z) = \bar{R}_{0,z}\{C(z)\} = \frac{q\pi_0 R_{0,z}[p\{C(z)\}]}{p\{C(z)\}} \quad (27)$$

in probability. Consequently,

$$\lim_{N \rightarrow \infty} \mathbb{E}_{\Omega_N} [\text{FDR}_{\Omega_N}(\text{BH}_q)] = \mathbb{E}[D(Z)], \quad (28)$$

where

$$\text{FDR}_{\Omega_N}(\text{BH}_q) = \mathbb{E}_{Z,\varepsilon} [\text{FDP}_N \mid \Omega_N]. \quad (29)$$

Proof. Fix a regular z and $\rho, \tau > 0$. Since $D(z) = \bar{R}_{0,z}\{C(z)\}$ and $c \mapsto \bar{R}_{0,z}(c)$ is continuous at $C(z)$, choose

$$0 < \rho(z) < \min\{\rho, C(z) - u_q\}$$

so that

$$\sup_{C(z)-\rho(z) \leq c \leq C(z)+\rho(z)} |\bar{R}_{0,z}(c) - D(z)| \leq \tau. \quad (30)$$

Set $A(z) = C(z) - \rho(z)$. Equation (17) gives $\sup_{u_q \leq c \leq A(z)} \bar{R}_z(c) < 1$, and (18) permits a choice of $B(z) \in (C(z), C(z) + \rho(z))$ with $\bar{R}_z\{B(z)\} > 1$. Set

$$\gamma(z) = \frac{1}{2} \min \left\{ 1 - \sup_{u_q \leq c \leq A(z)} \bar{R}_z(c), \quad \bar{R}_z\{B(z)\} - 1 \right\} > 0.$$

Put

$$E_N(z) = \max \left\{ \|\hat{R}_N - R_z\|_{\infty, [0,1]}, \|\hat{R}_{0,N} - \pi_0 R_{0,z}\|_{\infty, [0,1]} \right\}.$$

Lemma 2.6 gives $E_N(z) \rightarrow 0$ in probability conditional on $Z = z$. For every $u_q \leq c \leq B(z)$,

$$\begin{aligned} \left| \frac{q\hat{R}_N\{p(c)\}}{p(c)} - \bar{R}_z(c) \right| &\leq \frac{qE_N(z)}{p\{B(z)\}}, \\ \left| \frac{q\hat{R}_{0,N}\{p(c)\}}{p(c)} - \bar{R}_{0,z}(c) \right| &\leq \frac{qE_N(z)}{p\{B(z)\}}. \end{aligned} \quad (31)$$

By the empirical curve we mean

$$c \mapsto \frac{q\hat{R}_N\{p(c)\}}{p(c)}.$$

On the event $qE_N(z)/p\{B(z)\} < \gamma(z)$, the definition of $\gamma(z)$ and (31) give

$$\begin{aligned} \sup_{u_q \leq c \leq A(z)} \frac{q\hat{R}_N\{p(c)\}}{p(c)} &\leq \sup_{u_q \leq c \leq A(z)} \bar{R}_z(c) + \frac{qE_N(z)}{p\{B(z)\}} \\ &< 1 - 2\gamma(z) + \gamma(z) < 1, \end{aligned}$$

whereas

$$\frac{q\hat{R}_N\{p(B(z))\}}{p(B(z))} \geq \bar{R}_z\{B(z)\} - \frac{qE_N(z)}{p\{B(z)\}} > 1 + 2\gamma(z) - \gamma(z) > 1.$$

Lemma 2.7 and (30) therefore give

$$A(z) < \hat{C}_N \leq B(z), \quad |\text{FDP}_N - D(z)| \leq \tau + \frac{qE_N(z)}{p\{B(z)\}}.$$

Because $\rho, \tau > 0$ were arbitrary, this proves (27) at every regular z .

For completeness, the quantities in the expectation can be taken to be measurable without making a measurable selection of $B(z)$. The maps $(z, c) \mapsto \bar{R}_z(c)$ and $(z, c) \mapsto \bar{R}_{0,z}(c)$ are jointly Borel measurable, and the first is left-continuous in c . Define

$$C^\circ(z) = \inf\{c \in \mathbb{Q} : c \geq u_q, \bar{R}_z(c) > 1\}.$$

At every regular z , left continuity and (18) imply $C^\circ(z) = C(z)$. Thus C has a measurable version. Moreover, for $c < C(z)$, $\bar{R}_{0,z}(c) \leq \bar{R}_z(c) < 1$; continuity of the null contribution at the crossing gives $0 \leq D(z) \leq 1$. Hence

$$D^\circ(z) = \begin{cases} \min[1, \bar{R}_{0,z}\{C^\circ(z)\}], & C^\circ(z) < \infty, \\ 0, & C^\circ(z) = \infty, \end{cases}$$

is a measurable $[0, 1]$ -valued version of $D(z)$ at almost every z . Use these versions below. Conditional convergence in probability, together with boundedness, now yields

$$\mathbb{E}[|\text{FDP}_N - D(z)| \mid Z = z] \longrightarrow 0 \quad \text{for almost every } z.$$

Dominated convergence and the tower property give

$$\mathbb{E}_{\Omega_N, Z, \varepsilon}[|\text{FDP}_N - D(Z)|] \longrightarrow 0,$$

and a second application of the tower property yields (28). $\square$

Lastly, we show that Σ can be made positive definite by an independent Gaussian perturbation. For a fixed realization $\omega = ((\theta_i, b_i))_{i=1}^N$, let $\xi \sim N(0, I_N)$ be independent and define

$$T_i^{(\eta)} = \theta_i + \sqrt{1 - \eta^2} \left(b_i Z + \sqrt{1 - b_i^2} \varepsilon_i \right) + \eta \xi_i, \quad 0 < \eta < 1. \quad (32)$$

Then

$$\Sigma_{\omega, \eta} = (1 - \eta^2) \left[b b^\top + \text{diag}(1 - b_1^2, \dots, 1 - b_N^2) \right] + \eta^2 I_N. \quad (33)$$

The following lemma shows that the perturbation makes the correlation matrix positive definite while preserving the FDR in the limit.

Lemma 2.9. *For every fixed ω ,*

$$\text{diag}(\Sigma_{\omega, \eta}) = \mathbf{1}_N, \quad \Sigma_{\omega, \eta} \succeq \eta^2 I_N, \quad (34)$$

and

$$\text{FDR}_\omega^{(\eta)}(\text{BH}_q) \longrightarrow \text{FDR}_\omega(\text{BH}_q) \quad (\eta \downarrow 0). \quad (35)$$

Proof. Equation (34) is immediate from (33). Under the coupling (32), $T^{(\eta)} \rightarrow T$ almost surely. Define $P_i^{(\eta)} = p(|T_i^{(\eta)}|)$. Then $P_i^{(\eta)} \rightarrow P_i = p(|T_i|)$ almost surely. Each P_i has a continuous marginal law, so

$$\mathbb{P} \left\{ P_i \in \left\{ \frac{qk}{N} : 1 \leq k \leq N \right\} \text{ for some } i \right\} = 0.$$

Off this event, the truth values of all finitely many comparisons $P_i^{(\eta)} \leq qk/N$ agree with those of $P_i \leq qk/N$ for all sufficiently small η . Hence the BH rejection count and rejected-null set are eventually constant. Bounded convergence proves (35). $\square$

2.5 Construction of $\nu_{q,r}$

We construct a family of laws $\nu_{q,r}$ indexed by $r \downarrow 0$. For $a, y \geq 0$, put

$$L_a(y) = e^{-a^2/2} \cosh(ay).$$

For $y > 1$, let

$$a(y) = \arg \max_{\alpha \geq 0} L_\alpha(y), \quad y(\alpha) = a^{-1}(\alpha), \quad \alpha > 0.$$

Set

$$a_q = a(y_q). \tag{36}$$

Appendix A establishes the properties of these functions that we use below.

The quantities $b_r, \tau_r, \mu_r, \pi_{0,r}$, and c_r are defined below. The law $\nu_{q,r}$ has the three components displayed in Table 3.

Table 3: Two-sided population construction defining $\nu_{q,r}$.			
Type	Mass	Θ	B
Nulls	$\pi_{0,r}$	0	r
Primary nonnulls	r	$c_r + y(b_r)$	1
Secondary nonnulls	$\tau_r \mu_r(\mathrm{d}a)$	$a/r + y(a)$	1, $a \in [b_r, a_q]$

Equivalently, for every bounded measurable $f : \mathbb{R} \times [-1, 1] \rightarrow \mathbb{R}$,

$$\int f \, \mathrm{d}\nu_{q,r} = \pi_{0,r} f(0, r) + r f(c_r + y(b_r), 1) + \tau_r \int_{b_r}^{a_q} f\left(\frac{a}{r} + y(a), 1\right) \mu_r(\mathrm{d}a). \tag{37}$$

Definition of secondary nonnulls. For small r , set

$$b_r = r p^{-1}(r^2).$$

For the remaining definitions, take r small enough that $0 < b_r < a_q$. For $a \in [b_r, a_q]$, define

$$W_r(a) = \frac{[1 - qM\{y(a)\}]p(a/r)}{q}. \tag{38}$$

By Lemma A.1 and (198), the factor $1 - qM\{y(a)\}$ is nonnegative and decreasing on $[b_r, a_q]$; the factor $p(a/r)$ is decreasing in a . Hence W_r is continuous and decreasing, with $W_r(a_q) = 0$. Set

$$\tau_r := W_r(b_r) = \frac{[1 - qM\{y(b_r)\}]r^2}{q} = O(r^2). \tag{39}$$

We therefore call this the secondary nonnull block; its total mass is $O(r^2) = o(r)$. Let μ_r be the probability measure on $[b_r, a_q]$ determined by

$$\mu_r([a, a_q]) = \frac{W_r(a)}{\tau_r}, \quad b_r \leq a \leq a_q. \tag{40}$$

In particular, $\mu_r([b_r, a_q]) = W_r(b_r)/\tau_r = 1$. **Definition of nulls.** Put

$$\pi_{0,r} = 1 - r - \tau_r.$$

Since $\pi_{0,r} + r + \tau_r = 1$, the masses in (37) define a probability measure.

Definition of primary nonnulls. For all sufficiently small r , $\pi_{0,r} > 0$. Define c_r by

$$p(c_r) = \frac{q(r + \tau_r)}{1 - q\pi_{0,r}}. \quad (41)$$

The numerator is positive. Since $\tau_r = O(r^2)$ and $\pi_{0,r} = 1 - r - \tau_r$,

$$1 - q\pi_{0,r} = 1 - q + qr + q\tau_r \longrightarrow 1 - q > 0, \quad q(r + \tau_r) \longrightarrow 0.$$

Thus the right side of (41) is positive and smaller than one for small r , so c_r is well-defined.

2.6 Population FDR analysis under $\nu_{q,r}$

2.6.1 Preliminaries

Apply Subsection 2.3 with $\nu_q = \nu_{q,r}$. Denote the resulting quantities by

$$R_{r,z}, \quad R_{0,r,z}, \quad \bar{R}_{r,z}, \quad \bar{R}_{0,r,z}, \quad C_r(z).$$

For $c \geq u_q$,

$$\bar{R}_{r,z}(c) = \bar{R}_{0,r,z}(c) + \bar{R}_{1,r,z}(c) + \bar{R}_{2,r,z}(c), \quad (42)$$

where

$$\begin{aligned} \bar{R}_{0,r,z}(c) &= q\pi_{0,r} \frac{R_{0,r,z}\{p(c)\}}{p(c)}, \\ \bar{R}_{1,r,z}(c) &= \frac{qr}{p(c)} \mathbf{1}\{|c_r + y(b_r) + z| \geq c\}, \\ \bar{R}_{2,r,z}(c) &= \frac{q\tau_r}{p(c)} \int_{b_r}^{a_q} \mathbf{1}\left\{\left|\frac{a}{r} + y(a) + z\right| \geq c\right\} \mu_r(da). \end{aligned} \quad (43)$$

For a measurable definition at every z , put

$$C_r^\circ(z) = \inf\{c \in \mathbb{Q} : c \geq u_q, \bar{R}_{r,z}(c) > 1\}$$

and define

$$D_r(z) = \begin{cases} \min[1, \bar{R}_{0,r,z}\{C_r^\circ(z)\}], & C_r^\circ(z) < \infty, \\ 0, & C_r^\circ(z) = \infty. \end{cases}$$

The joint Borel measurability of $(z, c) \mapsto \bar{R}_{r,z}(c)$ and $(z, c) \mapsto \bar{R}_{0,r,z}(c)$ makes both C_r° and D_r Borel measurable. At every regular crossing, the rational-cutoff argument in the proof of Theorem 2.8 gives $C_r^\circ(z) = C_r(z)$, and $\bar{R}_{0,r,z}\{C_r(z)\} \leq 1$. Hence

$$D_r(z) = \bar{R}_{0,r,z}\{C_r(z)\}. \quad (44)$$

Moreover, $0 \leq D_r(z) \leq 1$ for every z .

We next state two asymptotic lemmas that will be applied in the following analysis. The first gives the order of b_r and c_r as $r \downarrow 0$.

Lemma 2.10. *As $r \downarrow 0$,*

$$b_r \downarrow 0, \quad \frac{b_r}{r} = p^{-1}(r^2) \rightarrow \infty, \quad p(b_r/r) = r^2, \quad y(b_r) \downarrow 1. \quad (45)$$

Moreover,

$$p(c_r) \sim \frac{qr}{1-q}, \quad c_r \rightarrow \infty, \quad rc_r \rightarrow 0, \quad \frac{b_r}{r} - c_r \rightarrow \infty. \quad (46)$$

Proof. The identity $p(b_r/r) = r^2$ follows directly from $b_r = r p^{-1}(r^2)$. Applying (207) in Lemma A.2 gives

$$\frac{b_r}{r} = p^{-1}(r^2) \sim \sqrt{4 \log(1/r)} \rightarrow \infty, \quad b_r \sim 2r \sqrt{\log(1/r)} \rightarrow 0.$$

The last limit in (45) follows from Lemma A.1: $a(y)$ is continuous and strictly increasing from $(1, \infty)$ onto $(0, \infty)$, with $\lim_{y \downarrow 1} a(y) = 0$. Since $y(\cdot)$ is its inverse and $b_r \downarrow 0$, we have $y(b_r) \downarrow 1$.

By (41) and (39),

$$p(c_r) = \frac{q(r + \tau_r)}{1 - q\pi_{0,r}} \sim \frac{qr}{1 - q}.$$

For every fixed $\delta \in (0, q/(1 - q))$, for all sufficiently small r ,

$$\left\{ \frac{q}{1 - q} - \delta \right\} r \leq p(c_r) \leq \left\{ \frac{q}{1 - q} + \delta \right\} r.$$

Since p^{-1} is decreasing,

$$p^{-1}\left(\left\{ \frac{q}{1 - q} + \delta \right\} r\right) \leq c_r \leq p^{-1}\left(\left\{ \frac{q}{1 - q} - \delta \right\} r\right).$$

Equation (207) in Lemma A.2, applied to the two fixed positive constants, shows that both endpoints are asymptotic to $\sqrt{2 \log(1/r)}$. Hence

$$c_r \sim \sqrt{2 \log(1/r)}.$$

Hence $c_r \rightarrow \infty$ and $rc_r \rightarrow 0$. Combining this with

$$\frac{b_r}{r} = p^{-1}(r^2) \sim \sqrt{4 \log(1/r)}$$

gives

$$\frac{b_r}{r} - c_r \sim (2 - \sqrt{2}) \sqrt{\log(1/r)} \rightarrow \infty.$$

□

The next lemma gives a uniform approximation to the normalized conditional FDP contribution from nulls.

Lemma 2.11. *For every $A > 0$ and $T \geq 0$, as $r \downarrow 0$,*

$$\sup_{|z| \leq T} \sup_{0 \leq c \leq A/r} \left| \frac{R_{0,r,z}\{p(c)\}}{p(c)} - L_{rc}(|z|) \right| \rightarrow 0. \quad (47)$$

Proof. For $0 < r < 1$, the null pair $(\Theta, B) = (0, r)$ gives

$$R_{0,r,z}\{p(c)\} = \mathbb{P}\{|rz + \sqrt{1 - r^2} \varepsilon| \geq c\}.$$

Write

$$x_{\pm} = \frac{c \mp rz}{\sqrt{1 - r^2}}.$$

Then

$$R_{0,r,z}\{p(c)\} = \bar{\Phi}(x_+) + \bar{\Phi}(x_-).$$

Fix $C > 0$. Uniformly for $|z| \leq T$ and $C \leq c \leq A/r$,

$$\begin{aligned} x_{\pm} - c &= c \left\{ \frac{1}{\sqrt{1 - r^2}} - 1 \right\} \mp \frac{rz}{\sqrt{1 - r^2}} \\ &= \frac{cr^2}{\sqrt{1 - r^2}\{1 + \sqrt{1 - r^2}\}} \mp \frac{rz}{\sqrt{1 - r^2}}. \end{aligned}$$

Therefore,

$$|x_{\pm} - c| \leq \frac{Ar}{\sqrt{1-r^2}\{1 + \sqrt{1-r^2}\}} + \frac{Tr}{\sqrt{1-r^2}} = O_{A,T}(r).$$

Consequently, for all sufficiently small r , the shifts $x_{\pm} - c$ are uniformly bounded and $x_{\pm} \geq C/2$. The uniform expansion (206) in Lemma A.2 therefore gives remainders $\rho_{r,C,\pm}(c, z)$ such that

$$\begin{aligned} \frac{\bar{\Phi}(x_{\pm})}{\bar{\Phi}(c)} &= \frac{c}{x_{\pm}} \exp \left\{ -\frac{x_{\pm}^2 - c^2}{2} \right\} \{1 + \rho_{r,C,\pm}(c, z)\}, \\ -\frac{x_{\pm}^2 - c^2}{2} &= -\frac{(rc)^2}{2} \pm rcz + O_{A,T}(r^2). \end{aligned} \quad (48)$$

and

$$\lim_{C \rightarrow \infty} \limsup_{r \downarrow 0} \sup_{|z| \leq T} \sup_{C \leq c \leq A/r} \max_{\pm} |\rho_{r,C,\pm}(c, z)| = 0. \quad (49)$$

For each fixed $C > 0$, $c/x_{\pm} \rightarrow 1$ uniformly for $|z| \leq T$ and $C \leq c \leq A/r$ as $r \downarrow 0$. To identify the leading term, put $t = rc$, so that $0 \leq t \leq A$. Then

$$\frac{1}{2} \left[\exp \left\{ -\frac{t^2}{2} + tz \right\} + \exp \left\{ -\frac{t^2}{2} - tz \right\} \right] = e^{-t^2/2} \cosh(tz) = L_t(|z|).$$

Because t and z range over compact sets, these exponential factors are uniformly bounded. Thus, after summing the two tails in (48) and using $p(c) = 2\bar{\Phi}(c)$, the errors from $c/x_{\pm} - 1$ and from the $O_{A,T}(r^2)$ term vanish uniformly for each fixed C . Consequently, there is a constant $K_{A,T} < \infty$, independent of C , such that, for

$$\eta(C) = \limsup_{r \downarrow 0} \sup_{|z| \leq T} \sup_{C \leq c \leq A/r} \left| \frac{R_{0,r,z}\{p(c)\}}{p(c)} - L_{rc}(|z|) \right|,$$

we have

$$\eta(C) \leq K_{A,T} \limsup_{r \downarrow 0} \sup_{|z| \leq T} \sup_{C \leq c \leq A/r} \max_{\pm} |\rho_{r,C,\pm}(c, z)|.$$

The right side tends to zero as $C \rightarrow \infty$ by (49). Hence $\eta(C) \rightarrow 0$.

It remains to control $0 \leq c \leq C$. On this range, $p(c) \geq p(C) > 0$. Also, uniformly over $0 \leq c \leq C$ and $|z| \leq T$,

$$R_{0,r,z}\{p(c)\} = \bar{\Phi}\left(\frac{c - rz}{\sqrt{1-r^2}}\right) + \bar{\Phi}\left(\frac{c + rz}{\sqrt{1-r^2}}\right) = p(c) + O_{C,T}(r),$$

because the two arguments differ from c by $O_{C,T}(r)$ and $\bar{\Phi}$ has bounded derivative. Hence

$$\sup_{|z| \leq T} \sup_{0 \leq c \leq C} \left| \frac{R_{0,r,z}\{p(c)\}}{p(c)} - 1 \right| \rightarrow 0.$$

Moreover, $L_{rc}(|z|) = e^{-(rc)^2/2} \cosh(rc|z|) \rightarrow 1$ uniformly for $|z| \leq T$ and $0 \leq c \leq C$. Thus

$$\limsup_{r \downarrow 0} \sup_{|z| \leq T} \sup_{0 \leq c \leq A/r} \left| \frac{R_{0,r,z}\{p(c)\}}{p(c)} - L_{rc}(|z|) \right| \leq \eta(C).$$

Letting $C \rightarrow \infty$ proves the claim. $\square$

The following lemma provides a generic criterion for deriving the limit of $C_r(z)$.

Lemma 2.12. *Let K be compact. For $z \in K$, let $m_r(z) > u_q$ and $w_r(z) > 0$. Suppose that, for every fixed $H > 0$,*

$$\inf_{z \in K} \{m_r(z) - Hw_r(z)\} > u_q \quad (50)$$

for all sufficiently small r , and

$$\sup_{z \in K} \sup_{|h| \leq H} \left| \bar{R}_{r,z} \{m_r(z) + w_r(z)h\} - d(z) - \{1 - d(z)\}e^{\xi(z)h} \right| \longrightarrow 0, \quad (51)$$

where $d : K \rightarrow [0, 1]$ and $\xi : K \rightarrow (0, \infty)$ are continuous. Suppose also that, for every $h_0 > 0$,

$$\limsup_{r \downarrow 0} \sup_{z \in K} \sup_{u_q \leq c \leq m_r(z) - w_r(z)h_0} \bar{R}_{r,z}(c) < 1. \quad (52)$$

Then

$$\sup_{z \in K} \left| \frac{C_r(z) - m_r(z)}{w_r(z)} \right| \longrightarrow 0. \quad (53)$$

Proof. Fix $h_0 > 0$. Compactness gives

$$\inf_{z \in K} \{1 - d(z)\} \{e^{\xi(z)h_0} - 1\} > 0.$$

Hence (51) gives $\bar{R}_{r,z} \{m_r(z) + w_r(z)h_0\} > 1$ uniformly on K , while (52) excludes a crossing at or before $m_r(z) - w_r(z)h_0$. Thus

$$m_r(z) - w_r(z)h_0 < C_r(z) \leq m_r(z) + w_r(z)h_0$$

uniformly on K . Letting $h_0 \downarrow 0$ proves (53). $\square$

The next lemma gives regularity and the limiting null contribution once the localized cutoff has been identified.

Lemma 2.13. *Under the assumptions and notation of Lemma 2.12, assume in addition that, for every fixed $H > 0$,*

$$\sup_{z \in K} \sup_{|h| \leq H} \left| \bar{R}_{0,r,z} \{m_r(z) + w_r(z)h\} - d(z) \right| \longrightarrow 0. \quad (54)$$

If, for some $H_ > 0$ and all sufficiently small r , the map*

$$h \longmapsto \bar{R}_{r,z} \{m_r(z) + w_r(z)h\}$$

is differentiable with positive derivative on $(-H_, H_*)$, and the map $h \mapsto \bar{R}_{0,r,z} \{m_r(z) + w_r(z)h\}$ is continuous there for every $z \in K$, then $C_r(z)$ is regular and*

$$\sup_{z \in K} |D_r(z) - d(z)| \longrightarrow 0. \quad (55)$$

Proof. For all sufficiently small r , (53) gives

$$m_r(z) - w_r(z)H_* < C_r(z) < m_r(z) + w_r(z)H_*.$$

With $c = m_r(z) + w_r(z)h$, the chain rule gives

$$\frac{d}{dh} \bar{R}_{r,z} \{m_r(z) + w_r(z)h\} = w_r(z) \bar{R}'_{r,z}(c).$$

Since $w_r(z) > 0$,

$$0 < \frac{d}{dh} \bar{R}_{r,z} \{m_r(z) + w_r(z)h\} = w_r(z) \bar{R}'_{r,z}(c) \implies \bar{R}'_{r,z}(c) > 0.$$

Thus

$$\bar{R}'_{r,z}(c) > 0, \quad m_r(z) - w_r(z)H_* < c < m_r(z) + w_r(z)H_*.$$

Equations (53) and (50), together with the derivative bound above, verify the hypotheses of Proposition 2.5, which gives regularity. Finally, put

$$h_r(z) = \frac{C_r(z) - m_r(z)}{w_r(z)}.$$

Equation (53) gives $h_r(z) \rightarrow 0$ uniformly. By (44),

$$D_r(z) = \bar{R}_{0,r,z}\{C_r(z)\} = \bar{R}_{0,r,z}\{m_r(z) + w_r(z)h_r(z)\}.$$

Evaluating (54) at $h = h_r(z)$ therefore proves (55). $\square$

The next lemma derives the derivatives of the null and secondary nonnull contributions.

Lemma 2.14. *For $c > 0$, put*

$$x_{\pm} = \frac{c \mp rz}{\sqrt{1-r^2}}, \quad Q_{\pm}(c, z) = \frac{\bar{\Phi}(x_{\pm})}{\bar{\Phi}(c)}.$$

The null contribution $\bar{R}_{0,r,z}(c)$ satisfies

$$\bar{R}'_{0,r,z}(c) = \frac{q\pi_{0,r}}{2} \sum_{\sigma \in \{-, +\}} Q_{\sigma}(c, z) \left\{ \lambda(c) - \frac{\lambda(x_{\sigma})}{\sqrt{1-r^2}} \right\}, \quad (56)$$

$$\frac{d}{dc} \log \bar{R}_{0,r,z}(c) = \lambda(c) - \frac{Q_+(c, z)\lambda(x_+) + Q_-(c, z)\lambda(x_-)}{\sqrt{1-r^2}\{Q_+(c, z) + Q_-(c, z)\}}. \quad (57)$$

Define

$$g_r(a) = a + ry(a). \quad (58)$$

For $0 < a < a_q$,

$$g'_r(a) = 1 + ry'(a) > 1,$$

so g_r is strictly increasing on $[b_r, a_q]$. Finally, fix $y > 1$ and $c > 0$, and suppose that

$$g_r(b_r) > r(y - c). \quad (59)$$

If

$$g_r(b_r) < r(c + y) < g_r(a_q),$$

let $\beta_{r,y}(c) \in (b_r, a_q)$ be the unique solution of

$$g_r\{\beta_{r,y}(c)\} = r(c + y). \quad (60)$$

Conditional on $Z = -y$, a secondary nonnull indexed by $a \in [b_r, a_q]$ is then rejected at cutoff c if and only if

$$a \in [\beta_{r,y}(c), a_q]. \quad (61)$$

Whenever $1 - qM\{y(\beta_{r,y}(c))\} > 0$, the secondary nonnull contribution $\bar{R}_{2,r,-y}(c)$ satisfies

$$\frac{\partial \beta_{r,y}(c)}{\partial c} = \frac{r}{1 + ry'\{\beta_{r,y}(c)\}}, \quad (62)$$

$$\begin{aligned} \frac{d}{dc} \log \bar{R}_{2,r,-y}(c) = & -\frac{qM'\{y(\beta_{r,y}(c))\}y'\{\beta_{r,y}(c)\}}{1 - qM\{y(\beta_{r,y}(c))\}} \frac{r}{1 + ry'\{\beta_{r,y}(c)\}} \\ & - \frac{\lambda\{\beta_{r,y}(c)/r\}}{1 + ry'\{\beta_{r,y}(c)\}} + \lambda(c), \end{aligned} \quad (63)$$

$$\bar{R}'_{2,r,-y}(c) = \bar{R}_{2,r,-y}(c) \frac{d}{dc} \log \bar{R}_{2,r,-y}(c). \quad (64)$$

If $r(c+y) \leq g_r(b_r)$, the secondary non-null block is fully rejected. Under the strict inequality $r(c+y) < g_r(b_r)$,

$$\bar{R}_{2,r,-y}(c) = \frac{q\tau_r}{p(c)}, \quad \bar{R}'_{2,r,-y}(c) = -\frac{q\tau_r p'(c)}{p(c)^2} = \lambda(c)\bar{R}_{2,r,-y}(c).$$

If $r(c+y) \geq g_r(a_q)$, the secondary non-null contribution vanishes: at equality the rejected set is $\{a_q\}$, which has μ_r -measure zero by (40) and $W_r(a_q) = 0$. Under the strict inequality $r(c+y) > g_r(a_q)$,

$$\bar{R}_{2,r,-y}(c) = \bar{R}'_{2,r,-y}(c) = 0.$$

As $r(c+y) \downarrow g_r(b_r)$, $\beta_{r,y}(c) \downarrow b_r$; as $r(c+y) \uparrow g_r(a_q)$, $\beta_{r,y}(c) \uparrow a_q$. Since W_r is continuous, $W_r(b_r) = \tau_r$, and $W_r(a_q) = 0$, the interior expression

$$\bar{R}_{2,r,-y}(c) = \frac{qW_r\{\beta_{r,y}(c)\}}{p(c)}$$

matches the endpoint values. Thus the contribution is continuous at both equalities. Its one-sided derivatives need not agree there, so a two-sided derivative is not asserted.

Proof. Since

$$\frac{dx_{\pm}}{dc} = \frac{1}{\sqrt{1-r^2}}, \quad \frac{d}{dt} \log \bar{\Phi}(t) = -\lambda(t),$$

we have

$$\frac{d}{dc} \log Q_{\pm}(c, z) = -\frac{\lambda(x_{\pm})}{\sqrt{1-r^2}} + \lambda(c).$$

Multiplying by $Q_{\pm}(c, z)$ and summing gives

$$\begin{aligned} \bar{R}'_{0,r,z}(c) &= \frac{q\pi_{0,r}}{2} \{Q'_+(c, z) + Q'_-(c, z)\} \\ &= \frac{q\pi_{0,r}}{2} \sum_{\sigma \in \{-, +\}} Q_{\sigma}(c, z) \left\{ \lambda(c) - \frac{\lambda(x_{\sigma})}{\sqrt{1-r^2}} \right\}, \end{aligned}$$

which is (56), and

$$\begin{aligned} \frac{\bar{R}'_{0,r,z}(c)}{\bar{R}_{0,r,z}(c)} &= \frac{Q'_+(c, z) + Q'_-(c, z)}{Q_+(c, z) + Q_-(c, z)} \\ &= \lambda(c) - \frac{Q_+(c, z)\lambda(x_+) + Q_-(c, z)\lambda(x_-)}{\sqrt{1-r^2}\{Q_+(c, z) + Q_-(c, z)\}}, \end{aligned}$$

which is (57).

For a secondary nonnull indexed by a , conditional on $Z = -y$,

$$T(a) = \frac{a}{r} + y(a) - y, \quad r\{T(a) + y\} = g_r(a).$$

By (58) and the positivity of y' in Lemma A.1, $g'_r(a) = 1 + ry'(a) > 1$, proving the asserted strict monotonicity. Equation (59) gives

$$g_r(a) \geq g_r(b_r) > r(y - c) \implies T(a) > -c,$$

so rejection can occur only through $T(a) \geq c$. In the interior case, (60) and strict monotonicity give

$$T(a) \geq c \iff g_r(a) \geq r(c+y) \iff a \geq \beta_{r,y}(c).$$

This proves (61). Thus

$$\begin{aligned}\bar{R}_{2,r,-y}(c) &= \frac{q\tau_r}{p(c)}\mu_r([\beta_{r,y}(c), a_q]) \\ &= \frac{qW_r\{\beta_{r,y}(c)\}}{p(c)} \\ &= [1 - qM\{y(\beta_{r,y}(c))\}] \frac{p\{\beta_{r,y}(c)/r\}}{p(c)},\end{aligned}$$

by (40) and (38).

Differentiating the threshold equation yields

$$[1 + ry'\{\beta_{r,y}(c)\}] \frac{\partial \beta_{r,y}(c)}{\partial c} = r,$$

which proves (62). Since $p'(t)/p(t) = -\lambda(t)$,

$$\begin{aligned}\frac{d}{dc} \log \bar{R}_{2,r,-y}(c) &= -\frac{qM'\{y(\beta_{r,y}(c))\}y'\{\beta_{r,y}(c)\}}{1 - qM\{y(\beta_{r,y}(c))\}} \frac{\partial \beta_{r,y}(c)}{\partial c} \\ &\quad - \lambda\{\beta_{r,y}(c)/r\} \frac{1}{r} \frac{\partial \beta_{r,y}(c)}{\partial c} + \lambda(c).\end{aligned}$$

Substitution of (62) gives (63), and

$$\bar{R}'_{2,r,-y}(c) = \bar{R}_{2,r,-y}(c) \frac{d}{dc} \log \bar{R}_{2,r,-y}(c)$$

gives (64). The same equivalence $T(a) \geq c \iff g_r(a) \geq r(c+y)$ gives the endpoint cases:

$$\bar{R}_{2,r,-y}(c) = \begin{cases} q\tau_r/p(c), & r(c+y) \leq g_r(b_r), \\ 0, & r(c+y) \geq g_r(a_q), \end{cases} \quad \bar{R}'_{2,r,-y}(c) = \begin{cases} \lambda(c)\bar{R}_{2,r,-y}(c), & r(c+y) < g_r(b_r), \\ 0, & r(c+y) > g_r(a_q). \end{cases}$$

Because W_r is continuous, (40) implies that μ_r has no atoms. Thus the displayed endpoint values agree with the limits from the interior, so the contribution is continuous at both thresholds. $\square$

2.6.2 The regime $z > -1$

Lemma 2.15. *For every compact $K \subset (-1, \infty)$ and every $H > 0$, uniformly for $z \in K$ and $|h| \leq H$,*

$$\bar{R}_{r,z}\left(c_r + \frac{h}{c_r}\right) \longrightarrow q + (1-q)e^h. \quad (65)$$

Moreover, for every $h_0 > 0$,

$$\limsup_{r \downarrow 0} \sup_{z \in K} \sup_{u_q \leq c \leq c_r - h_0/c_r} \bar{R}_{r,z}(c) \leq q + (1-q)e^{-h_0} < 1. \quad (66)$$

Consequently,

$$C_r(z) = c_r + o(c_r^{-1}) \quad (67)$$

and

$$D_r(z) \longrightarrow q, \quad (68)$$

uniformly on K . Moreover, for sufficiently small r , $C_r(z)$ is regular for every $z \in K$.

Proof. Proof of (65). Fix $H > 0$, take $z \in K$, and take $|h| \leq H$. The primary non-null has $B = 1$ and $\Theta = c_r + y(b_r)$, so conditional on $Z = z$ its value is $c_r + y(b_r) + z$. Since $K \subset (-1, \infty)$ is compact,

$$\eta_K := \inf_{z \in K} (1 + z) > 0.$$

By Lemma 2.10, $c_r \rightarrow \infty$ and $y(b_r) \downarrow 1$. For the fixed H , this implies $H/c_r \rightarrow 0$. Combining these limits, for sufficiently small r ,

$$|y(b_r) - 1| \leq \frac{\eta_K}{4}, \quad \frac{H}{c_r} \leq \frac{\eta_K}{4}. \quad (69)$$

For $z \in K$ and $|h| \leq H$, the definition of η_K gives $z \geq -1 + \eta_K$, while (69) gives $y(b_r) \geq 1 - \eta_K/4$ and $-h/c_r \geq -H/c_r \geq -\eta_K/4$. Therefore

$$y(b_r) + z - \frac{h}{c_r} \geq \left(1 - \frac{\eta_K}{4}\right) + (-1 + \eta_K) - \frac{\eta_K}{4} = \frac{\eta_K}{2}, \quad z \in K, \quad |h| \leq H.$$

Equivalently, $y(b_r) + z \geq h/c_r + \eta_K/2$. Hence the conditional value of the primary non-null satisfies

$$\begin{aligned} |c_r + y(b_r) + z| &= c_r + y(b_r) + z \\ &= c_r + \frac{h}{c_r} + \left\{ y(b_r) + z - \frac{h}{c_r} \right\} \\ &\geq c_r + \frac{h}{c_r} + \frac{\eta_K}{2} > c_r + \frac{h}{c_r}, \end{aligned} \quad (70)$$

so the primary non-null is rejected at cutoff $c_r + h/c_r$, uniformly for $z \in K$ and $|h| \leq H$.

Next consider a secondary non-null indexed by $a \in [b_r, a_q]$. Its conditional value at $Z = z$ is $a/r + y(a) + z$. Since $a \geq b_r$, we have $a/r \geq b_r/r$. Also, Lemma A.1 shows that $y(\cdot)$, the inverse of the strictly increasing map $a(\cdot)$, is increasing; hence $y(a) \geq y(b_r)$. Together with $z \geq \inf_{u \in K} u$ and $h \leq H$, this gives

$$\frac{a}{r} + y(a) + z - \left(c_r + \frac{h}{c_r}\right) \geq \left(\frac{b_r}{r} - c_r\right) + y(b_r) + \inf_{u \in K} u - \frac{H}{c_r}. \quad (71)$$

Here $\frac{b_r}{r} - c_r \rightarrow \infty$ by (46), while

$$y(b_r) + \inf_{u \in K} u - \frac{H}{c_r} \longrightarrow 1 + \inf_{u \in K} u \quad (72)$$

because $y(b_r) \downarrow 1$ and $H/c_r \rightarrow 0$. Equations (46) and (72) show that the right side of (71) tends to $+\infty$. Hence it is positive for sufficiently small r , uniformly in $z \in K$, $|h| \leq H$, and $a \in [b_r, a_q]$. Thus every secondary non-null is rejected at cutoff $c_r + h/c_r$.

Lastly, we analyze the nulls. Since $|h| \leq H$ and $rc_r \rightarrow 0$ by (46),

$$r \left(c_r + \frac{h}{c_r}\right) = rc_r + \frac{rh}{c_r} \longrightarrow 0. \quad (73)$$

Let $T_K = \sup_{z \in K} |z| < \infty$. Since $c_r \rightarrow \infty$ by (46) in Lemma 2.10, while (73) gives a vanishing rescaled cutoff, we have

$$u_q \leq c_r + \frac{h}{c_r} \leq \frac{1}{r}$$

for all sufficiently small r , uniformly for $z \in K$ and $|h| \leq H$. Applying (47) in Lemma 2.11 with $A = 1$ and $T = T_K$ gives

$$\frac{R_{0,r,z}\{p(c_r + h/c_r)\}}{p(c_r + h/c_r)} = L_{r(c_r + h/c_r)}(|z|) + o(1).$$

Moreover,

$$L_{r(c_r+h/c_r)}(|z|) = \exp \left\{ -\frac{r^2(c_r + h/c_r)^2}{2} \right\} \cosh(r(c_r + h/c_r)|z|) = 1 + o(1),$$

uniformly for $z \in K$ and $|h| \leq H$. Since $\pi_{0,r} = 1 - r - \tau_r \rightarrow 1$, it follows that

$$\bar{R}_{0,r,z}(c_r + h/c_r) = q + o(1).$$

By (41),

$$\frac{q(r + \tau_r)}{p(c_r + h/c_r)} = (1 - q\pi_{0,r}) \frac{p(c_r)}{p(c_r + h/c_r)} \rightarrow (1 - q)e^h$$

by (211) in Lemma A.2. This proves (65).

Proof of (66). Fix $h_0 > 0$. Equation (70), evaluated at $h = h_0$, gives

$$|c_r + y(b_r) + z| > c_r + \frac{h_0}{c_r},$$

so the primary non-null is rejected at cutoff $c_r + h_0/c_r$. Equations (71), (46), and (72) give

$$\frac{a}{r} + y(a) + z > c_r + \frac{h_0}{c_r}, \quad a \in [b_r, a_q],$$

for sufficiently small r , uniformly in $z \in K$. Thus every secondary non-null is also rejected at cutoff $c_r + h_0/c_r$. Since rejection at cutoff c means $|T_i| \geq c$, all non-nulls are rejected for every $u_q \leq c \leq c_r + h_0/c_r$. Therefore, on this interval,

$$\bar{R}_{r,z}(c) = \bar{R}_{0,r,z}(c) + \frac{q(r + \tau_r)}{p(c)}.$$

For $u_q \leq c \leq c_r - h_0/c_r$, Lemma 2.11 and $rc_r \rightarrow 0$ from (46) give

$$\sup_{z \in K} \sup_{u_q \leq c \leq c_r - h_0/c_r} \bar{R}_{0,r,z}(c) \leq q + o(1).$$

Also $p(c) \geq p(c_r - h_0/c_r)$, so (41) and (211) in Lemma A.2 give

$$\sup_{u_q \leq c \leq c_r - h_0/c_r} \frac{q(r + \tau_r)}{p(c)} \leq \frac{q(r + \tau_r)}{p(c_r - h_0/c_r)} = (1 - q\pi_{0,r}) \frac{p(c_r)}{p(c_r - h_0/c_r)} \rightarrow (1 - q)e^{-h_0}.$$

Taking the supremum over $z \in K$ and then the limsup in r proves (66).

Proof of (67). Apply Lemma 2.12, with $u = z$ and

$$m_r(z) = c_r, \quad w_r(z) = c_r^{-1}, \quad d(z) = q, \quad \xi(z) = 1.$$

The set K is compact by hypothesis. Since $c_r \rightarrow \infty$, for all sufficiently small r , $m_r(z) = c_r > u_q$ and $w_r(z) = c_r^{-1} > 0$. Moreover, $d \equiv q$ and $\xi \equiv 1$ are continuous on K , with $d(z) \in [0, 1]$ and $\xi(z) > 0$. For every fixed $H > 0$,

$$\inf_{z \in K} \{m_r(z) - Hw_r(z)\} = c_r - \frac{H}{c_r} > u_q$$

for all sufficiently small r , which verifies (50). Under these choices, (51) becomes

$$\sup_{z \in K} \sup_{|h| \leq H} \left| \bar{R}_{r,z} \left(c_r + \frac{h}{c_r} \right) - q - (1 - q)e^h \right| \rightarrow 0,$$

which is exactly (65). Its left-of-crossing assumption (52) becomes, for every $h_0 > 0$,

$$\limsup_{r \downarrow 0} \sup_{z \in K} \sup_{u_q \leq c \leq c_r - h_0/c_r} \bar{R}_{r,z}(c) < 1.$$

This follows from (66), because $q + (1 - q)e^{-h_0} < 1$. Hence all assumptions of Lemma 2.12 hold, and (53) gives

$$\sup_{z \in K} \left| \frac{C_r(z) - c_r}{c_r^{-1}} \right| = \sup_{z \in K} c_r |C_r(z) - c_r| \longrightarrow 0,$$

which is (67) uniformly on K .

Proof of (68). Fix $H > 0$. On $|c - c_r| \leq H/c_r$, all non-nulls are rejected, uniformly over $z \in K$, and hence

$$\bar{R}_{r,z}(c) = \bar{R}_{0,r,z}(c) + \frac{q(r + \tau_r)}{p(c)}. \quad (74)$$

In the notation of Lemma 2.14, uniformly for $|c - c_r| \leq H/c_r$ and $z \in K$, (46) gives

$$c \longrightarrow \infty, \quad rc \longrightarrow 0.$$

For all sufficiently small r , this range lies in $[C, 1/r]$, with $C \rightarrow \infty$. Thus (48)–(49) in Lemma 2.11 give, for either sign,

$$\begin{aligned} Q_{\pm}(c, z) &= \frac{c}{x_{\pm}} \exp \left\{ -\frac{(rc)^2}{2} \pm rcz + O(r^2) \right\} \{1 + o(1)\} \\ &= 1 + o(1), \end{aligned}$$

because $c/x_{\pm} \rightarrow 1$, $rc \rightarrow 0$, and z ranges over the compact set K . Also $x_{\pm}/c_r \rightarrow 1$, $c/c_r \rightarrow 1$, and $\sqrt{1 - r^2} \rightarrow 1$. Thus (202) in Lemma A.2, applied directly to (56) in Lemma 2.14, gives

$$\sup_{z \in K} \sup_{|c - c_r| \leq H/c_r} |\bar{R}'_{0,r,z}(c)| = o(c_r).$$

For $|c - c_r| \leq H/c_r$, write $c = c_r + h/c_r$, where $|h| \leq H$. By (41),

$$\frac{q(r + \tau_r)}{p(c)} = (1 - q\pi_{0,r}) \frac{p(c_r)}{p(c)},$$

and (211) in Lemma A.2 gives

$$\frac{p(c_r)}{p(c)} = \frac{p(c_r)}{p(c_r + h/c_r)} \longrightarrow e^h,$$

uniformly for $|h| \leq H$. Since $e^h \geq e^{-H}$ on this range and $1 - q\pi_{0,r} \rightarrow 1 - q > 0$, for all sufficiently small r ,

$$\inf_{|c - c_r| \leq H/c_r} \frac{q(r + \tau_r)}{p(c)} \geq \frac{1 - q}{2} \cdot \frac{e^{-H}}{2}.$$

Thus we may take $m_H = (1 - q)e^{-H}/4 > 0$. Since all nonnulls are fully rejected for $|c - c_r| \leq H/c_r$, their combined contribution is $q(r + \tau_r)/p(c)$, whose derivative is $\lambda(c)q(r + \tau_r)/p(c)$. Together with $\lambda(c) \sim c_r$ uniformly for $|c - c_r| \leq H/c_r$, this gives

$$\inf_{|c - c_r| \leq H/c_r} \frac{d}{dc} \frac{q(r + \tau_r)}{p(c)} \geq \frac{m_H}{2} c_r$$

for all sufficiently small r . By contrast, $\bar{R}'_{0,r,z}(c) = o(c_r)$ uniformly over $z \in K$ and $|c - c_r| \leq H/c_r$. Differentiating (74) therefore shows that

$$\inf_{z \in K} \inf_{|c - c_r| \leq H/c_r} \bar{R}'_{r,z}(c) > 0 \quad (75)$$

for all sufficiently small r . By (67), uniformly for $z \in K$,

$$c_r - \frac{H}{c_r} < C_r(z) < c_r + \frac{H}{c_r}$$

for all sufficiently small r ; moreover, $c_r - H/c_r > u_q$. Equation (75) and the smoothness of the null contribution on this interval therefore verify the hypotheses of Proposition 2.5, so $C_r(z)$ is regular.

Equation (44) gives

$$D_r(z) = \bar{R}_{0,r,z}\{C_r(z)\}.$$

Put $h_r(z) = c_r\{C_r(z) - c_r\}$. Equation (67) gives $h_r(z) \rightarrow 0$ uniformly on K . The null limit established in the proof of (65), evaluated at $h = h_r(z)$, now gives

$$\sup_{z \in K} |D_r(z) - q| = \sup_{z \in K} |\bar{R}_{0,r,z}\{C_r(z)\} - q| \rightarrow 0,$$

which is (68). □

2.6.3 The regime $z \in (-y_q, -1)$

Let $y = -z$. For $1 < y < y_q$, put

$$\kappa(y) = a(y)y'\{a(y)\} > 0. \quad (76)$$

The positivity follows from Lemma A.1, since $a(y) > 0$ and $y'\{a(y)\} > 0$ for $1 < y < y_q$. For the remainder of the population analysis, define

$$s_r(y) = c_r + y(b_r) - y. \quad (77)$$

The next lemma separates the primary nonnull value from the cutoffs on the r^{-1} scale.

Lemma 2.16. *For every compact $I \subset (1, \infty)$ and every $H > 0$, if $a_I = \inf_{y \in I} a(y) > 0$, then*

$$\begin{aligned} \sup_{y \in I} r|s_r(y)| &\leq rc_r + r \sup_{y \in I} |y(b_r) - y| \rightarrow 0, \\ \inf_{\substack{y \in I \\ |h| \leq H}} r \left\{ \frac{a(y)}{r} + h \right\} &= a_I - rH \rightarrow a_I > 0. \end{aligned} \quad (78)$$

Consequently, the primary nonnull is not rejected at cutoff $a(y)/r + h$ for all sufficiently small r , uniformly for $y \in I$ and $|h| \leq H$.

Proof. Conditional on $Z = -y$, the primary nonnull value is $s_r(y)$ by (77). The two limits in (78) follow from (45) and (46) in Lemma 2.10. Their limiting bounds are separated by $a_I > 0$, which proves the rejection claim. □

Lemma 2.17. *For every compact $I \subset (1, \infty)$,*

$$\sup_{y \in I} \sup_{u_q \leq s \leq s_r(y)} \bar{R}_{r,-y}(s) \leq q + o(1) < 1. \quad (79)$$

Proof. Put $y_I = \inf I > 1$ and $\delta_I^{\text{shift}} = (y_I - 1)/2 > 0$. Since $y(b_r) \downarrow 1$,

$$\delta_I^{\text{shift}} \leq y - y(b_r) \leq \sup_{u \in I} u - 1 =: \Delta_I < \infty \quad (80)$$

for all sufficiently small r , uniformly for $y \in I$. Hence $s_r(y) \leq c_r - \delta_I^{\text{shift}}$. From (42) and the bound by one on each nonnull rejection probability,

$$\bar{R}_{r,-y}(s) \leq \bar{R}_{0,r,-y}(s) + \frac{q(r + \tau_r)}{p(s)}, \quad u_q \leq s \leq s_r(y).$$

Uniformly on this range, $0 \leq rs \leq rc_r \rightarrow 0$. Therefore (47) in Lemma 2.11 gives

$$\bar{R}_{0,r,-y}(s) = q + o(1).$$

Since p is decreasing, (41) yields

$$\begin{aligned} \frac{q(r + \tau_r)}{p(s)} &\leq \frac{q(r + \tau_r)}{p\{s_r(y)\}} \\ &= (1 - q\pi_{0,r}) \frac{p(c_r)}{p\{c_r + y(b_r) - y\}}. \end{aligned}$$

Here $s_r(y) \rightarrow \infty$ uniformly and the shift $y - y(b_r)$ lies in $[\delta_I^{\text{shift}}, \Delta_I]$. Thus (212) in Lemma A.2 makes the last ratio converge uniformly to zero. Combining the bounds proves (79). $\square$

We first prove a lemma that is useful for the key result in this regime.

Lemma 2.18. *For every compact $I \subset (1, y_q)$ and every $H > 0$, there is an interval U_I whose closure is contained in $(0, a_q)$ and contains $\{a(y) : y \in I\}$. For all sufficiently small r , with g_r as in (58), the equation*

$$g_r\{\tilde{\alpha}_r(y, h)\} = g_r\{a(y)\} + rh \quad (81)$$

has a unique solution $\tilde{\alpha}_r(y, h) \in U_I$, uniformly for $y \in I$ and $|h| \leq H$, and

$$\tilde{\alpha}_r(y, h) = a(y) + O(r). \quad (82)$$

Conditional on $Z = -y$, a secondary non-null with $a = \tilde{a}$ is rejected at cutoff $a(y)/r + h$ if and only if

$$\tilde{a} \in [\tilde{\alpha}_r(y, h), a_q].$$

Finally,

$$\log \frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)} = \kappa(y)h + o(1), \quad (83)$$

uniformly for $y \in I$ and $|h| \leq H$.

Proof. Fix a compact $I \subset (1, y_q)$ and $H > 0$ as in the statement. Let

$$A_I = \{a(y) : y \in I\}.$$

Because $I \subset (1, y_q)$ is compact, A_I is a compact subset of $(0, a_q)$. Choose an interval U_I whose closure is contained in $(0, a_q)$ and such that $A_I \subset U_I$. By (45) in Lemma 2.10, $b_r \rightarrow 0$, so $U_I \subset (b_r, a_q)$ for all sufficiently small r . Lemma 2.14 then shows that g_r is strictly increasing on U_I .

Put $d_I = \text{dist}(A_I, U_I^c) > 0$. If $rH < d_I/2$, then $a(y) \pm rH \in U_I$ for all $y \in I$. Since $g'_r \geq 1$ on U_I , the intermediate value theorem gives a unique solution to (81) in U_I , and

$$|\tilde{\alpha}_r(y, h) - a(y)| \leq rH.$$

Hence, (82) is proved.

Put $c = a(y)/r + h$ and $\beta = \tilde{\alpha}_r(y, h)$. If $a_I = \inf_{u \in I} a(u) > 0$, then

$$r(y - c) = ry - a(y) - rh \leq r \sup_{u \in I} u - a_I + rH < 0 < g_r(b_r)$$

for all sufficiently small r , uniformly for $y \in I$ and $|h| \leq H$. Thus (59) holds uniformly. Moreover, (81) and $y\{a(y)\} = y$ give

$$g_r(\beta) = g_r\{a(y)\} + rh = r(c + y).$$

This is (60), so (61) shows that a secondary non-null is rejected exactly when its index lies in $[\tilde{\alpha}_r(y, h), a_q]$.

Write $\beta = \tilde{\alpha}_r(y, h)$. Differentiating (81) with respect to h gives

$$\frac{\partial \beta}{\partial h} = \frac{r}{1 + ry'(\beta)}.$$

Since $p'(x)/p(x) = -\lambda(x)$,

$$\frac{\partial}{\partial h} \log \frac{p(\beta/r)}{p\{a(y)/r + h\}} = \lambda\left(\frac{a(y)}{r} + h\right) - \frac{\lambda(\beta/r)}{1 + ry'(\beta)}. \quad (84)$$

Moreover, (81) and $y\{a(y)\} = y$ imply

$$\frac{\beta}{r} = \frac{a(y)}{r} + h + y - y(\beta). \quad (85)$$

The function $a(y)$ is bounded away from zero on I , and (82) gives $\beta = a(y) + O(r)$ uniformly. Hence both arguments of λ in (84) are of order r^{-1} . Using $\lambda(x) = x + O(x^{-1})$ from (202) in Lemma A.2 and then (85), we obtain

$$\begin{aligned} \lambda\left(\frac{a(y)}{r} + h\right) - \frac{\lambda(\beta/r)}{1 + ry'(\beta)} &= \frac{a(y)}{r} + h - \frac{\beta/r}{1 + ry'(\beta)} + O(r) \\ &= \frac{\{a(y) + rh\}y'(\beta) + y(\beta) - y}{1 + ry'(\beta)} + O(r) \\ &\longrightarrow a(y)y'\{a(y)\} = \kappa(y) \end{aligned}$$

uniformly for $y \in I$ and $|h| \leq H$. At $h = 0$, uniqueness in (81) gives $\beta = a(y)$, so the logarithm in (84) is zero. Integrating from 0 to h proves (83). $\square$

Lemma 2.19. *For every compact $I \subset (1, y_q)$ and every $H > 0$, uniformly for $y \in I$ and $|h| \leq H$,*

$$\bar{R}_{r,-y}\left(\frac{a(y)}{r} + h\right) \longrightarrow qM(y) + \{1 - qM(y)\}e^{\kappa(y)h}. \quad (86)$$

Moreover, for every $h_0 > 0$, there is $\gamma > 0$ such that, for all sufficiently small r ,

$$\sup_{y \in I} \sup_{u_q \leq c \leq a(y)/r - h_0} \bar{R}_{r,-y}(c) \leq 1 - \gamma. \quad (87)$$

Consequently, uniformly for $y \in I$,

$$C_r(-y) - \frac{a(y)}{r} \longrightarrow 0 \quad (88)$$

and

$$D_r(-y) \longrightarrow qM(y). \quad (89)$$

Moreover, for sufficiently small r , $C_r(-y)$ is regular for every $y \in I$.

Proof. Proof of (86). Fix $H > 0$, take $y \in I$, and take $|h| \leq H$. By Lemma 2.18, a secondary non-null with $a = \tilde{a}$ is rejected at cutoff $a(y)/r + h$ if and only if

$$\tilde{a} \in [\tilde{\alpha}_r(y, h), a_q].$$

By (40), their mass under the secondary component is

$$\tau_r \mu_r([\tilde{\alpha}_r(y, h), a_q]) = W_r\{\tilde{\alpha}_r(y, h)\}.$$

The definition (38) therefore gives

$$\frac{qW_r\{\tilde{\alpha}_r(y, h)\}}{p(a(y)/r + h)} = [1 - qM\{y(\tilde{\alpha}_r(y, h))\}] \frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)}.$$

By (82), $y(\tilde{\alpha}_r(y, h)) \rightarrow y$ uniformly for $y \in I$ and $|h| \leq H$. Equation (83) gives

$$\frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)} \rightarrow e^{\kappa(y)h},$$

uniformly for $y \in I$ and $|h| \leq H$. Hence

$$\frac{qW_r\{\tilde{\alpha}_r(y, h)\}}{p(a(y)/r + h)} = [1 - qM\{y(\tilde{\alpha}_r(y, h))\}] \frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)} \rightarrow \{1 - qM(y)\}e^{\kappa(y)h} \quad (90)$$

Lemma 2.16, through (78), shows that the primary nonnull is not rejected at cutoff $a(y)/r + h$, uniformly for $y \in I$ and $|h| \leq H$.

Since $a(y)$ is bounded away from zero on I and $a(y) \leq a_q$, for all sufficiently small r ,

$$u_q \leq \frac{a(y)}{r} + h \leq \frac{a_q + 1}{r}$$

uniformly for $y \in I$ and $|h| \leq H$. Thus (47) in Lemma 2.11, applied with $A = a_q + 1$ and $T = \sup_{u \in I} u$, gives

$$\bar{R}_{0,r,-y}(a(y)/r + h) \rightarrow qL_{a(y)}(y) = qM(y). \quad (91)$$

Combining (90) and (91) proves (86).

Proof of (87). Lemma 2.17, through (79), excludes a crossing on $u_q \leq s \leq s_r(y)$, uniformly for $y \in I$. Fix $h_0 > 0$. The complementary range $s_r(y) < s \leq a(y)/r - h_0$ is nonempty uniformly on I , because

$$\inf_{y \in I} \left\{ \frac{a(y)}{r} - h_0 - s_r(y) \right\} \geq \frac{\inf_{y \in I} a(y)}{r} - c_r - O_I(1) \rightarrow \infty,$$

where $rc_r \rightarrow 0$ by (46) in Lemma 2.10. It remains to prove that there is $\gamma > 0$ such that, for all sufficiently small r ,

$$\sup_{y \in I} \sup_{s_r(y) < s \leq a(y)/r - h_0} \bar{R}_{r,-y}(s) \leq 1 - \gamma \quad (92)$$

For $s > s_r(y)$, the identity $T_i = s_r(y) > 0$ gives $|T_i| = T_i < s$. For sufficiently small r , every secondary non-null value is positive uniformly over $y \in I$ and $\tilde{a} \in [b_r, a_q]$. For every $y > 1$ and $s > s_r(y)$, define

$$\mathcal{A}_{r,y,s} := \{\tilde{a} \in [b_r, a_q] : g_r(\tilde{a}) \geq r(s + y)\}, \quad \beta_{r,y,s} := \inf(\mathcal{A}_{r,y,s} \cup \{a_q\}). \quad (93)$$

On the range $s_r(y) < s \leq a(y)/r - h_0$, the set is nonempty for sufficiently small r , since $\tilde{a} = a_q$ satisfies the inequality uniformly over $y \in I$. By (58) and Lemma 2.14, g_r is strictly increasing. Thus a secondary non-null with $a = \tilde{a}$ is rejected if and only if $\tilde{a} \in [\beta_{r,y,s}, a_q]$. Therefore, by (42),

$$\bar{R}_{r,-y}(s) = \bar{R}_{0,r,-y}(s) + \frac{q\tau_r}{p(s)}\mu_r([\beta_{r,y,s}, a_q]). \quad (94)$$

Suppose (92) fails for this h_0 . Then there are

$$r_n \downarrow 0, \quad y_n \in I, \quad s_{r_n}(y_n) < s_n \leq a(y_n)/r_n - h_0, \quad \liminf_{n \rightarrow \infty} \bar{R}_{r_n, -y_n}(s_n) \geq 1. \quad (95)$$

After taking a subsequence,

$$y_n \rightarrow y \in I, \quad a(y_n) \rightarrow a(y), \quad x_n = r_n s_n \rightarrow x \in [0, a(y)]. \quad (96)$$

Define

$$h_n := s_n - \frac{a(y_n)}{r_n}.$$

Then $h_n \leq -h_0$ by (95). If $h_n = O(1)$, then after passing to a further subsequence $h_n \rightarrow h \leq -h_0$. Equation (86) gives

$$\bar{R}_{r_n, -y_n}(s_n) \rightarrow qM(y) + \{1 - qM(y)\}e^{\kappa(y)h} \leq qM(y) + \{1 - qM(y)\}e^{-\kappa(y)h_0} < 1,$$

contradicting $\liminf_{n \rightarrow \infty} \bar{R}_{r_n, -y_n}(s_n) \geq 1$. Hence, after passing to a further subsequence if necessary, it remains to consider

$$h_n \rightarrow -\infty. \quad (97)$$

We next identify the rejected secondary non-nulls at cutoff s_n . A secondary non-null with index $\tilde{a} \in [b_{r_n}, a_q]$ has $(\Theta, B) = (\tilde{a}/r_n + y(\tilde{a}), 1)$. Conditional on $Z = -y_n$, its value is

$$T_{\tilde{a}, n} = \frac{\tilde{a}}{r_n} + y(\tilde{a}) - y_n.$$

For all large n , this value is positive uniformly in $\tilde{a} \in [b_{r_n}, a_q]$, because $b_{r_n}/r_n \rightarrow \infty$ and $y(\tilde{a}) - y_n$ is bounded below. Hence $|T_{\tilde{a}, n}| = T_{\tilde{a}, n}$, and the rejection rule $|T_{\tilde{a}, n}| \geq s_n$ is equivalent to

$$\frac{\tilde{a}}{r_n} + y(\tilde{a}) - y_n \geq s_n.$$

The rejected set is nonempty because the endpoint $\tilde{a} = a_q$ is rejected. Indeed, compactness of $I \subset (1, y_q)$ and continuity of $a(\cdot)$ give

$$\Delta_I := a_q - \sup_{u \in I} a(u) > 0.$$

Since $s_n \leq a(y_n)/r_n - h_0$ by (95),

$$\frac{a_q}{r_n} + y_q - y_n - s_n \geq \frac{a_q - a(y_n)}{r_n} + y_q - y_n + h_0 > 0$$

for large n , because $a_q - a(y_n) \geq \Delta_I$ and $y_n \leq \sup I < y_q$. Define

$$G_n(\tilde{a}) := g_{r_n}(\tilde{a}) - r_n(s_n + y_n), \quad \tilde{a} \in [b_{r_n}, a_q],$$

where g_r is defined in (58). Multiplying the rejection inequality by r_n , a secondary non-null with index $\tilde{a}$ is rejected if and only if

$$G_n(\tilde{a}) \geq 0.$$

By (58) and Lemma 2.14, G_n is continuous and strictly increasing on $[b_{r_n}, a_q]$ for all large n . Therefore a secondary non-null with $a = \tilde{a}$ is rejected if and only if $\tilde{a} \in [\beta_n, a_q]$, where

$$\beta_n := \inf \{\tilde{a} \in [b_{r_n}, a_q] : G_n(\tilde{a}) \geq 0\}. \quad (98)$$

Put $d_n := y_n - y(\beta_n)$. If $\beta_n > b_{r_n}$, then $G_n(\beta_n) = 0$. Thus, in this case,

$$\frac{\beta_n}{r_n} - s_n = d_n. \quad (99)$$

If $\beta_n = b_{r_n}$, then $G_n(\beta_n) \geq 0$, so (99) is replaced by

$$\frac{\beta_n}{r_n} - s_n \geq d_n. \quad (100)$$

Moreover, $s_n \rightarrow \infty$, because $s_n > s_{r_n}(y_n) = c_{r_n} + y(b_{r_n}) - y_n$, $c_{r_n} \rightarrow \infty$, $y(b_{r_n}) \rightarrow 1$, and I is compact. Also $d_n > 0$ eventually. If $\beta_n = b_{r_n}$, this follows from the definition of d_n and (80) in Lemma 2.17, which give

$$d_n = y_n - y(b_{r_n}) \geq \delta_I^{\text{shift}} > 0.$$

If $\beta_n > b_{r_n}$, then

$$g_{r_n}(\beta_n) = r_n(s_n + y_n) = g_{r_n}\{a(y_n)\} + r_n h_n \leq g_{r_n}\{a(y_n)\} - r_n h_0.$$

The strict monotonicity of g_{r_n} therefore gives $\beta_n < a(y_n)$, and hence $d_n = y_n - y(\beta_n) > 0$. The null contribution term in (94) already has a limit along this subsequence:

$$q\pi_{0,r_n} \frac{R_{0,r_n,-y_n}\{p(s_n)\}}{p(s_n)} \longrightarrow qL_x(y) \leq qM(y) < 1. \quad (101)$$

Indeed, (47) in Lemma 2.11 applies because $r_n s_n \rightarrow x$, $y_n \rightarrow y$, and $\pi_{0,r_n} \rightarrow 1$. The inequality follows from $M(y) = \sup_{\alpha \geq 0} L_\alpha(y)$, the strict monotonicity of M on $(1, \infty)$, $y \in I \subset (1, y_q)$, and $qM(y_q) = 1$. Equations (94) and (101), together with the threshold characterization in (98), imply

$$\bar{R}_{r_n,-y_n}(s_n) = qL_x(y) + o(1) + \frac{q\tau_{r_n}}{p(s_n)}\mu_{r_n}([\beta_n, a_q]). \quad (102)$$

Moreover, (40), (38), (198) in Lemma A.1, and the bound from (99)–(100) give

$$\frac{q\tau_{r_n}}{p(s_n)}\mu_{r_n}([\beta_n, a_q]) = \frac{qW_{r_n}(\beta_n)}{p(s_n)} = [1 - qM\{y(\beta_n)\}] \frac{p(\beta_n/r_n)}{p(s_n)} \leq \frac{p(s_n + d_n)}{p(s_n)}. \quad (103)$$

It remains only to prove that

$$s_n d_n \longrightarrow \infty. \quad (104)$$

Equations (103), (104), and (213) in Lemma A.2 imply

$$\frac{q\tau_{r_n}}{p(s_n)}\mu_{r_n}([\beta_n, a_q]) = o(1).$$

Substituting this into (102) gives

$$\bar{R}_{r_n,-y_n}(s_n) \longrightarrow qL_x(y) < 1,$$

which yields a contradiction to (95).

To prove (104), we now consider three cases under (95), (96), and (97).

Case (a): $0 < x < a(y)$. Suppose $x > 0$. Since $b_{r_n} \rightarrow 0$ by (45) in Lemma 2.10 and $r_n s_n \rightarrow x$, the endpoint case $\beta_n = b_{r_n}$ cannot occur for large n : multiplying (100) by r_n would give

$$b_{r_n} - r_n s_n \geq r_n \{y_n - y(b_{r_n})\},$$

whose left side tends to $-x < 0$, while the right side is nonnegative for large n . Hence (99) holds eventually. Multiplying it by r_n gives

$$\beta_n - r_n s_n = r_n d_n.$$

The factor d_n is bounded. Indeed, I is compact, so $|y_n| \leq \sup_{u \in I} |u|$. Also $\beta_n \in [b_{r_n}, a_q]$, and $y(\cdot)$ is increasing by Lemma A.1; hence

$$y(b_{r_n}) \leq y(\beta_n) \leq y(a_q) = y_q. \quad (105)$$

Since $y(b_{r_n}) \downarrow 1$ by (45) in Lemma 2.10, the sequence $y(\beta_n)$ is bounded. Thus $r_n d_n \rightarrow 0$, so $\beta_n - r_n s_n \rightarrow 0$. Since $r_n s_n \rightarrow x$, we have $\beta_n \rightarrow x$. Since $x < a(y)$ and $y(\cdot)$ is strictly increasing by Lemma A.1, $y(x) < y$.

Together, $y_n \rightarrow y$ from (96) and $\beta_n \rightarrow x$ imply, by continuity of $y(\cdot)$, that $d_n \rightarrow d := y - y(x) > 0$. Also, (99) gives $\beta_n/r_n = s_n + d_n$. Therefore, for all large n ,

$$\frac{\beta_n}{r_n} = s_n + d_n, \quad 0 < \frac{d}{2} \leq d_n \leq 2d, \quad s_n \rightarrow \infty.$$

As a result, (104) is proved.

Case (b): $x = 0$. Then $r_n s_n \rightarrow 0$. Since

$$s_n > s_{r_n}(y_n) = c_{r_n} + y(b_{r_n}) - y_n,$$

(46) gives $c_{r_n} \rightarrow \infty$, while (45) gives $y(b_{r_n}) \downarrow 1$ and compactness of I keeps y_n bounded. Hence $s_n \rightarrow \infty$. If $\beta_n = b_{r_n}$, then $\beta_n \rightarrow 0$ by (45). If $\beta_n > b_{r_n}$, then (99) holds; multiplying it by r_n , using compactness of I , and applying (105) gives

$$\beta_n - r_n s_n = r_n d_n \rightarrow 0.$$

Since $r_n s_n \rightarrow 0$, this also gives $\beta_n \rightarrow 0$. Thus $\beta_n \rightarrow 0$ in both cases. By (96), $y_n \rightarrow y$, and by the inverse relation together with (196) in Lemma A.1, $y(\beta_n) \rightarrow 1$. Hence

$$d_n \rightarrow y - 1 > 0.$$

Moreover, (99) in the interior case and (100) in the endpoint case give $\beta_n/r_n \geq s_n + d_n$. Since $d_n \rightarrow d := y - 1 > 0$ and $s_n \rightarrow \infty$, (104) is proved.

Case (c): $x = a(y)$. By (97), this is the only remaining subcase. Since $r_n s_n \rightarrow a(y)$ and $a(y_n) \rightarrow a(y)$, we have

$$r_n h_n = r_n s_n - a(y_n) \rightarrow a(y) - a(y) = 0.$$

The threshold equation $G_n(\tilde{a}) = 0$ is

$$g_{r_n}(\tilde{a}) = r_n(s_n + y_n) = a(y_n) + r_n y_n + r_n h_n = g_{r_n}\{a(y_n)\} + r_n h_n,$$

where $y\{a(y_n)\} = y_n$. Since $h_n < 0$ and g_{r_n} is strictly increasing by (58) and Lemma 2.14, the corresponding threshold lies below $a(y_n)$. To justify the existence and location of this threshold, choose $\eta > 0$ such that

$$[a(y_n) - \eta, a(y_n)] \subset (0, a_q)$$

for all large n ; this is possible because $a(y_n) \rightarrow a(y) \in (0, a_q)$. Since $-r_n h_n \rightarrow 0$ and $g'_{r_n} \geq 1$, for all large n ,

$$g_{r_n}\{a(y_n) - \eta\} < g_{r_n}\{a(y_n)\} + r_n h_n < g_{r_n}\{a(y_n)\}.$$

Continuity and strict monotonicity of g_{r_n} therefore give a unique $\tilde{\beta}_n \in (a(y_n) - \eta, a(y_n))$ satisfying

$$g_{r_n}(\tilde{\beta}_n) = g_{r_n}\{a(y_n)\} + r_n h_n.$$

Moreover, the mean-value theorem and $g'_{r_n} \geq 1$ give

$$0 < a(y_n) - \tilde{\beta}_n \leq -r_n h_n \rightarrow 0.$$

Because $a(y_n)$ remains bounded away from 0 and a_q , and $b_{r_n} \rightarrow 0$ by (45) in Lemma 2.10, it follows that

$$b_{r_n} < \tilde{\beta}_n < a_q$$

eventually. Thus $\tilde{\beta}_n$ is the threshold β_n in (98), so $G_n(\beta_n) = 0$. The mean-value theorem applied to g_{r_n} now gives, for some ξ_n between β_n and $a(y_n)$,

$$-r_n h_n = g_{r_n}\{a(y_n)\} - g_{r_n}(\beta_n) = (a(y_n) - \beta_n)\{1 + r_n y'(\xi_n)\}.$$

Therefore $\beta_n < a(y_n)$ and

$$a(y_n) - \beta_n = \frac{-r_n h_n}{1 + r_n y'(\xi_n)}. \quad (106)$$

For some ζ_n between β_n and $a(y_n)$,

$$\frac{\beta_n}{r_n} - s_n = d_n = y'(\zeta_n)\{a(y_n) - \beta_n\}. \quad (107)$$

Combining (106) and (107) gives

$$s_n d_n = (r_n s_n) \frac{y'(\zeta_n)}{1 + r_n y'(\xi_n)} (-h_n). \quad (108)$$

Here $r_n s_n \rightarrow a(y) > 0$. Also $\beta_n \rightarrow a(y)$ and $a(y_n) \rightarrow a(y)$, so $\xi_n, \zeta_n \rightarrow a(y)$; hence $y'(\zeta_n) \rightarrow y'\{a(y)\} > 0$, while $1 + r_n y'(\xi_n) \rightarrow 1$. Since $h_n \rightarrow -\infty$ by (97), (108) yields

$$s_n d_n \rightarrow \infty.$$

Thus (104) is proved.

This proves (92). Combining (79) and (92), and reducing γ if necessary, proves (87).

Proof of (88). Apply Lemma 2.12 to $\bar{R}_{r,-y}$, with

$$m_r(y) = \frac{a(y)}{r}, \quad w_r(y) = 1, \quad d(y) = qM(y), \quad \xi(y) = \kappa(y).$$

The set I is compact, and $a_I = \inf_{y \in I} a(y) > 0$; hence, for every fixed $H > 0$,

$$\inf_{y \in I} \left\{ \frac{a(y)}{r} - H \right\} \geq \frac{a_I}{r} - H > u_q$$

for all sufficiently small r . Because $I \subset (1, y_q)$ and M is continuous and strictly increasing there,

$$0 < d(y) = qM(y) < 1, \quad \xi(y) = \kappa(y) > 0,$$

and both functions are continuous on I . Equation (86) is exactly (51), while (87) is (52). Thus all hypotheses of Lemma 2.12 hold, and (53) gives (88) uniformly on I .

Proof of (89). Fix $H = 1$. For $y \in I$ and $|h| \leq H$, let $\tilde{\alpha}_r(y, h)$ be the threshold defined by (81). Write

$$\bar{R}_{2,r,-y} \left(\frac{a(y)}{r} + h \right) = \frac{qW_r\{\tilde{\alpha}_r(y, h)\}}{p(a(y)/r + h)} = [1 - qM\{y(\tilde{\alpha}_r(y, h))\}] \frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)}.$$

Equation (78) shows that the primary non-null is not rejected for $y \in I$ and $|h| \leq H$. Consequently, (42) gives

$$\bar{R}_{r,-y} \left(\frac{a(y)}{r} + h \right) = \bar{R}_{0,r,-y} \left(\frac{a(y)}{r} + h \right) + \bar{R}_{2,r,-y} \left(\frac{a(y)}{r} + h \right).$$

We first differentiate the secondary non-null contribution $\bar{R}_{2,r,-y}\{a(y)/r + h\}$. Equation (59) holds uniformly, since

$$g_r(b_r) > 0 > ry - a(y) - rh = r \left\{ y - \left(\frac{a(y)}{r} + h \right) \right\}$$

for all sufficiently small r . Apply (63) in Lemma 2.14 with $c = a(y)/r + h$ and $\beta = \tilde{\alpha}_r(y, h)$. The term in (63),

$$-\frac{qM'\{y(\beta)\}y'(\beta)}{1 - qM\{y(\beta)\}} \frac{r}{1 + ry'(\beta)},$$

is $O(r)$ uniformly by (82) and (198) in Lemma A.1. Indeed, (82) places β , uniformly over $y \in I$ and $|h| \leq H$, in a fixed compact subinterval of $(0, a_q)$. On this subinterval,

$$\frac{|M'\{y(\beta)\}|y'(\beta)}{1 - qM\{y(\beta)\}} = O(1), \quad \frac{r}{1 + ry'(\beta)} = O(r),$$

by (198) in Lemma A.1; hence their product is $O(r)$ uniformly.

The last two terms in (63) are

$$-\frac{\lambda(\beta/r)}{1 + ry'(\beta)} + \lambda(c).$$

For these terms, (60) implies

$$\frac{\beta}{r} = c + y - y(\beta).$$

Because $a(y)$ is bounded away from zero on I , (82) gives $c \asymp \beta/r \asymp r^{-1}$ uniformly. Thus (202) in Lemma A.2 gives

$$\begin{aligned} \lambda(c) - \frac{\lambda(\beta/r)}{1 + ry'(\beta)} &= c - \frac{\beta/r}{1 + ry'(\beta)} + O(r) \\ &= \frac{rcy'(\beta) + y(\beta) - y}{1 + ry'(\beta)} + O(r). \end{aligned}$$

Now $rc = a(y) + rh \rightarrow a(y)$, while (82) and continuity of y and y' give

$$y(\beta) - y = o(1), \quad y'(\beta) = y'\{a(y)\} + o(1)$$

uniformly. Consequently,

$$\lambda(c) - \frac{\lambda(\beta/r)}{1 + ry'(\beta)} \rightarrow a(y)y'\{a(y)\} = \kappa(y)$$

uniformly, by (76). Combining this with (90) yields

$$\frac{\partial}{\partial h} \bar{R}_{2,r,-y} \left(\frac{a(y)}{r} + h \right) \rightarrow \kappa(y) \{1 - qM(y)\} e^{\kappa(y)h} \quad (109)$$

uniformly for $y \in I$ and $|h| \leq H$.

It remains to differentiate the null contribution. Use the notation of Lemma 2.14 at $c = a(y)/r + h$. Since $a(y)$ is bounded away from zero on I and $a(y) \leq a_q$, for all sufficiently small r ,

$$u_q \leq \frac{a(y)}{r} + h \leq \frac{a_q + 1}{r}$$

uniformly for $y \in I$ and $|h| \leq H$. Applying (47) in Lemma 2.11 with $A = a_q + 1$ and $T = \sup_{u \in I} u$ shows that

$$\sup_{y \in I} \sup_{|h| \leq H} \bar{R}_{0,r,-y} \left(\frac{a(y)}{r} + h \right) = O(1). \quad (110)$$

Equation (57) in Lemma 2.14 can be written as

$$\frac{\partial}{\partial h} \log \bar{R}_{0,r,-y}(c) = \sum_{\sigma \in \{-, +\}} \frac{Q_\sigma}{Q_+ + Q_-} \left\{ \lambda(c) - \frac{\lambda(x_\sigma)}{\sqrt{1 - r^2}} \right\},$$

because $\partial c / \partial h = 1$. Moreover, (202) in Lemma A.2 gives, for either sign,

$$\lambda(c) - \frac{\lambda(x_\pm)}{\sqrt{1 - r^2}} = c - \frac{c \pm ry}{1 - r^2} + O(r) = O(r)$$

uniformly for $y \in I$ and $|h| \leq H$. Since the nonnegative weights $Q_\sigma/(Q_+ + Q_-)$ sum to one,

$$\sup_{y \in I} \sup_{|h| \leq H} \left| \frac{\partial}{\partial h} \log \bar{R}_{0,r,-y} \left(\frac{a(y)}{r} + h \right) \right| = O(r). \quad (111)$$

Multiplying (110) and (111) gives

$$\sup_{y \in I} \sup_{|h| \leq H} \left| \frac{\partial}{\partial h} \bar{R}_{0,r,-y} \left(\frac{a(y)}{r} + h \right) \right| \rightarrow 0. \quad (112)$$

Since I is compact and $\kappa(y)\{1 - qM(y)\}e^{\kappa(y)h}$ is strictly positive on $I \times [-H, H]$, (109) and (112) imply

$$\inf_{y \in I} \inf_{|h| \leq H} \frac{\partial}{\partial h} \bar{R}_{r,-y} \left(\frac{a(y)}{r} + h \right) > 0 \quad (113)$$

for all sufficiently small r . For every fixed $H > 0$, (91) verifies (54) with $d(y) = qM(y)$. The null contribution is smooth, and (113), with $H = 1$, gives the positive derivative required by Lemma 2.13. That lemma therefore shows that $C_r(-y)$ is regular for every $y \in I$ when r is sufficiently small, and (55) gives

$$\sup_{y \in I} |D_r(-y) - qM(y)| \rightarrow 0,$$

which is (89). □

2.6.4 The regime $z < -y_q$

Let $y = -z$. For $y > y_q$, let $\alpha_q(y) \in (0, a_q)$ be the unique solution of

$$qL_{\alpha_q(y)}(y) = 1. \quad (114)$$

For fixed y , the map $a \mapsto L_a(y)$ is continuous and

$$\frac{\partial}{\partial a} \log L_a(y) = -a + y \tanh(ay). \quad (115)$$

By Lemma A.1, this derivative is positive on $(0, a(y))$. Since $y > y_q$, the same lemma gives $a(y) > a_q$, so $a \mapsto L_a(y)$ is strictly increasing on $[0, a_q]$. Moreover, by the definition of a_q in (36),

$$qL_0(y) = q < 1, \quad qL_{a_q}(y) > qL_{a_q}(y_q) = qM(y_q) = 1,$$

where the strict inequality follows from $a_q > 0$, $y > y_q$, and the monotonicity of $\cosh(a_q y)$ in $y > 0$. Thus continuity and strict monotonicity give exactly one solution to (114).

Lemma 2.20. *For every compact $I \subset (y_q, \infty)$, uniformly for $y \in I$,*

$$rC_r(-y) \rightarrow \alpha_q(y) \quad (116)$$

and

$$D_r(-y) \rightarrow 1. \quad (117)$$

Proof. Proof of (116). Fix a compact $I \subset (y_q, \infty)$, and put

$$\delta_I = \inf_{y \in I} (y - y_q) > 0.$$

Use $s_r(y)$ from (77) and $(\mathcal{A}_{r,y,s}, \beta_{r,y,s})$ from (93). By (58) and Lemma 2.14, $g'_r(\tilde{a}) = 1 + ry'(\tilde{a}) > 1$. For all sufficiently small r , the primary non-null is not rejected when $s > s_r(y)$. To verify the corresponding

sign assertion for the secondary non-nulls, note that, conditional on $Z = -y$, the value associated with $\tilde{a} \in [b_r, a_q]$ is

$$\frac{\tilde{a}}{r} + y(\tilde{a}) - y \geq \frac{b_r}{r} + y(b_r) - \sup_{u \in I} u. \quad (118)$$

The monotonicity of $y(\cdot)$ gives (118). Its right-hand side tends to infinity as $r \downarrow 0$, by (45) in Lemma 2.10. Hence every secondary non-null value is positive for all sufficiently small r , uniformly over $y \in I$ and $\tilde{a} \in [b_r, a_q]$. Thus the rejected secondary non-nulls are indexed by $\mathcal{A}_{r,y,s}$. If this set is nonempty, continuity and strict monotonicity of g_r show that it is the closed interval $[\beta_{r,y,s}, a_q]$ and that

$$g_r(\beta_{r,y,s}) \geq r(s + y).$$

Consequently, in the nonempty case,

$$\frac{\beta_{r,y,s}}{r} - s \geq y - y(\beta_{r,y,s}) \geq y - y_q \geq \delta_I. \quad (119)$$

If $\mathcal{A}_{r,y,s} = \emptyset$, its μ_r -mass is zero. Since $\mu_r(\{a_q\}) = 0$ by (40), the convention in (93) gives in both cases

$$\mu_r(\mathcal{A}_{r,y,s}) = \mu_r([\beta_{r,y,s}, a_q]).$$

Using (40), (198) in Lemma A.1, and (119) when the set is nonempty, we obtain

$$0 \leq \frac{q\tau_r}{p(s)} \mu_r([\beta_{r,y,s}, a_q]) \leq \frac{p(s + \delta_I)}{p(s)}.$$

Moreover,

$$\inf_{y \in I} s_r(y) = c_r + y(b_r) - \sup_{y \in I} y \longrightarrow \infty,$$

because $c_r \rightarrow \infty$, $y(b_r) \rightarrow 1$, and I is compact. Therefore (212) in Lemma A.2 gives

$$\sup_{y \in I} \sup_{s > s_r(y)} \frac{q\tau_r}{p(s)} \mu_r([\beta_{r,y,s}, a_q]) \longrightarrow 0. \quad (120)$$

The function $(a, y) \mapsto L_a(y)$ is C^∞ . Equation (115), together with $\alpha_q(y) < a_q < a(y)$ and Lemma A.1, shows that

$$\left. \frac{\partial}{\partial a} \log L_a(y) \right|_{a=\alpha_q(y)} > 0.$$

The implicit function theorem therefore shows that $y \mapsto \alpha_q(y)$ is continuous on (y_q, ∞) . Since I is compact,

$$m_I := \inf_{y \in I} \min\{\alpha_q(y), a_q - \alpha_q(y)\} > 0. \quad (121)$$

Fix any $0 < \delta < m_I/2$. For every $y \in I$, continuity and strict monotonicity in a give

$$qL_{\alpha_q(y)-\delta}(y) < qL_{\alpha_q(y)}(y) = 1 < qL_{\alpha_q(y)+\delta}(y).$$

Both strict gaps are continuous functions of y . Their minima over the compact set I are positive, so there is $\gamma > 0$ such that, uniformly on I ,

$$qL_{\alpha_q(y)-\delta}(y) \leq 1 - 2\gamma, \quad qL_{\alpha_q(y)+\delta}(y) \geq 1 + 2\gamma \quad (122)$$

The choice $\delta < m_I/2$ also gives

$$\alpha_q(y) + \delta \leq \alpha_q(y) + \frac{m_I}{2} \leq \frac{\alpha_q(y) + a_q}{2} < a_q < a(y), \quad (123)$$

uniformly for $y \in I$.

Write $\psi_y(x) = \log L_x(y)$. The proof of Lemma A.1, through (199), shows that $\psi'_y(x) > 0$ for $0 < x < a(y)$. By (123), every $x \in (0, \alpha_q(y) + \delta)$ lies in $(0, a(y))$. Hence $\partial_x L_x(y) = L_x(y)\psi'_y(x) > 0$ throughout $(0, \alpha_q(y) + \delta)$, and therefore $x \mapsto L_x(y)$ is increasing on $[0, \alpha_q(y) + \delta]$.

On $u_q \leq s \leq s_r(y)$, (79) in Lemma 2.17 gives

$$\sup_{y \in I} \sup_{u_q \leq s \leq s_r(y)} \bar{R}_{r,-y}(s) \leq q + o(1).$$

For $s > s_r(y)$, the term in (42) that corresponds to the primary non-nulls is zero, and (120) and (47) in Lemma 2.11 give, uniformly for $y \in I$ and $s_r(y) < s \leq (\alpha_q(y) + \delta)/r$,

$$\bar{R}_{r,-y}(s) = \bar{R}_{0,r,-y}(s) + \frac{q\tau_r}{p(s)}\mu_r([\beta_{r,y,s}, a_q]) = qL_{rs}(y) + o(1).$$

Combining these two ranges gives

$$\sup_{u_q \leq s \leq (\alpha_q(y) + \delta)/r} \bar{R}_{r,-y}(s) \leq qL_{\alpha_q(y) - \delta}(y) + o(1) \leq 1 - \gamma. \quad (124)$$

Since $rs_r(y) \rightarrow 0$ uniformly on I and $\alpha_q(y) + \delta$ is bounded away from zero, we have

$$\frac{\alpha_q(y) + \delta}{r} > s_r(y)$$

for all sufficiently small r , uniformly on I . At the cutoff $(\alpha_q(y) + \delta)/r$, the term in (42) that corresponds to the primary non-nulls is zero, while the secondary term is nonnegative. Therefore (47) in Lemma 2.11 gives, uniformly for $y \in I$,

$$\bar{R}_{r,-y}\left(\frac{\alpha_q(y) + \delta}{r}\right) \geq \bar{R}_{0,r,-y}\left(\frac{\alpha_q(y) + \delta}{r}\right) = qL_{\alpha_q(y) + \delta}(y) + o(1). \quad (125)$$

By (122), the right side is at least $1 + \gamma$ for all sufficiently small r . Hence

$$\alpha_q(y) - \delta < rC_r(-y) \leq \alpha_q(y) + \delta.$$

Letting $\delta \downarrow 0$ gives

$$rC_r(-y) \longrightarrow \alpha_q(y),$$

uniformly for $y \in I$.

Proof of (117). We now verify regularity for each fixed sufficiently small r . Fix one $0 < \delta < m_I/2$ as in (121) and consider the interval

$$\frac{\alpha_q(y) - \delta}{r} \leq s \leq \frac{\alpha_q(y) + \delta}{r}.$$

The primary non-null has conditional value

$$T_i = c_r + y(b_r) - y = s_r(y).$$

It is positive for sufficiently small r , uniformly in $y \in I$, and it is rejected at cutoff s exactly when $s \leq s_r(y)$. Since $\delta < m_I/2$ and $m_I \leq \alpha_q(y)$, we have $\alpha_q(y) - \delta \geq m_I/2$. On the other hand,

$$rs_r(y) = rc_r + r\{y(b_r) - y\} \longrightarrow 0$$

uniformly for $y \in I$, by (46), $y(b_r) \rightarrow 1$, and compactness of I . Therefore, for all sufficiently small r , for every $y \in I$ and every cutoff s satisfying

$$\frac{\alpha_q(y) - \delta}{r} \leq s \leq \frac{\alpha_q(y) + \delta}{r},$$

we have

$$s_r(y) < \frac{\alpha_q(y) - \delta}{r} \leq s.$$

Thus every cutoff in this interval lies strictly to the right of the primary value $s_r(y)$, so the primary non-null is not rejected there. Moreover, uniformly for these y and s ,

$$\begin{aligned} r(s + y) - g_r(b_r) &\geq \alpha_q(y) - \delta + r\{y - y(b_r)\} - b_r \geq \frac{m_I}{2} - o(1) > 0, \\ g_r(a_q) - r(s + y) &\geq a_q - \alpha_q(y) - \delta + r(y_q - y) \geq \frac{m_I}{2} - o(1) > 0. \end{aligned}$$

Strict monotonicity of g_r therefore implies that $\mathcal{A}_{r,y,s} \neq \emptyset$, that $\beta_{r,y,s} \in (b_r, a_q)$. By (93), continuity and strict monotonicity of g_r give

$$g_r(\beta_{r,y,s}) = r(s + y).$$

Using $g_r(a) = a + ry(a)$ from (58), this equality yields

$$\beta_{r,y,s} = rs + r\{y - y(\beta_{r,y,s})\} = rs + O(r)$$

uniformly. Hence, for all sufficiently small r ,

$$\frac{m_I}{4} \leq \beta_{r,y,s} \leq a_q - \frac{m_I}{4},$$

so the thresholds remain in a fixed compact subinterval of $(0, a_q)$.

For $y \in I$ and cutoffs s in the local crossing interval

$$\frac{\alpha_q(y) - \delta}{r} \leq s \leq \frac{\alpha_q(y) + \delta}{r},$$

write the secondary non-null contribution as

$$\bar{R}_{2,r,-y}(s) = [1 - qM\{y(\beta_{r,y,s})\}] \frac{p(\beta_{r,y,s}/r)}{p(s)}.$$

Then

$$\bar{R}_{r,-y}(s) = \bar{R}_{0,r,-y}(s) + \bar{R}_{2,r,-y}(s).$$

Abbreviate $\beta = \beta_{r,y,s}$. Equation (59) holds uniformly because $g_r(b_r) > 0 > r(y - s)$ for all sufficiently small r . The term in (63) of Lemma 2.14, evaluated at $c = s$ and $\beta_{r,y}(s) = \beta_{r,y,s} = \beta$, is

$$-\frac{qM'\{y(\beta)\}y'(\beta)}{1 - qM\{y(\beta)\}} \frac{r}{1 + ry'(\beta)}.$$

Because

$$\frac{m_I}{4} \leq \beta \leq a_q - \frac{m_I}{4},$$

(198) in Lemma A.1 and continuity give a constant $C_I < \infty$ such that

$$\sup_{m_I/4 \leq a \leq a_q - m_I/4} \left| \frac{qM'\{y(a)\}y'(a)}{1 - qM\{y(a)\}} \right| \leq C_I.$$

Since $y'(\beta) > 0$,

$$0 < \frac{r}{1 + ry'(\beta)} \leq r,$$

and therefore

$$\left| \frac{qM'\{y(\beta)\}y'(\beta)}{1 - qM\{y(\beta)\}} \frac{r}{1 + ry'(\beta)} \right| \leq C_I r.$$

The expressions in (63) are

$$-\frac{\lambda(\beta/r)}{1+ry'(\beta)} \quad \text{and} \quad \lambda(s).$$

To bound their sum, write

$$\lambda(t) = t + \varepsilon(t), \quad 0 < \varepsilon(t) < t^{-1},$$

by (202) in Lemma A.2. On the cutoff range,

$$s \geq \frac{m_I}{2r}, \quad \frac{\beta}{r} \geq \frac{m_I}{4r},$$

and hence $\varepsilon(s) = O(r)$ and $\varepsilon(\beta/r) = O(r)$ uniformly. Using $\beta/r = s + y - y(\beta)$, we obtain

$$\begin{aligned} \lambda(s) - \frac{\lambda(\beta/r)}{1+ry'(\beta)} &= s - \frac{\beta/r}{1+ry'(\beta)} + O(r) \\ &= \frac{rsy'(\beta) + y(\beta) - y}{1+ry'(\beta)} + O(r) = O(1), \end{aligned}$$

uniformly, because rs , y , $y(\beta)$, and $y'(\beta)$ are uniformly bounded. Hence

$$\sup_{y \in I} \sup_{|rs - \alpha_q(y)| \leq \delta} \left| \frac{\partial}{\partial s} \log \bar{R}_{2,r,-y}(s) \right| = O(1). \quad (126)$$

Moreover, $\beta_{r,y,s}/r - s \geq \delta_I$ by (119), so $\beta_{r,y,s}/r \geq s + \delta_I$. Since p is decreasing on $[0, \infty)$, and since (198) in Lemma A.1 gives $0 \leq 1 - qM\{y(\beta_{r,y,s})\} \leq 1$, we have

$$0 \leq \bar{R}_{2,r,-y}(s) \leq \frac{p(s + \delta_I)}{p(s)}.$$

Applying (209) in Lemma A.2 with $x = s$ and $\delta = \delta_I$ gives

$$0 \leq \bar{R}_{2,r,-y}(s) \leq \frac{p(s + \delta_I)}{p(s)} \leq (1 + s^{-2}) \exp \left\{ -s\delta_I - \frac{\delta_I^2}{2} \right\} = o(r), \quad (127)$$

uniformly on the interval. The last $o(r)$ follows because (121) and $0 < \delta < m_I/2$ imply

$$s \geq \frac{\alpha_q(y) - \delta}{r} \geq \frac{m_I}{2r}$$

whenever $y \in I$ and $|rs - \alpha_q(y)| \leq \delta$, and therefore

$$r^{-1}(1 + s^{-2}) \exp \left\{ -s\delta_I - \frac{\delta_I^2}{2} \right\} \leq r^{-1}(1 + o(1)) \exp \left\{ -\frac{m_I\delta_I}{2r} - \frac{\delta_I^2}{2} \right\} \rightarrow 0$$

uniformly. Since $\partial_s \bar{R}_{2,r,-y}(s) = \bar{R}_{2,r,-y}(s) \partial_s \log \bar{R}_{2,r,-y}(s)$, (126) and (127) imply

$$\sup_{y \in I} \sup_{|rs - \alpha_q(y)| \leq \delta} \left| \frac{\partial}{\partial s} \bar{R}_{2,r,-y}(s) \right| = o(r). \quad (128)$$

For the null contribution term, put

$$x_+ = \frac{s + ry}{\sqrt{1 - r^2}}, \quad x_- = \frac{s - ry}{\sqrt{1 - r^2}}, \quad Q_{\pm} = \frac{\bar{\Phi}(x_{\pm})}{\bar{\Phi}(s)}.$$

Because $\alpha_q(y) - \delta \geq m_I/2$, the range in (126) implies

$$s \geq \frac{m_I}{2r} \longrightarrow \infty,$$

while rs and y range over compact sets. Therefore (48) in Lemma 2.11, with $z = -y$, gives uniformly

$$Q_+ = e^{-(rs)^2/2 - rsy} + o(1), \quad Q_- = e^{-(rs)^2/2 + rsy} + o(1).$$

Indeed, $s/x_{\pm} = 1 + o(1)$, the $O(r^2)$ term in (48) is uniform, and its remainder tends to zero because $s \rightarrow \infty$ uniformly.

Next, (203) in Lemma A.2 gives

$$\begin{aligned} \frac{1}{r} \left\{ \lambda(s) - \frac{\lambda(x_+)}{\sqrt{1-r^2}} \right\} &= -rs - y + o(1), \\ \frac{1}{r} \left\{ \lambda(s) - \frac{\lambda(x_-)}{\sqrt{1-r^2}} \right\} &= -rs + y + o(1). \end{aligned}$$

More precisely, for either sign,

$$\lambda(s) - \frac{\lambda(x_{\pm})}{\sqrt{1-r^2}} = s - \frac{s \pm ry}{1-r^2} + \frac{1}{s} - \frac{1}{s \pm ry} + O(r^3),$$

where

$$\frac{1}{s} - \frac{1}{s \pm ry} = O(r^3), \quad \frac{1}{r} \left\{ s - \frac{s \pm ry}{1-r^2} \right\} = \frac{-rs \mp y}{1-r^2}.$$

This proves the two scaled hazard expansions above.

Substitution of these four expansions into (56) in Lemma 2.14, together with $\pi_{0,r} \rightarrow 1$, yields

$$\begin{aligned} \frac{1}{r} \frac{\partial}{\partial s} \bar{R}_{0,r,-y}(s) &= \frac{q}{2} e^{-(rs)^2/2} [e^{-rsy}(-rs - y) + e^{rsy}(-rs + y)] + o(1) \\ &= q e^{-(rs)^2/2} \{-rs \cosh(rsy) + y \sinh(rsy)\} + o(1). \end{aligned}$$

Finally, differentiating $L_a(y) = e^{-a^2/2} \cosh(ay)$ gives

$$\frac{\partial}{\partial a} L_a(y) = e^{-a^2/2} \{-a \cosh(ay) + y \sinh(ay)\}.$$

Evaluating at $a = rs$ proves

$$\frac{1}{r} \frac{\partial}{\partial s} \bar{R}_{0,r,-y}(s) = q e^{-(rs)^2/2} \{-rs \cosh(rsy) + y \sinh(rsy)\} + o(1) = q \frac{\partial}{\partial a} L_a(y) \Big|_{a=rs} + o(1). \quad (129)$$

By (121) and (123), the pairs (rs, y) under consideration range over a compact subset of $\{(a, y) : y \in I, 0 < a < a(y)\}$. Continuity, (115), and Lemma A.1 therefore show that $\partial_a L_a(y)|_{a=rs}$ is bounded away from zero uniformly. Combining (128) and (129) yields

$$\inf_{y \in I} \inf_{|rs - \alpha_q(y)| \leq \delta} \frac{\partial}{\partial s} \bar{R}_{r,-y}(s) > 0 \quad (130)$$

for all sufficiently small r . Equation (116) gives, uniformly for $y \in I$,

$$\frac{\alpha_q(y) - \delta}{r} < C_r(-y) < \frac{\alpha_q(y) + \delta}{r}$$

for all sufficiently small r . Equation (130) and the smoothness of the null contribution for $|rs - \alpha_q(y)| \leq \delta$ verify the hypotheses of Proposition 2.5; hence $C_r(-y)$ is regular.

Equation (44) gives

$$D_r(-y) = \bar{R}_{0,r,-y}\{C_r(-y)\}.$$

Because $rC_r(-y) \rightarrow \alpha_q(y)$ uniformly and $\alpha_q(y)$ stays in a compact subset of $(0, a_q)$, (47) in Lemma 2.11 applies at $c = C_r(-y)$ and gives

$$D_r(-y) = qL_{rC_r(-y)}(y) + o(1) \longrightarrow qL_{\alpha_q(y)}(y) = 1,$$

uniformly for $y \in I$, where the equality follows from (114). This proves (117). $\square$

2.6.5 Conditional limit of D_r

The following lemma establishes the limit of $D_r(z)$ as $r \downarrow 0$, with D_* defined in Section 2.2.

Lemma 2.21. *For every $z \notin \{-1, -y_q\}$,*

$$D_r(z) \longrightarrow D_*(z). \quad (131)$$

Consequently,

$$\mathbb{E}[D_r(Z)] \longrightarrow \mathbb{E}[D_*(Z)] = \ell_=(q). \quad (132)$$

Proof. If $z > -1$, Lemma 2.15 gives $D_r(z) \rightarrow q$. If $-y_q < z < -1$, put $y = -z$ and apply Lemma 2.19. If $z < -y_q$, apply Lemma 2.20. This proves (131).

At every regular crossing, $0 \leq D_r \leq 1$; the value assigned at nonregular points also lies in $[0, 1]$. Dominated convergence gives the convergence in (132). Finally,

$$\begin{aligned} \mathbb{E}[D_*(Z)] &= q\mathbb{P}(Z > -1) + q \int_{-y_q}^{-1} M(-z)\phi(z) dz + \mathbb{P}(Z < -y_q) \\ &= q\Phi(1) + q \int_1^{y_q} M(y)\phi(y) dy + \bar{\Phi}(y_q) \\ &= \ell_=(q). \end{aligned}$$

$\square$

2.7 Proofs of Theorem 2.1 and Propositions 2.2 and 2.3

Proof of Theorem 2.1. Fix $\delta > 0$. Choose a compact set

$$E \subset \mathbb{R} \setminus \{-1, -y_q\}$$

such that

$$\int_E D_*(z)\phi(z) dz > \ell_=(q) - \frac{\delta}{4}.$$

Because E is compact and separated from -1 and $-y_q$, it is the union of at most three compact subsets of the regimes $z > -1$, $-y_q < z < -1$, and $z < -y_q$. Equations (65)–(66), (86)–(87), and (124)–(125), together with the corresponding null limits (91) and (47), therefore give, for all sufficiently small r , brackets

$$A_r(z) < B_r(z), \quad z \in E,$$

and a constant $\gamma > 0$, such that

$$\sup_{z \in E} \sup_{u_q \leq c \leq A_r(z)} \bar{R}_{r,z}(c) \leq 1 - \gamma, \quad \inf_{z \in E} \bar{R}_{r,z}\{B_r(z)\} \geq 1 + \gamma,$$

$$\sup_{z \in E} \sup_{A_r(z) \leq c \leq B_r(z)} |\bar{R}_{0,r,z}(c) - D_*(z)| \leq \frac{\delta}{8}.$$

Fix such an r and put

$$C_* = \sup_{z \in E} B_r(z) < \infty.$$

For $z \in E$, define

$$\mathcal{E}_N(z) = \max \left\{ \|\hat{R}_N - R_{r,z}\|_{\infty,[0,1]}, \|\hat{R}_{0,N} - \pi_{0,r} R_{0,r,z}\|_{\infty,[0,1]} \right\}$$

and

$$t_* = \frac{p(C_*)}{q} \min \left\{ \frac{\gamma}{2}, \frac{\delta}{16} \right\} > 0.$$

Following the argument in the proof of Theorem 2.8, on the event $\mathcal{E}_N(z) < t_*$, for $u_q \leq c \leq C_*$,

$$\begin{aligned} \left| \frac{q\hat{R}_N\{p(c)\}}{p(c)} - \bar{R}_{r,z}(c) \right| &\leq \frac{q\mathcal{E}_N(z)}{p(C_*)}, \\ \left| \frac{q\hat{R}_{0,N}\{p(c)\}}{p(c)} - \bar{R}_{0,r,z}(c) \right| &\leq \frac{q\mathcal{E}_N(z)}{p(C_*)}. \end{aligned}$$

The first bound and the two crossing margins imply

$$A_r(z) < \hat{C}_N \leq B_r(z).$$

The second bound, (25), and the null bracket then give

$$\text{FDP}_N \geq \bar{R}_{0,r,z}(\hat{C}_N) - \frac{\delta}{16} \geq D_*(z) - \frac{3\delta}{16}.$$

Lemma 2.6 gives, uniformly for $z \in E$,

$$\mathbb{P}\{\mathcal{E}_N(z) \geq t_* \mid Z = z\} \leq \eta_N := 4e^{-2Nt_*^2} \longrightarrow 0.$$

Since $\text{FDP}_N \geq 0$ and $0 \leq D_* \leq 1$, conditional expectation and integration over E yield

$$\begin{aligned} \mathbb{E}_{\Omega_N, Z, \varepsilon}[\text{FDP}_N] &\geq (1 - \eta_N) \int_E \left(D_*(z) - \frac{3\delta}{16} \right)_+ \phi(z) \, dz \\ &\geq \int_E D_*(z) \phi(z) \, dz - \frac{3\delta}{16} - \eta_N. \end{aligned}$$

Choose N large enough that $\eta_N < \delta/16$. Using the choice of E and (29), we obtain

$$\mathbb{E}_{\Omega_N}[\text{FDR}_{\Omega_N}(\text{BH}_q)] > \ell_=(q) - \frac{\delta}{2}.$$

For such an N , some realization ω^* satisfies

$$\text{FDR}_{\omega^*}(\text{BH}_q) > \ell_=(q) - \frac{\delta}{2}.$$

By Lemma 2.9, choose $\eta > 0$ so small that

$$\text{FDR}_{\omega^*}^{(\eta)}(\text{BH}_q) > \text{FDR}_{\omega^*}(\text{BH}_q) - \frac{\delta}{2} > \ell_=(q) - \delta. \quad (133)$$

The perturbed vector $T^{(\eta)}$ has finite dimension N , mean vector $(\theta_i)_{i=1}^N \in \mathbb{R}^N$, and covariance matrix $\Sigma_{\omega^*, \eta}$, which has unit diagonal and is strictly positive definite by (34) in Lemma 2.9. Thus the bound (133) establishes (6). Since $\delta > 0$ was arbitrary, taking the supremum proves (7). $\square$

Proof of Proposition 2.2. Recall from Lemma A.1 that, for $y > 1$, $a(y)$ is the unique positive maximizer of $a \mapsto L_a(y)$, equivalently the unique positive solution of

$$a(y) = y \tanh\{a(y)y\}.$$

By the monotonicity of $a(\cdot)$ established in that lemma, $a(y) \geq a(2) > 0$ for $y \geq 2$, so $a(y)y \rightarrow \infty$. Moreover,

$$0 \leq y - a(y) = y[1 - \tanh\{a(y)y\}] \leq 2ye^{-2a(y)y} \rightarrow 0.$$

Because $a(y)$ is the maximizer defining $M(y)$, we have $M(y) = L_{a(y)}(y)$. Substituting the definitions of $L_{a(y)}(y)$ and $\phi(y)$, expanding $\cosh(t) = (e^t + e^{-t})/2$, and completing the two squares gives

$$\begin{aligned} M(y)\phi(y) &= e^{-a(y)^2/2} \cosh\{a(y)y\} \frac{e^{-y^2/2}}{\sqrt{2\pi}} \\ &= \frac{1}{2\sqrt{2\pi}} \left[e^{-\{a(y)^2+y^2\}/2+a(y)y} + e^{-\{a(y)^2+y^2\}/2-a(y)y} \right] \\ &= \frac{1}{2\sqrt{2\pi}} \left[e^{-\{y-a(y)\}^2/2} + e^{-\{y+a(y)\}^2/2} \right]. \end{aligned} \tag{134}$$

Consequently, $M(y)\phi(y) \rightarrow 1/(2\sqrt{2\pi})$. The difference $M(y)\phi(y) - 1/(2\sqrt{2\pi})$ is integrable on $[1, \infty)$. Indeed, for all sufficiently large y , $a(y) \geq y/2$, so

$$y - a(y) \leq 2ye^{-y^2},$$

while

$$\left| e^{-\{y-a(y)\}^2/2} - 1 \right| \leq \frac{\{y-a(y)\}^2}{2}, \quad e^{-\{y+a(y)\}^2/2} \leq e^{-y^2/2}.$$

It follows from the definition of c_ℓ that, as $Y \rightarrow \infty$,

$$\int_1^Y M(y)\phi(y) dy = \frac{Y}{2\sqrt{2\pi}} + c_\ell - \Phi(1) + o(1). \tag{135}$$

Next, (134) gives

$$M(y) = \frac{1}{2}e^{y^2/2} \left[e^{-\{y-a(y)\}^2/2} + e^{-\{y+a(y)\}^2/2} \right] = \frac{1}{2}e^{y^2/2}\{1 + o(1)\}.$$

Since $M(y_q) = 1/q \rightarrow \infty$, (196) in Lemma A.1 implies $y_q \rightarrow \infty$. The identity $qM(y_q) = 1$ therefore gives

$$y_q^2 = 2\log(1/q) + 2\log 2 + o(1), \quad y_q = \sqrt{2\log(1/q) + o(1)}. \tag{136}$$

Equations (214) in Lemma A.2 and (136) give

$$\bar{\Phi}(y_q) \leq \frac{\phi(y_q)}{y_q} = O\left(\frac{q}{y_q}\right) = o(q). \tag{137}$$

Substituting (135), (136), and (137) into (5) yields

$$\begin{aligned} \ell_=(q) &= q\Phi(1) + q \left\{ \frac{y_q}{2\sqrt{2\pi}} + c_\ell - \Phi(1) + o(1) \right\} + o(q) \\ &= \frac{q\sqrt{\log(1/q)}}{2\sqrt{\pi}} + c_\ell q + o(q). \end{aligned}$$

Numerical quadrature of the absolutely convergent integral defining c_ℓ gives $c_\ell = 0.6492828\dots$ □

Proof of Proposition 2.3. Since $q \in (0, 1)$, (4) gives $M(y_q) = 1/q > 1$. By Lemma A.1, $M = 1$ on $[0, 1]$ and M is strictly increasing on $(1, \infty)$, so $y_q > 1$ and $M(y) \geq 1$ for $1 \leq y \leq y_q$. Using (5),

$$\begin{aligned}\ell_=(q) - q &= q\Phi(1) + q \int_1^{y_q} M(y)\phi(y) dy + \bar{\Phi}(y_q) \\ &\quad - q \left\{ \Phi(1) + \int_1^{y_q} \phi(y) dy + \bar{\Phi}(y_q) \right\} \\ &= q \int_1^{y_q} \{M(y) - 1\}\phi(y) dy + (1 - q)\bar{\Phi}(y_q).\end{aligned}$$

The integral is nonnegative, and $(1 - q)\bar{\Phi}(y_q) > 0$ because $q < 1$ and $y_q < \infty$. Hence $\ell_=(q) > q$. $\square$

3 Lower bound for one-sided tests

3.1 Main result

For $0 < q < 1$, define

$$y_q^+ = \sqrt{2 \log(1/q)} \quad (138)$$

and

$$\ell_{\leq}(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + \bar{\Phi}(y_q^+). \quad (139)$$

The following theorem gives the finite-dimensional one-sided Gaussian lower bound.

Theorem 3.1. *For every $q \in (0, 1)$ and $\delta > 0$, there are $m < \infty$, $\theta \in \mathbb{R}^m$, and a positive definite correlation matrix Σ such that BH at level q , applied to the one-sided p -values $P_i = \bar{\Phi}(T_i)$ from $T \sim N(\theta, \Sigma)$, satisfies*

$$\text{FDR}_{\theta, \Sigma}(\text{BH}_q) > \ell_{\leq}(q) - \delta.$$

Every true null in the construction has $\theta_i = 0$, and every non-null has $\theta_i > 0$.

The first proposition gives the small- q expansion of the one-sided lower bound.

Proposition 3.2. *As $q \downarrow 0$,*

$$\ell_{\leq}(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + o(q).$$

In particular, $\ell_{\leq}(q)/q \rightarrow \infty$.

The next proposition shows strict anticonservativeness at every nontrivial nominal level.

Proposition 3.3. *For every $q \in (0, 1)$, $\ell_{\leq}(q) > q$.*

Figure 2 in the Introduction plots $\ell_{\leq}(q) - q$ and $\ell_{\leq}(q)/q$ over the grid $q = 0.005k$, $k = 1, \dots, 200$. Table 2 gives the absolute inflation $\ell_{\leq}(q) - q$ and relative inflation $\ell_{\leq}(q)/q$ for commonly used nominal levels.

3.2 Roadmap

Fix q . For a small $r > 0$, let $(\Theta_i, B_i)_{i=1}^N$ be i.i.d. from a law $\nu_{q,r}^+$, independently of $Z, \varepsilon_1, \dots, \varepsilon_N$, which are independent standard normal variables, and set

$$T_i = \Theta_i + B_i Z + \sqrt{1 - B_i^2} \varepsilon_i. \quad (140)$$

The law $\nu_{q,r}^+$ has one null component and two non-null components. We first analyze the population BH crossing conditional on $Z = z$. Its conditional FDP converges, as $r \downarrow 0$, to

$$D_+^*(z) = \begin{cases} q, & z \leq 0, \\ qe^{z^2/2}, & 0 < z < y_q^+, \\ 1, & z \geq y_q^+. \end{cases} \quad (141)$$

The definition (141) specifies the values at $\{0, y_q^+\}$; these values do not affect the expectation because the boundary points have zero probability. Therefore

$$\mathbb{E}\{D_+^*(Z)\} = \frac{q}{2} + q \int_0^{y_q^+} e^{z^2/2} \phi(z) dz + \bar{\Phi}(y_q^+) = \ell_{\leq}(q).$$

We reuse the machinery and use similar notation in Subsections 2.3 and 2.4. Throughout this section, put $p_+(x) = \bar{\Phi}(x)$ and $u_s^+ = p_+^{-1}(s)$. Use the symbols

$$\pi_0^+, R_z^+, R_{0,z}^+, \bar{R}_z^+, \bar{R}_{0,z}^+, C^+(z), D^+(z)$$

as the one-sided analogues of the quantities in (8) and (11)–(16). In particular, the one-sided population cutoff is

$$C^+(z) = \inf\{c \geq u_q^+ : \bar{R}_z^+(c) \geq 1\},$$

and every one-sided use of Definition 2.4 has lower endpoint u_q^+ in place of u_q . The only change to (9)–(10) is

$$F_{\theta,b,z}^+(s) = \bar{\Phi}\left(\frac{u_s^+ - \theta - bz}{\sqrt{1-b^2}}\right), \quad |b| < 1, \quad (142)$$

while $F_{\theta,b,z}^+(s) = \mathbf{1}\{\theta + bz \geq u_s^+\}$ for $|b| = 1$. We retain the empty-crossing convention $C^+(z) = \infty$, $D^+(z) = 0$, and the regularity definition in Definition 2.4.

The proofs of Lemmas 2.6 and 2.7 and Theorem 2.8 are p-value agnostic and apply after (142); the one-sided rejection set in Lemma 2.7 is $\{i : T_i \geq \hat{C}_N\}$. We also use the localized DKW bracketing in the proof of Theorem 2.8. Finally, the perturbation (32)–(33) and Lemma 2.9 apply to the one-sided p-values as well, because their continuity argument does not use the two-sided form of p .

3.3 Construction of $\nu_{q,r}^+$

This is the one-sided, sign-reversed specialization of the three-component construction in Subsection 2.4: the signal offset $y(a)$ is replaced by a , the envelope term $M\{y(a)\}$ by $e^{a^2/2}$, and the non-null loading $+1$ by -1 .

Take $r > 0$, and set

$$b_r^+ = r p_+^{-1}(r^2). \quad (143)$$

We henceforth assume r is sufficiently small that $r^2 < 1/2$ and $b_r^+ < y_q^+$. For $b_r^+ \leq a \leq y_q^+$, define

$$W_r^+(a) = \frac{\{1 - qe^{a^2/2}\}p_+(a/r)}{q}. \quad (144)$$

Both factors in the numerator are nonnegative and decreasing in a , and the first vanishes at $a = y_q^+$. Hence W_r^+ is continuous and decreasing. Put

$$\tau_r^+ = W_r^+(b_r^+), \quad \mu_r^+([a, y_q^+]) = \frac{W_r^+(a)}{\tau_r^+}, \quad b_r^+ \leq a \leq y_q^+. \quad (145)$$

Thus μ_r^+ is a probability measure on $[b_r^+, y_q^+]$. Since $p_+(b_r^+/r) = r^2$, the quantile expansion (207) in Lemma A.2 gives $b_r^+ \rightarrow 0$, and hence

$$\tau_r^+ \sim \frac{1-q}{q} r^2. \quad (146)$$

Finally, set

$$\pi_{0,r}^+ = 1 - r - \tau_r^+, \quad p_+(c_r^+) = \frac{q(r + \tau_r^+)}{1 - q\pi_{0,r}^+}. \quad (147)$$

For all sufficiently small r , these quantities are well-defined and $\pi_{0,r}^+ > 0$.

The law $\nu_{q,r}^+$ is displayed in Table 4. Equivalently, for bounded measurable f ,

$$\int f \, d\nu_{q,r}^+ = \pi_{0,r}^+ f(0, r) + r f(c_r^+ + b_r^+, -1) + \tau_r^+ \int_{b_r^+}^{y_q^+} f(a/r + a, -1) \mu_r^+(da). \quad (148)$$

Table 4: One-sided population construction. The opposite loading signs are essential.

Type	Mass	Θ	B
Nulls	$\pi_{0,r}^+$	0	r
Primary non-nulls	r	$c_r^+ + b_r^+$	-1
Secondary non-nulls	$\tau_r^+ \mu_r^+(da)$	$a/r + a$	$-1, a \in [b_r^+, y_q^+]$

The nulls become more significant when Z is positive. Every non-null statistic is deterministic conditional on Z and decreases with Z . The construction is not PRDS. To see this directly, pair a null i with a non-null j , and fix $v \in (0, 1)$. Conditional on $P_i = s$, the event $\{P_j \geq v\}$, which is increasing in the p-values, has probability

$$\Phi\left(\frac{p_+^{-1}(v) - \theta_j + r p_+^{-1}(s)}{\sqrt{1 - r^2}}\right),$$

strictly decreasing in s . This also exhibits the negative null-to-non-null covariance, opposite to the nonnegative covariance assumption that guarantees Gaussian one-sided PRDS [Benjamini and Yekutieli, 2001, Chi et al., 2025]. On the other hand, all null-null correlations equal r^2 , so the null p-values are positively regression dependent and the full vector is PRDN in the terminology of Su [2018].

3.4 Population analysis

3.4.1 Preliminaries

The next lemma gives the one-sided counterparts of (45)–(46).

Lemma 3.4. *As $r \downarrow 0$,*

$$b_r^+ \rightarrow 0, \quad b_r^+/r \rightarrow \infty, \quad c_r^+ \rightarrow \infty, \quad r c_r^+ \rightarrow 0, \quad b_r^+/r - c_r^+ \rightarrow \infty. \quad (149)$$

Proof. By (143) and the quantile expansion (207) in Lemma A.2,

$$\frac{b_r^+}{r} = p_+^{-1}(r^2) \sim 2\sqrt{\log(1/r)} \rightarrow \infty, \quad b_r^+ \sim 2r\sqrt{\log(1/r)} \rightarrow 0.$$

Moreover, (146) gives $\tau_r^+ = O(r^2)$ and hence $\pi_{0,r}^+ \rightarrow 1$. Therefore (147) yields

$$p_+(c_r^+) \sim \frac{qr}{1-q}.$$

Another application of (207) in Lemma A.2 gives

$$c_r^+ \sim \sqrt{2 \log(1/r)}.$$

Thus $c_r^+ \rightarrow \infty$ and $rc_r^+ \rightarrow 0$, while

$$\frac{b_r^+}{r} - c_r^+ \sim (2 - \sqrt{2})\sqrt{\log(1/r)} \rightarrow \infty.$$

□

Apply the reused population notation to $\nu_{q,r}^+$, and write the resulting quantities as $R_{r,z}^+$, $R_{0,r,z}^+$, $\bar{R}_{r,z}^+$, $\bar{R}_{0,r,z}^+$, $\bar{R}_{1,r,z}^+$, $\bar{R}_{2,r,z}^+$, $C_r^+(z)$, and $D_r^+(z)$. Then

$$\bar{R}_{r,z}^+(c) = \bar{R}_{0,r,z}^+(c) + \bar{R}_{1,r,z}^+(c) + \bar{R}_{2,r,z}^+(c), \quad (150)$$

$$\bar{R}_{0,r,z}^+(c) = q\pi_{0,r}^+ \frac{p_+\{(c-rz)/\sqrt{1-r^2}\}}{p_+(c)}. \quad (151)$$

$$\bar{R}_{1,r,z}^+(c) = \frac{qr}{p_+(c)} \mathbf{1}\{c \leq c_r^+ + b_r^+ - z\}, \quad (152)$$

$$\bar{R}_{2,r,z}^+(c) = \frac{q\tau_r^+}{p_+(c)} \mu_r^+ \left(\left[\frac{r(c+z)}{1+r}, y_q^+ \right] \right). \quad (153)$$

The next lemma derives the derivatives of the null and secondary non-null contributions.

Lemma 3.5. *For $c > 0$, put*

$$u = \frac{c-rz}{\sqrt{1-r^2}}.$$

The null contribution satisfies

$$\frac{d}{dc} \log \bar{R}_{0,r,z}^+(c) = \lambda(c) - \frac{\lambda(u)}{\sqrt{1-r^2}}, \quad (154)$$

$$\frac{d}{dc} \bar{R}_{0,r,z}^+(c) = \bar{R}_{0,r,z}^+(c) \left\{ \lambda(c) - \frac{\lambda(u)}{\sqrt{1-r^2}} \right\}. \quad (155)$$

Define

$$\beta_{r,z}(c) = \frac{r(c+z)}{1+r}.$$

If $\beta_{r,z}(c) \in (b_r^+, y_q^+)$, then

$$\bar{R}_{2,r,z}^+(c) = \left[1 - qe^{\{\beta_{r,z}(c)\}^2/2} \right] \frac{p_+\{\beta_{r,z}(c)/r\}}{p_+(c)} \quad (156)$$

and

$$\frac{d}{dc} \log \bar{R}_{2,r,z}^+(c) = -\frac{qr\beta_{r,z}(c)e^{\{\beta_{r,z}(c)\}^2/2}}{(1+r) \left[1 - qe^{\{\beta_{r,z}(c)\}^2/2} \right]} + \lambda(c) - \frac{\lambda\{\beta_{r,z}(c)/r\}}{1+r}. \quad (157)$$

Proof. Since

$$\frac{du}{dc} = \frac{1}{\sqrt{1-r^2}}, \quad \frac{d}{dt} \log p_+(t) = \frac{p'_+(t)}{p_+(t)} = -\lambda(t),$$

Equation (151) gives

$$\begin{aligned} \frac{d}{dc} \log \bar{R}_{0,r,z}^+(c) &= \frac{d}{dc} \left[\log(q\pi_{0,r}^+) + \log p_+(u) - \log p_+(c) \right] \\ &= -\frac{\lambda(u)}{\sqrt{1-r^2}} + \lambda(c), \end{aligned}$$

which is (154). Therefore

$$\begin{aligned}\frac{d}{dc}\bar{R}_{0,r,z}^+(c) &= \bar{R}_{0,r,z}^+(c)\frac{d}{dc}\log \bar{R}_{0,r,z}^+(c) \\ &= \bar{R}_{0,r,z}^+(c)\left\{\lambda(c) - \frac{\lambda(u)}{\sqrt{1-r^2}}\right\},\end{aligned}$$

which proves (155).

Now suppose that $\beta_{r,z}(c) \in (b_r^+, y_q^+)$. By (153), (145), and (144),

$$\begin{aligned}\bar{R}_{2,r,z}^+(c) &= \frac{q\tau_r^+}{p_+(c)}\mu_r^+([\beta_{r,z}(c), y_q^+]) \\ &= \frac{qW_r^+\{\beta_{r,z}(c)\}}{p_+(c)} \\ &= \left[1 - qe^{\{\beta_{r,z}(c)\}^2/2}\right] \frac{p_+\{\beta_{r,z}(c)/r\}}{p_+(c)}.\end{aligned}$$

This proves (156). Moreover,

$$\frac{\partial\beta_{r,z}(c)}{\partial c} = \frac{r}{1+r}.$$

Since $\beta_{r,z}(c) < y_q^+$, we have $1 - qe^{\{\beta_{r,z}(c)\}^2/2} > 0$, and hence

$$\begin{aligned}\frac{d}{dc}\log \bar{R}_{2,r,z}^+(c) &= -\frac{q\beta_{r,z}(c)e^{\{\beta_{r,z}(c)\}^2/2}}{1 - qe^{\{\beta_{r,z}(c)\}^2/2}}\frac{\partial\beta_{r,z}(c)}{\partial c} \\ &\quad - \lambda\{\beta_{r,z}(c)/r\}\frac{1}{r}\frac{\partial\beta_{r,z}(c)}{\partial c} + \lambda(c) \\ &= -\frac{qr\beta_{r,z}(c)e^{\{\beta_{r,z}(c)\}^2/2}}{(1+r)[1 - qe^{\{\beta_{r,z}(c)\}^2/2}]} + \lambda(c) - \frac{\lambda\{\beta_{r,z}(c)/r\}}{1+r}.\end{aligned}$$

This proves (157). □

The next lemma gives the limit of the null contribution.

Lemma 3.6. *For every compact $X \subset [0, \infty)$ and $K \subset \mathbb{R}$, as $r \downarrow 0$ and $c \rightarrow \infty$, uniformly over $z \in K$ subject to $rc \in X$,*

$$\bar{R}_{0,r,z}^+(c) = q \exp\left\{(rc)z - \frac{(rc)^2}{2}\right\} + o(1). \quad (158)$$

In particular, if $c = c(r) \rightarrow \infty$ and $rc \rightarrow x \geq 0$, then the left-hand side converges to $q \exp(xz - x^2/2)$, uniformly for z in compact sets.

Proof. Set

$$\Delta_r = \frac{c - rz}{\sqrt{1-r^2}} - c.$$

Because rc and z range over compact sets while $c \rightarrow \infty$, uniformly over the stated range, $\Delta_r = O(r) = o(1)$ and

$$\begin{aligned}c\Delta_r &= c^2\left\{\frac{1}{\sqrt{1-r^2}} - 1\right\} - \frac{rzc}{\sqrt{1-r^2}} \\ &= \frac{1}{2}(rc)^2 - z(rc) + o(1),\end{aligned}$$

where the remainder is uniform. The hazard expansion (203) in Lemma A.2, together with $c \rightarrow \infty$ and $\Delta_r = o(1)$, gives

$$\log \frac{p_+(c + \Delta_r)}{p_+(c)} = - \int_c^{c+\Delta_r} \lambda(t) dt = -c\Delta_r + o(1)$$

uniformly on the same range. Exponentiating, using $(c - rz)/\sqrt{1 - r^2} = c + \Delta_r$, and then applying (151) together with $\pi_{0,r}^+ \rightarrow 1$ proves (158). $\square$

The following lemma is the one-sided analogue of Lemma 2.12.

Lemma 3.7. *Let K be compact. For $z \in K$, let $m_r(z) > u_q^+$ and $w_r(z) > 0$. Suppose that, for every fixed $H > 0$,*

$$\inf_{z \in K} \{m_r(z) - Hw_r(z)\} > u_q^+ \quad (159)$$

for all sufficiently small r , and

$$\sup_{z \in K} \sup_{|h| \leq H} \left| \bar{R}_{r,z}^+ \{m_r(z) + w_r(z)h\} - d(z) - \{1 - d(z)\}e^{\xi(z)h} \right| \rightarrow 0, \quad (160)$$

where $d : K \rightarrow [0, 1)$ and $\xi : K \rightarrow (0, \infty)$ are continuous. Suppose also that, for every $h_0 > 0$,

$$\limsup_{r \downarrow 0} \sup_{z \in K} \sup_{u_q^+ \leq c \leq m_r(z) - w_r(z)h_0} \bar{R}_{r,z}^+(c) < 1. \quad (161)$$

Then

$$\sup_{z \in K} \left| \frac{C_r^+(z) - m_r(z)}{w_r(z)} \right| \rightarrow 0. \quad (162)$$

Proof. The proof of Lemma 2.12 applies with u_q , $\bar{R}_{r,z}$, and $C_r(z)$ replaced by u_q^+ , $\bar{R}_{r,z}^+$, and $C_r^+(z)$, respectively. $\square$

The following lemma is the one-sided analogue of Lemma 2.13.

Lemma 3.8. *Under the assumptions and notation of Lemma 3.7, assume in addition that, for every fixed $H > 0$,*

$$\sup_{z \in K} \sup_{|h| \leq H} \left| \bar{R}_{0,r,z}^+ \{m_r(z) + w_r(z)h\} - d(z) \right| \rightarrow 0. \quad (163)$$

If, for some $H_ > 0$ and all sufficiently small r , the map*

$$h \mapsto \bar{R}_{r,z}^+ \{m_r(z) + w_r(z)h\}$$

is differentiable with positive derivative on $(-H_, H_*)$, and the map*

$$h \mapsto \bar{R}_{0,r,z}^+ \{m_r(z) + w_r(z)h\}$$

is continuous there for every $z \in K$, then $C_r^+(z)$ is regular and

$$\sup_{z \in K} |D_r^+(z) - d(z)| \rightarrow 0. \quad (164)$$

Proof. The proof of Lemma 2.13 applies with u_q , $\bar{R}_{r,z}$, $\bar{R}_{0,r,z}$, $C_r(z)$, and $D_r(z)$ replaced by u_q^+ , $\bar{R}_{r,z}^+$, $\bar{R}_{0,r,z}^+$, $C_r^+(z)$, and $D_r^+(z)$, respectively. $\square$

The next lemma shows that the primary non-null contribution is negligible uniformly away from $z = 0$.

Lemma 3.9. *For every compact $K \subset (0, \infty)$, uniformly for $z \in K$, the primary non-null contribution in (150), whenever nonzero, is bounded by*

$$\frac{qr}{p_+(c_r^+ + b_r^+ - z)} = o(1). \quad (165)$$

Proof. Whenever the primary non-null contribution is nonzero, the indicator in (152) equals one, so $\bar{R}_{1,r,z}^+(c) = qr/p_+(c)$. Equation (152) also gives $c \leq c_r^+ + b_r^+ - z$. Since p_+ is decreasing,

$$\frac{qr}{p_+(c)} \leq \frac{qr}{p_+(c_r^+ + b_r^+ - z)}.$$

The right-hand side is the expression in (165). By (147), it can be written as

$$\frac{qr}{p_+(c_r^+ + b_r^+ - z)} = \frac{r(1 - q\pi_{0,r}^+)}{r + \tau_r^+} \frac{p_+(c_r^+)}{p_+(c_r^+ + b_r^+ - z)}.$$

Fix a compact $K \subset (0, \infty)$, and put $\delta_{r,z} = z - b_r^+$. By (149) in Lemma 3.4, $\delta_{r,z}$ remains in a fixed compact subset of $(0, \infty)$, uniformly for $z \in K$, and $c_r^+ - \delta_{r,z} \rightarrow \infty$ uniformly. Therefore (212) in Lemma A.2, applied with $x = c_r^+ - \delta_{r,z}$, gives

$$\frac{p_+(c_r^+)}{p_+(c_r^+ + b_r^+ - z)} = \frac{p_+(x + \delta_{r,z})}{p_+(x)} \rightarrow 0$$

uniformly for $z \in K$. Since $0 < r(1 - q\pi_{0,r}^+)/(r + \tau_r^+) \leq 1$, the expression in (165) is $o(1)$ uniformly. $\square$

The next lemma bounds the secondary non-null contribution in the two positive- z regimes.

Lemma 3.10. *For all $c, z \in \mathbb{R}$,*

$$0 \leq \frac{q\tau_r^+}{p_+(c)} \mu_r^+ \left(\left[\frac{r(c+z)}{1+r}, y_q^+ \right] \right) \leq \frac{p_+\{(c+z)/(1+r)\}}{p_+(c)}. \quad (166)$$

Proof. If $r(c+z)/(1+r) \leq b_r^+$, then

$$q\tau_r^+ \leq p_+(b_r^+/r) \leq p_+\{(c+z)/(1+r)\}.$$

If $b_r^+ < r(c+z)/(1+r) < y_q^+$, then

$$\begin{aligned} q\tau_r^+ \mu_r^+ \left(\left[\frac{r(c+z)}{1+r}, y_q^+ \right] \right) &= \left[1 - q \exp \left\{ \frac{r^2(c+z)^2}{2(1+r)^2} \right\} \right] p_+ \left(\frac{c+z}{1+r} \right) \\ &\leq p_+ \left(\frac{c+z}{1+r} \right). \end{aligned}$$

If $r(c+z)/(1+r) \geq y_q^+$, the contribution is zero. Dividing these bounds by $p_+(c)$ proves (166). $\square$

3.4.2 The regime $z < 0$

Lemma 3.11. *For every compact $K \subset (-\infty, 0)$, uniformly for $z \in K$,*

$$C_r^+(z) = c_r^+ + o\{(c_r^+)^{-1}\} \quad (167)$$

and

$$D_r^+(z) \rightarrow q. \quad (168)$$

Moreover, for all sufficiently small r , $C_r^+(z)$ is regular for every $z \in K$.

Proof. Proof of (167). Fix a compact $K \subset (-\infty, 0)$. Uniformly for $z \in K$ and bounded h , all primary and secondary non-nulls are rejected at the cutoff $c_r^+ + h/c_r^+$. Indeed, the primary statistic equals $c_r^+ + b_r^+ - z$, so its distance above the cutoff is $b_r^+ - z - h/c_r^+$. This converges to $-z > 0$ because $b_r^+ \rightarrow 0$, $c_r^+ \rightarrow \infty$, and h is bounded.

Among the secondary non-nulls, the smallest statistic is attained at $a = b_r^+$. Its distance above the cutoff is

$$\frac{b_r^+}{r} + b_r^+ - z - c_r^+ - \frac{h}{c_r^+} = \left(\frac{b_r^+}{r} - c_r^+ \right) + b_r^+ - z - \frac{h}{c_r^+},$$

which tends to infinity by (149) in Lemma 3.4. Thus both gaps are positive for all sufficiently small r , uniformly when z ranges over a compact subset of $(-\infty, 0)$ and h remains bounded.

Since every non-null is rejected,

$$\bar{R}_{r,z}^+ \left(c_r^+ + \frac{h}{c_r^+} \right) = \bar{R}_{0,r,z}^+ \left(c_r^+ + \frac{h}{c_r^+} \right) + \frac{q(r + \tau_r^+)}{p_+(c_r^+ + h/c_r^+)}.$$

For every fixed $H > 0$, (149) in Lemma 3.4 gives, uniformly for $z \in K$ and $|h| \leq H$,

$$c_r^+ + \frac{h}{c_r^+} \rightarrow \infty, \quad r \left(c_r^+ + \frac{h}{c_r^+} \right) \rightarrow 0.$$

Therefore (158) in Lemma 3.6 directly yields

$$\bar{R}_{0,r,z}^+ \left(c_r^+ + \frac{h}{c_r^+} \right) \rightarrow q$$

uniformly for $z \in K$ and $|h| \leq H$. For the combined non-null contribution, the calibration (147) gives the exact identity

$$\frac{q(r + \tau_r^+)}{p_+(c_r^+ + h/c_r^+)} = (1 - q\pi_{0,r}^+) \frac{p_+(c_r^+)}{p_+(c_r^+ + h/c_r^+)}.$$

The last ratio tends to e^h by (211) in Lemma A.2, whereas $1 - q\pi_{0,r}^+ \rightarrow 1 - q$. Combining the two contributions gives

$$\bar{R}_{r,z}^+ (c_r^+ + h/c_r^+) \rightarrow q + (1 - q)e^h, \quad \bar{R}_{0,r,z}^+ (c_r^+ + h/c_r^+) \rightarrow q. \quad (169)$$

The first limit is below one for $h < 0$, equals one at zero, and is above one for $h > 0$.

We next exclude earlier crossings. By (149) in Lemma 3.4, uniformly for $z \in K$, all non-nulls are rejected when $u_q^+ \leq c \leq c_r^+$, and $(c - rz)/\sqrt{1 - r^2} > c$ for small r . Hence

$$\bar{R}_{r,z}^+(c) < q\pi_{0,r}^+ + \frac{q(r + \tau_r^+)}{p_+(c)} \leq q\pi_{0,r}^+ + \frac{q(r + \tau_r^+)}{p_+(c_r^+)} = 1. \quad (170)$$

For every fixed $h_0 > 0$, (170) and the calibration (147) give, uniformly for $z \in K$,

$$\begin{aligned} \sup_{u_q^+ \leq c \leq c_r^+ - h_0/c_r^+} \bar{R}_{r,z}^+(c) &\leq q\pi_{0,r}^+ + (1 - q\pi_{0,r}^+) \frac{p_+(c_r^+)}{p_+(c_r^+ - h_0/c_r^+)} \\ &\rightarrow q + (1 - q)e^{-h_0} < 1. \end{aligned} \quad (171)$$

Thus Lemma 3.7 applies with

$$m_r(z) = c_r^+, \quad w_r(z) = \frac{1}{c_r^+}, \quad d(z) = q, \quad \xi(z) = 1.$$

Equations (169) and (171) verify (160) and (161), respectively. Equation (162) therefore gives (167).

Proof of (168). To apply Lemma 3.8, it remains to prove that the derivative of $\bar{R}_{r,z}^+(c_r^+ + h/c_r^+)$ with respect to h is positive there. Fix $H < \infty$. Uniformly for $z \in K$ and $|h| \leq H$, put

$$c = c_r^+ + \frac{h}{c_r^+}, \quad u = \frac{c - rz}{\sqrt{1 - r^2}}.$$

Equation (149) in Lemma 3.4 gives $c/c_r^+ \rightarrow 1$ and $u/c_r^+ \rightarrow 1$ uniformly. Hence (202) in Lemma A.2 gives

$$\begin{aligned} \lambda(c) - \frac{\lambda(u)}{\sqrt{1 - r^2}} &= c - \frac{u}{\sqrt{1 - r^2}} + O\{(c_r^+)^{-1}\} \\ &= \frac{rz - r^2 c}{1 - r^2} + O\{(c_r^+)^{-1}\} \\ &= O_{K,H}\left(r + r^2 c_r^+ + \frac{1}{c_r^+}\right) = o(c_r^+). \end{aligned}$$

Thus the hazard difference in (155) is $o(c_r^+)$ uniformly. Applying (155) in Lemma 3.5, differentiating the fully rejected non-null contribution directly using $p'_+(c) = -\lambda(c)p_+(c)$, and then using (169), the calibration (147), and (211) in Lemma A.2, gives

$$\frac{\partial}{\partial h} \bar{R}_{0,r,z}^+\left(c_r^+ + \frac{h}{c_r^+}\right) \rightarrow 0, \quad \frac{\partial}{\partial h} \frac{q(r + \tau_r^+)}{p_+(c_r^+ + h/c_r^+)} \rightarrow (1 - q)e^h, \quad \frac{\partial}{\partial h} \bar{R}_{r,z}^+\left(c_r^+ + \frac{h}{c_r^+}\right) \rightarrow (1 - q)e^h.$$

Since $\inf_{|h| \leq H} (1 - q)e^h = (1 - q)e^{-H} > 0$, for all sufficiently small r ,

$$\inf_{z \in K} \inf_{|h| \leq H} \frac{\partial}{\partial h} \bar{R}_{r,z}^+\left(c_r^+ + \frac{h}{c_r^+}\right) > 0.$$

The second limit in (169) verifies (163). The derivative bound above and continuity of the null contribution verify the remaining hypotheses of Lemma 3.8. That lemma therefore gives regularity, and (164) gives (168), uniformly on K . $\square$

3.4.3 The regime $z \in (0, y_q^+)$

Lemma 3.12. *For every compact $K \subset (0, y_q^+)$, uniformly for $z \in K$,*

$$C_r^+(z) = \frac{z}{r} + o(1) \tag{172}$$

and

$$D_r^+(z) \rightarrow qe^{z^2/2}. \tag{173}$$

Moreover, for all sufficiently small r , $C_r^+(z)$ is regular for every $z \in K$.

Proof. Proof of (172). Fix a compact $K \subset (0, y_q^+)$ and $H > 0$, and put $c = z/r + h$. On $z \in K$ and $|h| \leq H$, the primary non-null contribution in (150) is nonzero only if

$$c_r^+ + b_r^+ - z \geq \frac{z}{r} + h.$$

Let $\delta_K = \inf_{z \in K} z > 0$. Uniformly for $z \in K$ and $|h| \leq H$,

$$\begin{aligned} r \left\{ \frac{z}{r} + h - (c_r^+ + b_r^+ - z) \right\} &= z + rh - rc_r^+ - rb_r^+ + rz \\ &\geq \delta_K - rH - rc_r^+ - rb_r^+ + r\delta_K \rightarrow \delta_K > 0 \end{aligned}$$

by (149) in Lemma 3.4. Hence

$$c_r^+ + b_r^+ - z < \frac{z}{r} + h \quad (174)$$

for all sufficiently small r , uniformly for $z \in K$ and $|h| \leq H$, so the primary non-null contribution is absent.

A secondary non-null is rejected if and only if

$$a \geq \beta_{r,z}(c) = z + \frac{rh}{1+r}.$$

Compactness of $K \subset (0, y_q^+)$ implies that $\beta_{r,z}(c) \in (b_r^+, y_q^+)$ for all sufficiently small r , uniformly for $z \in K$ and $|h| \leq H$. In the formula for $\bar{R}_{2,r,z}^+(c)$ in Lemma 3.5,

$$\beta_{r,z}(c) \longrightarrow z, \quad \frac{\beta_{r,z}(c)}{r} = \frac{z}{r} + \frac{h}{1+r},$$

uniformly on $K \times [-H, H]$, so its first factor converges to $1 - qe^{z^2/2}$. With $x = z/r$, (206) in Lemma A.2 and the identity $p_+(x+u)/p_+(x+v) = p(x+u)/p(x+v)$ give

$$\begin{aligned} \log \frac{p_+\{x+h/(1+r)\}}{p_+(x+h)} &= \log \frac{x+h}{x+h/(1+r)} - x \left\{ \frac{h}{1+r} - h \right\} - \frac{1}{2} \left\{ \frac{h^2}{(1+r)^2} - h^2 \right\} + o(1) \\ &= \log \left\{ 1 + \frac{hr}{(1+r)x+h} \right\} + \frac{zh}{1+r} + \frac{h^2r(2+r)}{2(1+r)^2} + o(1) \\ &= zh + o(1) \quad \text{uniformly on } K \times [-H, H]. \end{aligned}$$

Consequently,

$$\bar{R}_{2,r,z}^+(z/r+h) \longrightarrow \{1 - qe^{z^2/2}\}e^{zh}. \quad (175)$$

For $c = z/r + h$, uniformly for $z \in K$ and $|h| \leq H$, we have $c \rightarrow \infty$ and $rc = z + rh$. By uniformity of (158) in Lemma 3.6,

$$\sup_{z \in K} \sup_{|h| \leq H} \left| \bar{R}_{0,r,z}^+(z/r+h) - qe^{z^2/2} \right| \longrightarrow 0.$$

Combining this with (175) and the absence of the primary non-null contribution for $z \in K$ and $|h| \leq H$ gives

$$\bar{R}_{r,z}^+(z/r+h) \longrightarrow qe^{z^2/2} + \{1 - qe^{z^2/2}\}e^{zh}, \quad \bar{R}_{0,r,z}^+(z/r+h) \longrightarrow qe^{z^2/2}. \quad (176)$$

Because $qe^{z^2/2} < 1$ uniformly away from one on K , the first limit crosses one strictly at $h = 0$.

We now verify (161) in Lemma 3.7. For every fixed $\eta > 0$, we show that

$$\limsup_{r \downarrow 0} \sup_{z \in K} \sup_{u_q^+ \leq c \leq z/r - \eta} \bar{R}_{r,z}^+(c) < 1. \quad (177)$$

Suppose instead that this assertion failed. Then there would be sequences satisfying

$$r_n \downarrow 0, \quad z_n \in K, \quad u_q^+ \leq c_n \leq \frac{z_n}{r_n} - \eta, \quad \bar{R}_{r_n, z_n}^+(c_n) \geq 1 - o(1).$$

After passing to subsequences,

$$z_n \longrightarrow z_0, \quad r_n c_n \longrightarrow x \in [0, z_0].$$

We consider three cases to get a contradiction.

Case (a). If $c_n = O(1)$, then, since $z_n \in K$,

$$\Delta_n := \frac{c_n - r_n z_n}{\sqrt{1 - r_n^2}} - c_n = c_n \left\{ \frac{1}{\sqrt{1 - r_n^2}} - 1 \right\} - \frac{r_n z_n}{\sqrt{1 - r_n^2}} = o(1).$$

Moreover, $\inf_n p_+(c_n) > 0$. Hence continuity of p_+ , $\pi_{0,r_n}^+ \rightarrow 1$, and (151) give

$$\frac{p_+(c_n + \Delta_n)}{p_+(c_n)} = 1 + o(1), \quad \bar{R}_{0,r_n,z_n}^+(c_n) = q\pi_{0,r_n}^+ \frac{p_+(c_n + \Delta_n)}{p_+(c_n)} = q + o(1).$$

By (150),

$$\begin{aligned} 0 &\leq \bar{R}_{r_n,z_n}^+(c_n) - \bar{R}_{0,r_n,z_n}^+(c_n) \\ &= \frac{qr_n}{p_+(c_n)} \mathbf{1}\{c_{r_n}^+ + b_{r_n}^+ - z_n \geq c_n\} + \frac{q\tau_{r_n}^+}{p_+(c_n)} \mu_{r_n}^+ \left(\left[\frac{r_n(c_n + z_n)}{1 + r_n}, y_q^+ \right] \right) \\ &\leq \frac{q(r_n + \tau_{r_n}^+)}{p_+(c_n)} = O(r_n) = o(1), \end{aligned}$$

where the last equality uses (146). Thus $\bar{R}_{r_n,z_n}^+(c_n) = q + o(1) < 1$, a contradiction.

Case (b). $c_n \rightarrow \infty$ and $x < z_0$. Put $d_n = (c_n + z_n)/(1 + r_n)$. Then

$$d_n - c_n = \frac{z_n - r_n c_n}{1 + r_n} \rightarrow z_0 - x > 0, \quad c_n(d_n - c_n) \rightarrow \infty.$$

By (166) in Lemma 3.10 and (202) in Lemma A.2, the secondary non-null contribution satisfies

$$0 \leq \frac{q\tau_{r_n}^+}{p_+(c_n)} \mu_{r_n}^+ \left(\left[\frac{r_n(c_n + z_n)}{1 + r_n}, y_q^+ \right] \right) \leq \frac{p_+(d_n)}{p_+(c_n)} = \exp \left\{ - \int_{c_n}^{d_n} \lambda(t) dt \right\} \leq e^{-c_n(d_n - c_n)} = o(1).$$

Equations (165) and (158) in Lemmas 3.9 and 3.6 therefore yield

$$\bar{R}_{r_n,z_n}^+(c_n) = q \exp \left(xz_0 - \frac{x^2}{2} \right) + o(1) \leq qe^{z_0^2/2} + o(1) < 1,$$

again a contradiction.

Case (c). $c_n \rightarrow \infty$ and $x = z_0$. Write $h_n = c_n - z_n/r_n \leq -\eta$. If h_n is bounded, (176) gives a limit strictly below one. If $h_n \rightarrow -\infty$, then

$$d_n - c_n = -\frac{r_n h_n}{1 + r_n} > 0, \quad c_n(d_n - c_n) \sim -z_0 h_n \rightarrow \infty.$$

The hazard bound (202) in Lemma A.2 gives

$$\frac{p_+(d_n)}{p_+(c_n)} = \exp \left\{ - \int_{c_n}^{d_n} \lambda(t) dt \right\} \leq \exp \{ -c_n(d_n - c_n) \} \rightarrow 0.$$

Hence (166) makes the secondary non-null contribution $o(1)$, while (165) makes the primary contribution $o(1)$. Since $r_n c_n \rightarrow x = z_0$ and $z_n \rightarrow z_0$, (158) gives

$$\bar{R}_{0,r_n,z_n}^+(c_n) \rightarrow q \exp \left(z_0^2 - \frac{z_0^2}{2} \right) = qe^{z_0^2/2} < 1.$$

Therefore

$$\bar{R}_{r_n, z_n}^+(c_n) = \bar{R}_{0, r_n, z_n}^+(c_n) + o(1) \longrightarrow qe^{z_0^2/2} < 1,$$

again a contradiction.

Since all three cases lead to a contradiction, (177) holds. Apply Lemma 3.7 with

$$m_r(z) = \frac{z}{r}, \quad w_r(z) = 1, \quad d(z) = qe^{z^2/2}, \quad \xi(z) = z.$$

Equations (176) and (177) verify (160) and (161), respectively. Therefore (162) yields (172).

Proof of (173). We apply Lemma 3.8. Fix $H > 0$, put

$$c = \frac{z}{r} + h, \quad \beta_{r,z}(c) = z + \frac{rh}{1+r}, \quad |h| \leq H.$$

Equation (174) shows that the primary non-null contribution is absent uniformly on this set. Put $u = (c - rz)/\sqrt{1 - r^2}$. The hazard expansion (203) in Lemma A.2 gives

$$\lambda(c) - \frac{\lambda(u)}{\sqrt{1 - r^2}} = o(1)$$

uniformly for $z \in K$ and $|h| \leq H$. In the secondary log derivative (157), the first term is $O(r)$, while the same hazard expansion gives

$$\lambda(c) - \frac{\lambda\{\beta_{r,z}(c)/r\}}{1+r} = z + o(1)$$

uniformly. Combining these two estimates with (175) yields

$$\sup_{z \in K} \sup_{|h| \leq H} \left| \frac{\partial}{\partial h} \bar{R}_{r,z}^+(z/r + h) - z\{1 - qe^{z^2/2}\}e^{zh} \right| \longrightarrow 0.$$

Since

$$\inf_{z \in K} \inf_{|h| \leq H} z\{1 - qe^{z^2/2}\}e^{zh} > 0,$$

the derivative is positive on every fixed local window for all sufficiently small r . Equation (172) verifies (162), and the second limit in (176) verifies (163). Continuity follows from (151). Lemma 3.8 therefore gives regularity, and (164) gives (173), uniformly on K . $\square$

3.4.4 The regime $z > y_q^+$

For $z > y_q^+$, define

$$\alpha_q^+(z) = z - \sqrt{z^2 - (y_q^+)^2} \in (0, z).$$

Lemma 3.13. *For every compact $K \subset (y_q^+, \infty)$, uniformly for $z \in K$,*

$$rC_r^+(z) \longrightarrow \alpha_q^+(z) \tag{178}$$

and

$$D_r^+(z) \longrightarrow 1. \tag{179}$$

Moreover, for all sufficiently small r , $C_r^+(z)$ is regular for every $z \in K$.

Proof. Proof of (178). Fix a compact $K \subset (y_q^+, \infty)$. For $z \in K$ and $0 < x < z$, (149) in Lemma 3.4 implies that the primary non-null is not rejected at $c = x/r$. For the secondary contribution, put $d = (x/r + z)/(1 + r)$. Then

$$d - \frac{x}{r} = \frac{z - x}{1 + r} > 0,$$

and (166) together with (202) in Lemma A.2 gives

$$0 \leq \bar{R}_{2,r,z}^+(x/r) \leq \frac{p_+(d)}{p_+(x/r)} \leq \exp\{-(x/r)(d - x/r)\} \longrightarrow 0.$$

Equation (158) in Lemma 3.6, applied with $c = x/r$, gives

$$\bar{R}_{r,z}^+(x/r) \longrightarrow q \exp\left(xz - \frac{x^2}{2}\right), \quad (180)$$

with the same limit for the null contribution. The equation

$$q \exp\left(xz - \frac{x^2}{2}\right) = 1$$

has smaller positive solution $\alpha_q^+(z)$. The limit in (180) is below one for $x < \alpha_q^+(z)$ and above one for $\alpha_q^+(z) < x < z$.

We now exclude earlier crossings. Fix $\eta > 0$ uniformly small enough that

$$0 < \alpha_q^+(z) - \eta < \alpha_q^+(z) + \eta < z, \quad z \in K.$$

If a crossing existed at some $u_q^+ \leq c \leq \{\alpha_q^+(z) - \eta\}/r$, pass to subsequences so that $z \rightarrow z_0$ and $rc \rightarrow x$, where $0 \leq x \leq \alpha_q^+(z_0) - \eta$. If $c = O(1)$, then $p_+(c)$ is bounded away from zero, (151) gives $\bar{R}_{0,r,z}^+(c) = q + o(1)$, and the primary and secondary contributions are respectively $O(r)$ and $O(\tau_r^+) = O(r^2)$. Thus the full curve tends to $q < 1$.

It remains to consider $c \rightarrow \infty$. Put $d = (c + z)/(1 + r)$. Since

$$d - c = \frac{z - rc}{1 + r} \longrightarrow z_0 - x > 0,$$

Lemma 3.10 and the hazard bound imply $\bar{R}_{2,r,z}^+(c) = o(1)$. Lemma 3.9 gives the same conclusion for the primary contribution, while the compact-uniform form of Lemma 3.6 gives

$$\bar{R}_{r,z}^+(c) \longrightarrow q \exp\left(xz_0 - \frac{x^2}{2}\right) < 1.$$

This includes the case $x = 0$. Hence no earlier crossing exists. At $c = \{\alpha_q^+(z) + \eta\}/r$, the null contribution alone is larger than one for all sufficiently small r , uniformly on K . Therefore

$$\alpha_q^+(z) - \eta < rC_r^+(z) < \alpha_q^+(z) + \eta.$$

Letting $\eta \downarrow 0$ proves

$$rC_r^+(z) \longrightarrow \alpha_q^+(z)$$

uniformly on K .

Proof of (179). It remains to prove regularity. Choose $\eta > 0$ such that, for every $z \in K$,

$$0 < \alpha_q^+(z) - 2\eta < \alpha_q^+(z) + 2\eta < \min\{z, y_q^+\}.$$

For any $x \in [\alpha_q^+(z) - \eta, \alpha_q^+(z) + \eta]$, uniformly over $z \in K$,

$$r\{x/r - (c_r^+ + b_r^+ - z)\} = x - rc_r^+ - rb_r^+ + rz > 0$$

by (149) in Lemma 3.4, so the primary term is zero. Applying (203) in Lemma A.2 directly to (154) in Lemma 3.5 gives, uniformly for $z \in K$ and $x \in [\alpha_q^+(z) - \eta, \alpha_q^+(z) + \eta]$,

$$\left. \frac{d}{dc} \log \bar{R}_{0,r,z}^+(c) \right|_{c=x/r} = r(z - x) + o(r).$$

Set

$$c_1 = \inf_{z \in K} \alpha_q^+(z) - \eta > 0, \quad \delta = \frac{1}{2} \left\{ \inf_{z \in K} [z - \alpha_q^+(z)] - \eta \right\} > 0.$$

The null limit (158) in Lemma 3.6 and the last display imply that, for some $c_0 > 0$,

$$\begin{aligned} \inf_{z \in K} \inf_{x \in [\alpha_q^+(z) - \eta, \alpha_q^+(z) + \eta]} \bar{R}_{0,r,z}^+(x/r) &\geq c_0, \\ \inf_{z \in K} \inf_{x \in [\alpha_q^+(z) - \eta, \alpha_q^+(z) + \eta]} \left. \frac{d}{dc} \bar{R}_{0,r,z}^+(c) \right|_{c=x/r} &\geq c_0 \delta r. \end{aligned} \tag{181}$$

for all sufficiently small r . For the secondary non-null contribution, put $d = (x/r + z)/(1 + r)$. Here $rd \in (b_r^+, y_q^+)$, $x/r \geq c_1/r$, and $d - x/r \geq \delta$. Thus (157) in Lemma 3.5 and (202) in Lemma A.2 yield

$$0 \leq \bar{R}_{2,r,z}^+(x/r) \leq \frac{p_+(d)}{p_+(x/r)} \leq e^{-(x/r)(d-x/r)} \leq e^{-c_1 \delta / r}, \quad \left| \left. \frac{d}{dc} \bar{R}_{2,r,z}^+(c) \right|_{c=x/r} \right| = O(r^{-1} e^{-c_1 \delta / r}) = o(r).$$

Combining this bound with (181),

$$\inf_{z \in K} \inf_{x \in [\alpha_q^+(z) - \eta, \alpha_q^+(z) + \eta]} \left. \frac{d}{dc} \bar{R}_{r,z}^+(c) \right|_{c=x/r} > 0.$$

Equation (178) gives

$$\frac{\alpha_q^+(z) - \eta}{r} < C_r^+(z) < \frac{\alpha_q^+(z) + \eta}{r}$$

for all sufficiently small r , uniformly for $z \in K$. Hence Proposition 2.5 proves regularity uniformly on K .

At the population cutoff, regularity gives $\bar{R}_{r,z}^+\{C_r^+(z)\} = 1$. Equation (149) in Lemma 3.4 implies that the primary non-null contribution is zero. With $d = \{C_r^+(z) + z\}/(1 + r)$, (178) gives

$$C_r^+(z) \longrightarrow \infty, \quad d - C_r^+(z) = \frac{z - rC_r^+(z)}{1 + r} \longrightarrow z - \alpha_q^+(z) > 0$$

uniformly on K . Hence Lemma 3.10 and (202) in Lemma A.2 give

$$0 \leq \bar{R}_{2,r,z}^+\{C_r^+(z)\} \leq \exp[-C_r^+(z)\{d - C_r^+(z)\}] \longrightarrow 0.$$

It follows that $\bar{R}_{0,r,z}^+\{C_r^+(z)\} \rightarrow 1$. By the definition of $D_r^+(z)$, uniformly on K ,

$$D_r^+(z) = \bar{R}_{0,r,z}^+\{C_r^+(z)\} \longrightarrow 1.$$

□

3.5 Proofs of Theorem 3.1 and Propositions 3.2 and 3.3

Proof of Theorem 3.1. Choose a compact set

$$E \subset \mathbb{R} \setminus \{0, y_q^+\}$$

such that

$$\mathbb{E}\{D_+^*(Z)\mathbf{1}\{Z \in E\}\} > \ell_{\leq}(q) - \frac{\delta}{4}.$$

The proofs of Lemmas 3.11, 3.12, and 3.13 give, for all sufficiently small r , brackets

$$A_r(z) < B_r(z), \quad z \in E,$$

and a constant $\gamma > 0$ such that

$$\sup_{z \in E} \sup_{u_q^+ \leq c \leq A_r(z)} \bar{R}_{r,z}^+(c) \leq 1 - \gamma, \quad \inf_{z \in E} \bar{R}_{r,z}^+\{B_r(z)\} \geq 1 + \gamma,$$

$$\sup_{z \in E} \sup_{A_r(z) \leq c \leq B_r(z)} \left| \bar{R}_{0,r,z}^+(c) - D_+^*(z) \right| \leq \frac{\delta}{8}.$$

Fix such an r , and put $C_* = \sup_{z \in E} B_r(z) < \infty$. For $z \in E$, define

$$\mathcal{E}_N(z) = \max \left\{ \|\hat{R}_N - R_{r,z}^+\|_{\infty}, \|\hat{R}_{0,N} - \pi_{0,r}^+ R_{0,r,z}^+\|_{\infty} \right\}$$

and

$$t_* = \frac{p_+(C_*)}{q} \min \left\{ \gamma, \frac{\delta}{8} \right\}.$$

Following the same argument in the proof of Theorem 2.8, Lemma 2.6 gives

$$\sup_{z \in E} \mathbb{P}\{\mathcal{E}_N(z) > t_* \mid Z = z\} \leq 4e^{-2Nt_*^2}.$$

On the complementary event, the empirical and population curves differ by at most $q\mathcal{E}_N(z)/p_+(C_*)$ throughout $[u_q^+, C_*]$. The two population brackets and Lemma 2.7 therefore imply

$$A_r(z) < \hat{C}_N \leq B_r(z), \quad \text{FDP}_N \geq D_+^*(z) - \frac{\delta}{4}.$$

Consequently,

$$\begin{aligned} \mathbb{E}_{\Omega_N}\{\text{FDR}_{\Omega_N}(\text{BH}_q)\} &= \mathbb{E}_{\Omega_N, Z, \varepsilon}(\text{FDP}_N) \\ &\geq \int_E \left\{ D_+^*(z) - \frac{\delta}{4} \right\} \phi(z) \, dz - 4e^{-2Nt_*^2} \\ &> \ell_{\leq}(q) - \frac{\delta}{2} - 4e^{-2Nt_*^2}. \end{aligned}$$

For all sufficiently large N , this gives

$$\mathbb{E}_{\Omega_N}\{\text{FDR}_{\Omega_N}(\text{BH}_q)\} > \ell_{\leq}(q) - \frac{3\delta}{4}.$$

Therefore some deterministic realization ω satisfies the same inequality. The one-sided analogue of Lemma 2.9 permits an $\eta > 0$ for which the perturbed model has a positive definite correlation matrix and

$$\text{FDR}_{\omega}^{(\eta)}(\text{BH}_q) > \ell_{\leq}(q) - \delta.$$

All means and null labels are unchanged by the perturbation. Table 4 gives the pre-perturbation sign pattern, and (33) shows that every off-diagonal covariance is multiplied by the positive factor $1 - \eta^2$. Thus the perturbation preserves that sign pattern. $\square$

Proof of Proposition 3.2. Since $e^{-(y_q^+)^2/2} = q$, the Mills expansion in Lemma A.2 gives

$$\bar{\Phi}(y_q^+) \sim \frac{q}{\sqrt{2\pi} y_q^+} = o(q).$$

Substitute $y_q^+ = \sqrt{2\log(1/q)}$ into (139). $\square$

Proof of Proposition 3.3. By (141), $D_+^*(z) = q$ for $z \leq 0$ and $D_+^*(z) > q$ for every $z > 0$. Hence $\ell_{\leq}(q) = \mathbb{E}\{D_+^*(Z)\} > q$. $\square$

4 Upper bounds

4.1 Preliminaries

The upper bounds concern the common-factor Gaussian model

$$T_i = \mu_i + a_i Z + \sqrt{1 - a_i^2} \varepsilon_i, \quad 1 \leq i \leq m, \quad (182)$$

where $Z, \varepsilon_1, \dots, \varepsilon_m$ are i.i.d. standard normal random variables and $|a_i| \leq 1$. Its covariance matrix is diagonal plus rank one. Conditional on Z , the coordinates and their corresponding p -values are independent.

The following conditional leave-one-out bound will be used for both types of tests. Let $P_1, \dots, P_m$ be conditionally independent given Z , let $\mathcal{H}_0$ denote the true-null indices, and apply BH at level q . For $i \in \mathcal{H}_0$, write $F_{i,z}(t) = \mathbb{P}(P_i \leq t \mid Z = z)$.

Lemma 4.1. *For almost every z ,*

$$\mathbb{E}(\text{FDP} \mid Z = z) \leq \frac{q}{m} \sum_{i \in \mathcal{H}_0} \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t} \quad (183)$$

Proof. Let R be the BH rejection count. For $i \in \mathcal{H}_0$, let R_i be the rejection count after replacing P_i by zero; in particular, $R_i \geq 1$. The standard leave-one-out argument [Benjamini and Yekutieli, 2001] starts from the pointwise identity

$$\frac{\mathbf{1}\{i \text{ is rejected}\}}{R \vee 1} = \frac{\mathbf{1}\{P_i \leq qR_i/m\}}{R_i}.$$

Indeed, if i is rejected, replacing P_i by zero does not change the rejection count. Conversely, if $P_i \leq qR_i/m$, restoring P_i leaves exactly R_i rejections by monotonicity and maximality of the BH step-up rule. Summing and taking expectations gives

$$\text{FDR} = \sum_{i \in \mathcal{H}_0} \mathbb{E} \left[\frac{\mathbf{1}\{P_i \leq qR_i/m\}}{R_i} \right]. \quad (184)$$

Conditional independence implies that P_i and R_i are independent given $Z = z$. Since $0 < qR_i/m \leq q$, conditioning gives, for every $i \in \mathcal{H}_0$ and almost every z ,

$$\begin{aligned} \mathbb{E} \left[\frac{\mathbf{1}\{P_i \leq qR_i/m\}}{R_i} \mid Z = z \right] &= \frac{q}{m} \mathbb{E} \left[\frac{F_{i,z}(qR_i/m)}{qR_i/m} \mid Z = z \right] \\ &\leq \frac{q}{m} \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t}. \end{aligned} \quad (185)$$

Summing the last bound over $i \in \mathcal{H}_0$ proves (183). $\square$

4.2 Upper bound for two-sided tests

In this subsection, we examine the common-factor class (182), whose covariance matrices are diagonal plus rank one and which contains the construction in Subsection 2.4. We show that the upper bound is $O(q\sqrt{\log(1/q)})$, which gives matching orders for small q .

Theorem 4.2. *Consider the common-factor model (182). Form the two-sided Gaussian p -values $P_i = p(|T_i|)$ and apply BH at level $0 < q \leq 2\bar{\Phi}(1) \approx 0.3173$. Let $\mathcal{H}_0 = \{i : \mu_i = 0\}$, $m_0 = |\mathcal{H}_0|$, and $\pi_0 = m_0/m$. If u_q is defined by $2\bar{\Phi}(u_q) = q$, then $\text{FDR}(\text{BH}_q) = 0$ when $m_0 = 0$, and otherwise*

$$\text{FDR}(\text{BH}_q) \leq q \left\{ 1 + \pi_0 \left(\frac{1-q}{2} + \frac{u_q}{\sqrt{2\pi}} \right) \right\}.$$

Consequently, the worst-case FDR in this common-factor class is $O\{q\sqrt{\log(1/q)}\}$, uniformly in m , the means $(\mu_1, \dots, \mu_m)$, and the loadings $(a_1, \dots, a_m)$, as $q \downarrow 0$.

Remark 4.3 (PRDN for the two-sided null p -values). The theorem imposes no sign restriction on the loadings a_i , so the null correlations $a_i a_j$ may be negative. Nevertheless, the two-sided null p -values satisfy PRDN. Let

$$T_0 = (T_i : i \in \mathcal{H}_0), \quad a_0 = (a_i : i \in \mathcal{H}_0), \quad \varepsilon_0 = (\varepsilon_i : i \in \mathcal{H}_0).$$

Thus T_0 is the m_0 -dimensional subvector containing precisely the test statistics for the true nulls; the subscript 0 denotes restriction to $\mathcal{H}_0$, not an additional coordinate. Put

$$D_0 = \text{diag}(1 - a_i^2 : i \in \mathcal{H}_0).$$

Since $\mu_i = 0$ for $i \in \mathcal{H}_0$,

$$T_0 = a_0 Z + D_0^{1/2} \varepsilon_0. \tag{186}$$

Hence T_0 is centered Gaussian with covariance matrix

$$\Sigma_0 = D_0 + a_0 a_0^\top.$$

Assume first that $|a_i| < 1$ for every null coordinate. Let

$$B = \text{diag}\{\text{sign}(a_i) : i \in \mathcal{H}_0\}, \quad r_0 = (|a_i| : i \in \mathcal{H}_0)^\top,$$

with $\text{sign}(0) = 1$. Then

$$B \Sigma_0 B = D_0 + r_0 r_0^\top.$$

By the Sherman–Morrison formula,

$$(D_0 + r_0 r_0^\top)^{-1} = D_0^{-1} - \frac{D_0^{-1} r_0 r_0^\top D_0^{-1}}{1 + r_0^\top D_0^{-1} r_0}.$$

Thus every off-diagonal entry of $(D_0 + r_0 r_0^\top)^{-1}$ is nonpositive. The Karlin–Rinott absolute-value multinormal MTP₂ criterion [Karlin and Rinott, 1981] therefore applies to BT_0 , and $(|(BT_0)_i| : i \in \mathcal{H}_0)$ is MTP₂. Since $|(BT_0)_i| = |T_i|$, coordinatewise sign changes do not alter the absolute null statistics. Applying the same decreasing map $x \mapsto p(x) = 2\bar{\Phi}(x)$ to every coordinate preserves MTP₂. Thus the vector of two-sided null p -values is MTP₂, which implies PRDN.

If some $|a_i| = 1$, define $a_i^{(\eta)} = (1 - \eta)a_i$ for all $i \in \mathcal{H}_0$, and let $\eta \downarrow 0$. The corresponding absolute null vectors converge weakly to $(|T_i| : i \in \mathcal{H}_0)$. Under the measure-theoretic definition of MTP₂, the class of MTP₂ probability measures is closed under weak convergence [Colangelo et al., 2006]. Hence the limiting absolute-value law remains MTP₂. Appendix A.1 of Su [2018] then implies that the induced vector of two-sided null p -values satisfies PRDN.

Proof. If $m_0 = 0$, then $V = 0$ identically and hence $\text{FDR} = 0$. Assume henceforth that $m_0 \geq 1$. Conditional on $Z = z$, each P_i is a function of ε_i alone, or is deterministic when $|a_i| = 1$. Thus the p -values are conditionally independent, and Lemma 4.1 gives (183).

To average (183) over Z , it remains to control

$$\int_{-u_q}^{u_q} \phi(z) \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t} dz$$

for each true null i . We do this on $|z| \leq u_q$; the event $|Z| > u_q$ is handled at the end by $\text{FDP} \leq 1$. Fix a true null index i . Put $r = |a_i|$ and $s = \sqrt{1 - r^2}$, and first suppose $s > 0$. Conditional on $Z = z$, $T_i \sim N(a_i z, s^2)$. The inequality $t = p(u) = 2\bar{\Phi}(u) \leq q$ holds if and only if $u \geq u_q$, and in this case

$$F_{i,z}(p(u)) = \int_u^\infty s^{-1} \phi\{(x - a_i z)/s\} dx + \int_u^\infty s^{-1} \phi\{(x + a_i z)/s\} dx.$$

Since $s^2 = 1 - a_i^2$,

$$\begin{aligned} \frac{s^{-1} \phi\{(x - a_i z)/s\}}{\phi(x)} &= \frac{1}{s} \exp \left\{ -\frac{(x - a_i z)^2}{2s^2} + \frac{x^2}{2} \right\} \\ &= \frac{1}{s} \exp \left\{ \frac{z^2}{2} - \frac{(a_i x - z)^2}{2s^2} \right\}. \end{aligned}$$

The second tail is the same calculation with z replaced by $-z$. Choose $\eta_+, \eta_- \in \{-1, 1\}$ with $\{\eta_+, \eta_-\} = \{-1, 1\}$ so that $\eta_+ a_i z \geq 0$ and $\eta_- a_i z \leq 0$. For $x \geq u$,

$$|a_i x - \eta_+ z| \geq (ru - |z|)_+, \quad |a_i x - \eta_- z| \geq ru + |z|.$$

Both right-hand sides are nondecreasing in u , so for $u \geq u_q$,

$$\begin{aligned} \phi(z) \frac{F_{i,z}(p(u))}{p(u)} &\leq \frac{\phi(z)}{p(u)} \frac{e^{z^2/2}}{s} \left[e^{-(ru_q - |z|)_+^2/(2s^2)} + e^{-(ru_q + |z|)^2/(2s^2)} \right] \int_u^\infty \phi(x) dx \\ &= \frac{1}{2s\sqrt{2\pi}} \left[e^{-(ru_q - |z|)_+^2/(2s^2)} + e^{-(ru_q + |z|)^2/(2s^2)} \right], \end{aligned}$$

because $\int_u^\infty \phi(x) dx = p(u)/2$. Taking the supremum gives

$$\phi(z) \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t} \leq \frac{1}{2s\sqrt{2\pi}} \left[e^{-(ru_q - |z|)_+^2/(2s^2)} + e^{-(ru_q + |z|)^2/(2s^2)} \right]. \quad (187)$$

Integrating (187) over $|z| \leq u_q$ and using symmetry gives

$$\int_{-u_q}^{u_q} \phi(z) \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t} dz \leq \frac{1}{s\sqrt{2\pi}} \left\{ \int_0^{u_q} e^{-(ru_q - z)^2/(2s^2)} dz + \int_0^{u_q} e^{-(ru_q + z)^2/(2s^2)} dz \right\}. \quad (188)$$

The two integrals on the right-hand side of (188) are

$$\begin{aligned} \int_0^{u_q} e^{-(ru_q - z)^2/(2s^2)} dz &= \int_0^{ru_q} e^{-y^2/(2s^2)} dy + (1 - r)u_q, \\ \int_0^{u_q} e^{-(ru_q + z)^2/(2s^2)} dz &= \int_{ru_q}^{(1+r)u_q} e^{-y^2/(2s^2)} dy. \end{aligned}$$

Consequently, writing $\gamma_r = \sqrt{(1 - r)/(1 + r)} \in (0, 1]$,

$$\begin{aligned} \int_{-u_q}^{u_q} \phi(z) \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t} dz &\leq \frac{1}{s\sqrt{2\pi}} \left\{ \int_0^{(1+r)u_q} e^{-y^2/(2s^2)} dy + (1 - r)u_q \right\} \\ &= \left\{ \Phi \left(\frac{(1 + r)u_q}{s} \right) - \frac{1}{2} \right\} + \frac{(1 - r)u_q}{s\sqrt{2\pi}} \\ &= \Phi(u_q/\gamma_r) - \frac{1}{2} + \frac{u_q \gamma_r}{\sqrt{2\pi}}, \end{aligned} \quad (189)$$

where the final equality in (189) uses $s^2 = 1 - r^2 = (1 - r)(1 + r)$, so $(1 + r)/s = 1/\gamma_r$ and $(1 - r)/s = \gamma_r$. The right-hand side of (189) is increasing in $\gamma_r \in (0, 1]$. Its derivative with respect to γ_r is

$$\frac{u_q}{\sqrt{2\pi}} \left[1 - \gamma_r^{-2} \exp\{-u_q^2/(2\gamma_r^2)\} \right] \geq 0,$$

because $u_q \geq 1$ and, on putting $w = \gamma_r^{-2} \geq 1$,

$$we^{-u_q^2 w/2} \leq we^{-w/2} \leq 2/e < 1.$$

Hence (189) is at most

$$\Phi(u_q) - \frac{1}{2} + \frac{u_q}{\sqrt{2\pi}} = \frac{1 - q}{2} + \frac{u_q}{\sqrt{2\pi}}. \quad (190)$$

When $r = 1$, the conditional p -value is deterministic. For $|z| < u_q$, it exceeds q , and the boundary $|z| = u_q$ has Lebesgue measure zero; hence (190) remains valid. When $r = 0$, the conditional null p -value is uniform and independent of Z ; (190) remains valid, although it is not sharp.

Let $A = \{|Z| \leq u_q\}$. Decomposing over A and A^c ,

$$\text{FDR} = \mathbb{E}(\text{FDP} \mathbf{1}_A) + \mathbb{E}(\text{FDP} \mathbf{1}_{A^c}).$$

For the first term, integrate (183) over $z \in [-u_q, u_q]$ and use (190). For the second term, no conditional tail bound is needed: since $0 \leq \text{FDP} \leq 1$,

$$\mathbb{E}(\text{FDP} \mathbf{1}_{A^c}) \leq \mathbb{P}(A^c) = \mathbb{P}(|Z| > u_q) = 2\bar{\Phi}(u_q) = q. \quad (191)$$

Combining (183), (190), and (191) yields

$$\text{FDR} \leq q + \frac{q}{m} \sum_{i \in \mathcal{H}_0} \left(\frac{1 - q}{2} + \frac{u_q}{\sqrt{2\pi}} \right) = q \left\{ 1 + \pi_0 \left(\frac{1 - q}{2} + \frac{u_q}{\sqrt{2\pi}} \right) \right\},$$

as claimed. $\square$

4.3 Upper bounds for one-sided tests

We derive an FDR upper bound for one-sided tests under the common-factor model and show that its leading constant cannot be improved within this class.

Theorem 4.4. *Consider the common-factor model (182), with its means denoted by $\theta_1, \dots, \theta_m$. Test $H_i : \theta_i \leq 0$ with $P_i = \bar{\Phi}(T_i)$. Let $\mathcal{H}_0 = \{i : \theta_i \leq 0\}$, $m_0 = |\mathcal{H}_0|$, and $\pi_0 = m_0/m$. For $0 < q < 1/2$, put $u_q^+ = \bar{\Phi}^{-1}(q) > 0$. The FDR is zero when $m_0 = 0$, and otherwise*

$$\text{FDR}(\text{BH}_q) \leq q \left\{ 2 + \pi_0 \left(\frac{1}{2} + \frac{u_q^+}{\sqrt{2\pi}} \right) \right\}. \quad (192)$$

Consequently, the worst-case FDR in this common-factor class is $O\{q\sqrt{\log(1/q)}\}$, uniformly in m , the means $(\theta_1, \dots, \theta_m)$, and the loadings $(a_1, \dots, a_m)$, as $q \downarrow 0$.

The following corollary identifies the exact leading constant for the common-factor class.

Corollary 4.5. *Over all common-factor models of the form (182), as $q \downarrow 0$,*

$$\sup \text{FDR}(\text{BH}_q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + O(q).$$

Thus the constant $1/\sqrt{\pi}$ in Proposition 3.2 is optimal in the common-factor class.

Proof of Theorem 4.4. If there are no true nulls, the FDR is zero. Partition the true nulls into the two sign groups

$$\mathcal{H}_{0,1} = \{i \in \mathcal{H}_0 : a_i \geq 0\}, \quad \mathcal{H}_{0,-1} = \{i \in \mathcal{H}_0 : a_i < 0\}.$$

Let $\mathcal{S} \subseteq \{-1, 1\}$ index the nonempty groups. The leave-one-out identity (184) and conditional reduction (185) are independent of the particular form of the p -values.

Suppose $i \in \mathcal{H}_{0,\sigma}$, where $\sigma \in \mathcal{S}$, and write $r = |a_i|$, $s = \sqrt{1 - r^2}$, and $Y_\sigma = \sigma Z \sim N(0, 1)$. Define the actual conditional CDF by

$$F_{i,y}^{\theta_i}(t) = \mathbb{P}_{\theta_i}(P_i \leq t \mid Y_\sigma = y)$$

and let $F_{r,y}^0$ denote the corresponding conditional CDF at the boundary mean $\theta_i = 0$. Conditional on $Y_\sigma = y$, $T_i \sim N(\theta_i + ry, s^2)$. Since $\theta_i \leq 0$ and rejection is increasing in T_i ,

$$F_{i,y}^{\theta_i}(t) \leq F_{r,y}^0(t), \quad 0 < t < 1.$$

Thus it suffices in the conditional leave-one-out bound to control the boundary CDF $F_{r,y}^0$.

For $s > 0$, the one-tail Gaussian density-ratio calculation gives

$$\phi(y) \sup_{0 < t \leq q} \frac{F_{r,y}^0(t)}{t} \leq \frac{1}{s\sqrt{2\pi}} \exp \left\{ -\frac{(ru_q^+ - y)_+^2}{2s^2} \right\}. \quad (193)$$

Indeed, write $t = p_+(u)$, where $u \geq u_q^+$. The bivariate normal density identity

$$\phi(y) \frac{1}{s} \phi\left(\frac{x - ry}{s}\right) = \phi(x) \frac{1}{s} \phi\left(\frac{y - rx}{s}\right)$$

and $p_+(u) = \int_u^\infty \phi(x) dx$ give

$$\begin{aligned} \phi(y) \frac{F_{r,y}^0\{p_+(u)\}}{p_+(u)} &= \frac{\int_u^\infty \phi(x) \frac{1}{s} \phi\left(\frac{y - rx}{s}\right) dx}{\int_u^\infty \phi(x) dx} \\ &\leq \sup_{x \geq u} \frac{1}{s} \phi\left(\frac{y - rx}{s}\right) \\ &\leq \frac{1}{s\sqrt{2\pi}} \exp \left\{ -\frac{(ru - y)_+^2}{2s^2} \right\}. \end{aligned}$$

The last inequality follows because, for every $x \geq u$,

$$(rx - y)^2 \geq (ru - y)_+^2.$$

Since $r \geq 0$, the map $u \mapsto (ru - y)_+$ is nondecreasing. Thus the last upper bound is nonincreasing in u and is maximized over $u \geq u_q^+$ at $u = u_q^+$, proving (193). Integrating over $y \leq u_q^+$ yields

$$\begin{aligned} I(r) &:= \int_{-\infty}^{u_q^+} \phi(y) \sup_{0 < t \leq q} \frac{F_{r,y}^0(t)}{t} dy \\ &\leq \frac{1}{2} + \frac{(1-r)u_q^+}{s\sqrt{2\pi}} \\ &\leq \frac{1}{2} + \frac{u_q^+}{\sqrt{2\pi}}. \end{aligned} \quad (194)$$

The last inequality uses $(1-r)/s = \sqrt{(1-r)/(1+r)} \leq 1$. If $r = 1$, then on $Y_\sigma < u_q^+$ the boundary-null p -value is larger than q , so (194) remains valid.

Let V_σ be the false rejections in sign group σ and $\text{FDP}_\sigma = V_\sigma/(R \vee 1)$. On $A_\sigma = \{Y_\sigma \leq u_q^+\}$, sum the per-null conditional bound over $\mathcal{H}_{0,\sigma}$ and apply (194). This gives

$$\mathbb{E}(\text{FDP}_\sigma \mathbf{1}_{A_\sigma}) \leq q \frac{|\mathcal{H}_{0,\sigma}|}{m} \left(\frac{1}{2} + \frac{u_q^+}{\sqrt{2\pi}} \right).$$

On A_σ^c , use $0 \leq \text{FDP}_\sigma \leq 1$ and $\mathbb{P}(A_\sigma^c) = q$. Summing over $\sigma \in \mathcal{S}$, using $|\mathcal{S}| \leq 2$ and $\text{FDP} = \sum_{\sigma \in \mathcal{S}} \text{FDP}_\sigma$, proves (192). $\square$

Proof of Corollary 4.5. Theorem 3.1 gives the lower bound within the common-factor class. Indeed, for the perturbed model (32), set

$$a_i^{(\eta)} = \sqrt{1 - \eta^2} b_i, \quad \tilde{\varepsilon}_i^{(\eta)} = \frac{\sqrt{(1 - \eta^2)(1 - b_i^2)} \varepsilon_i + \eta \xi_i}{\sqrt{1 - \{a_i^{(\eta)}\}^2}}.$$

The variables $\tilde{\varepsilon}_i^{(\eta)}$ are independent standard normal variables and are independent of Z , and

$$T_i^{(\eta)} = \theta_i + a_i^{(\eta)} Z + \sqrt{1 - \{a_i^{(\eta)}\}^2} \tilde{\varepsilon}_i^{(\eta)}.$$

Thus the perturbation is of the form (182). Together with Proposition 3.2, this gives

$$\sup \text{FDR}(\text{BH}_q) \geq \ell_{\leq}(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + o(q).$$

For the matching upper bound, note first that $u_q^+ = p^{-1}(2q)$, so (207) in Lemma A.2 gives $u_q^+ \sim \sqrt{2\log(1/q)}$. More precisely, (204) in Lemma A.2 and $q = \Phi(u_q^+)$ imply

$$(u_q^+)^2 = 2\log(1/q) - 2\log u_q^+ - \log(2\pi) + o(1) = 2\log(1/q) + O\{\log \log(1/q)\}.$$

Consequently,

$$u_q^+ = \sqrt{2\log(1/q)} + O\left\{\frac{\log \log(1/q)}{\sqrt{\log(1/q)}}\right\} = \sqrt{2\log(1/q)} + o(1).$$

Theorem 4.4, with $\pi_0 \leq 1$, now gives the reverse inequality up to an $O(q)$ term, proving the claim. $\square$

5 Discussion

For two-sided problems, Theorem 2.1 and Proposition 2.2 show that no bound of the form Cq , with C independent of q , can hold uniformly over all Gaussian correlation matrices. Because the lower-bound construction belongs to the common-factor class, Theorem 4.2 shows that the worst-case FDR within this class is of exact order $q\sqrt{\log(1/q)}$ as $q \downarrow 0$. The leading constants in the current two-sided upper and lower bounds do not match: they are respectively $1/\sqrt{\pi}$ and $1/(2\sqrt{\pi})$, leaving the exact two-sided constant open. For one-sided problems, by contrast, Theorem 3.1, Proposition 3.2, and Corollary 4.5 give the matching constant $1/\sqrt{\pi}$. Thus this constant is optimal in the common-factor class, with the upper bound allowing arbitrary means and scalar loadings.

There are also procedures with finite-sample FDR guarantees for dependent two-sided Gaussian tests. The conditional-calibration framework of Fithian and Lei [2022] decomposes FDR into hypothesis-wise contributions and, for each null i , conditions on a statistic under which the i th p -value remains conditionally valid and the relevant conditional law is tractable, often eliminating nuisance parameters. The resulting guarantees depend on the procedure and the dependence assumptions: under CPRDS, dBH₁ controls FDR and contains BH, whereas under arbitrary dependence, dBY controls FDR and contains the

Benjamini–Yekutieli procedure. Conditional calibration can therefore exploit known dependence without automatically paying the full harmonic-factor correction of Benjamini and Yekutieli [2001].

Another route is the fixed- X knockoff filter of Barber and Candès [2015]. In a Gaussian linear model, knockoffs augment the design with synthetic variables that mimic the correlation structure of the original variables, and the resulting original-versus-knockoff comparisons create null statistics with the symmetry needed for FDR control. Li and Fithian [2021] recast fixed- X knockoffs as adding user-generated Gaussian noise to a Gaussian estimator to obtain a whitened estimator, followed by conditional inference. This whitening can substantially reduce power when the covariance has large leading eigenvalues associated with dense leading eigenvectors, because it then requires enough added noise to erase much of the signal. Knockoffs also face a threshold phenomenon when the rejection set is small. The calibrated knockoff method of Luo et al. [2025] uses conditional calibration to uniformly improve knockoff procedures, with especially large gains in small-rejection regimes.

These findings point to two complementary directions: identifying structured dependence under which ordinary BH remains reliable, and developing procedures that use dependence information when the two-sided transformation destroys the monotonicity needed by standard positive-dependence arguments.

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References

- Rina Foygel Barber and Emmanuel J. Candès. Controlling the false discovery rate via knockoffs. *The Annals of Statistics*, 43(5):2055–2085, 2015. doi: 10.1214/15-AOS1337.
- Yoav Benjamini. Discovering the false discovery rate. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 72(4):405–416, 2010. doi: 10.1111/j.1467-9868.2010.00746.x. URL <https://doi.org/10.1111/j.1467-9868.2010.00746.x>.
- Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- Yoav Benjamini and Daniel Yekutieli. The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4):1165–1188, 2001. doi: 10.1214/aos/1013699998.
- Ziyu Chi, Aaditya Ramdas, and Ruodu Wang. Multiple testing under negative dependence. *Bernoulli*, 31(2):1230–1255, 2025.
- Antonio Colangelo, Alfred Müller, and Marco Scarsini. Positive dependence and weak convergence. *Journal of Applied Probability*, 43(1):48–59, 2006. doi: 10.1239/jap/1143936242.
- Edgar Dobriban. The Benjamini–Hochberg procedure can fail to control the FDR for correlated two-sided Gaussian tests, 2026. URL <https://arxiv.org/abs/2607.12208>.
- William Fithian and Lihua Lei. Conditional calibration for false discovery rate control under dependence. *The Annals of Statistics*, 50(6):3091–3118, 2022. doi: 10.1214/21-AOS2137.
- Yosef Hochberg and Dror Rom. Extensions of multiple testing procedures based on simes’ test. *Journal of Statistical Planning and Inference*, 48(2):141–152, 1995.

- Samuel Karlin and Yosef Rinott. Total positivity properties of absolute value multinormal variables with applications to confidence interval estimates and related probabilistic inequalities. *The Annals of Statistics*, 9(5), 1981. doi: 10.1214/aos/1176345583.
- Xiao Li and William Fithian. Whiteout: When do fixed-X knockoffs fail?, 2021. URL <https://arxiv.org/abs/2107.06388>.
- Yixiang Luo, William Fithian, and Lihua Lei. Improving knockoffs with conditional calibration. *The Annals of Statistics*, 53(5), 2025. doi: 10.1214/25-AOS2543.
- Anat Reiner-Benaim. FDR control by the BH procedure for two-sided correlated tests with implications to gene expression data analysis. *Biometrical Journal*, 49(1):107–126, 2007. doi: 10.1002/bimj.200510313.
- Marine Roux. *Inférence de graphes par une procédure de test multiple avec application en Neuroimagerie*. PhD thesis, Université Grenoble Alpes (ComUE), September 2018. URL <https://theses.fr/2018GREAT058>. Thesis no. 2018GREAT058.
- Ester Samuel-Cahn. Is the simes improved bonferroni procedure conservative? *Biometrika*, 83(4):928–933, 1996.
- Sanat K. Sarkar. On controlling the false discovery rate in multiple testing of the means of correlated normals against two-sided alternatives, 2023. URL <https://arxiv.org/abs/2304.05261>.
- Sanat K. Sarkar and Shiyu Zhang. Shifted BH methods for controlling false discovery rate in multiple testing of the means of correlated normals against two-sided alternatives. *Journal of Statistical Planning and Inference*, 236:106238, 2025. doi: 10.1016/j.jspi.2024.106238.
- Weijie J. Su. The FDR-linking theorem, 2018. URL <https://arxiv.org/abs/1812.08965>.

A Auxiliary lemmas

This appendix collects the properties of the envelope and Gaussian tails used to define and analyze $\nu_{q,r}$. The following lemma characterizes the likelihood-ratio envelope.

Lemma A.1. *For $0 \leq y \leq 1$, $M(y) = 1$. For $y > 1$, $a \mapsto L_a(y)$ has a unique maximizer $a(y) > 0$, characterized by*

$$a(y) = y \tanh\{a(y)y\}. \quad (195)$$

The maps $a : (1, \infty) \rightarrow (0, \infty)$ and $M : (1, \infty) \rightarrow (1, \infty)$ are continuous and strictly increasing, with

$$\lim_{y \downarrow 1} a(y) = 0, \quad \lim_{y \rightarrow \infty} M(y) = \infty. \quad (196)$$

The inverse $y = y(a)$ is continuously differentiable on $(0, \infty)$, and

$$y'(a) = \frac{1 - y(a)^2 \operatorname{sech}^2\{ay(a)\}}{\tanh\{ay(a)\} + ay(a) \operatorname{sech}^2\{ay(a)\}} > 0. \quad (197)$$

Moreover, with $a_q = a(y_q)$, the deficit in the secondary survival mass satisfies

$$1 - qM\{y(a)\} > 0 \quad (0 < a < a_q), \quad 1 - qM\{y(a_q)\} = 0, \quad \frac{d}{da}[1 - qM\{y(a)\}] < 0. \quad (198)$$

Proof. Let

$$g_y(a) = \log L_a(y) = -\frac{a^2}{2} + \log \cosh(ay).$$

Then

$$g'_y(a) = -a + y \tanh(ay). \quad (199)$$

If $y \leq 1$, then $y \tanh(ay) \leq ay^2 \leq a$, so $M(y) = L_0(y) = 1$.

For $y > 1$, put $t = ay$. Equation $g'_y(a) = 0$ is equivalent to

$$y^2 = \frac{t}{\tanh t}. \quad (200)$$

To verify the strict inequality needed below, let $H(t) = \tanh t - t \operatorname{sech}^2 t$. Then $H(0) = 0$ and $H'(t) = 2t \operatorname{sech}^2 t \tanh t > 0$ for $t > 0$. Hence

$$\frac{d}{dt} \left(\frac{t}{\tanh t} \right) = \frac{\tanh t - t \operatorname{sech}^2 t}{\tanh^2 t} > 0.$$

Moreover, $t/\tanh t \rightarrow 1$ as $t \downarrow 0$ and $t/\tanh t \rightarrow \infty$ as $t \rightarrow \infty$. Because $y^2 > 1$, (200) in Lemma A.1 therefore has a unique positive solution. The sign of g'_y changes from positive to negative there, so the solution is the unique maximizer. Moreover,

$$a(y) = \sqrt{t \tanh t}, \quad y = \sqrt{\frac{t}{\tanh t}}, \quad (201)$$

Both parameter functions in (201) in Lemma A.1 are smooth and strictly increasing in $t > 0$. They tend respectively to zero and one as $t \downarrow 0$, and both tend to infinity as $t \rightarrow \infty$. Thus $a(y)$ is continuous and strictly increasing on $(1, \infty)$, with $a(y) \rightarrow 0$ as $y \downarrow 1$; the inverse function theorem applied to the second parameter function also shows that $a(y)$ is continuously differentiable. Since $M(y) = L_{a(y)}(y)$, the chain rule and $\partial_a L_{a(y)}(y) = 0$ give

$$M'(y) = M(y)a(y) \tanh\{a(y)y\} > 0.$$

Thus M is continuous and strictly increasing. Also

$$M(y) \geq L_y(y) \geq \frac{1}{2}e^{y^2/2} \rightarrow \infty.$$

Finally, apply the implicit function theorem to

$$F(a, y) = a - y \tanh(ay) = 0.$$

At the maximizer, with $t = ay > 0$, equation (200) in Lemma A.1 gives

$$y^2 \operatorname{sech}^2(ay) = \frac{t \operatorname{sech}^2 t}{\tanh t} = \frac{2t}{\sinh(2t)} < 1.$$

Consequently,

$$F_a = 1 - y^2 \operatorname{sech}^2(ay) > 0, \quad F_y = -\tanh(ay) - ay \operatorname{sech}^2(ay) < 0.$$

Thus (197) in Lemma A.1 follows from $y' = -F_a/F_y$.

It remains to prove (198) in Lemma A.1. Since $a_q = a(y_q)$, we have $y(a_q) = y_q$, and therefore $1 - qM\{y(a_q)\} = 0$ by (4). If $0 < a < a_q$, then $y(a) < y_q$; since M is strictly increasing on $(1, \infty)$, $qM\{y(a)\} < qM(y_q) = 1$. Finally,

$$\frac{d}{da} [1 - qM\{y(a)\}] = -qM'\{y(a)\}y'(a) < 0,$$

because $M' > 0$ on $(1, \infty)$ and $y' > 0$ on $(0, \infty)$. □

The final lemma collects the Mills bounds and quantile asymptotics used to control Gaussian tails.

Lemma A.2. *The normal hazard λ is increasing on $\mathbb{R}$ and satisfies*

$$x < \lambda(x) < x + x^{-1} \quad (x > 0). \quad (202)$$

Moreover, as $x \rightarrow \infty$,

$$\lambda(x) = x + x^{-1} + O(x^{-3}). \quad (203)$$

As $x \rightarrow \infty$,

$$\bar{\Phi}(x) = \frac{\phi(x)}{x} \{1 + o(1)\}. \quad (204)$$

For every fixed $0 < \sigma < 1$, as $x \rightarrow \infty$,

$$\frac{\bar{\Phi}(\sigma x)}{\bar{\Phi}(x)} \sim \frac{1}{\sigma} \exp \left\{ \frac{(1 - \sigma^2)x^2}{2} \right\}. \quad (205)$$

For every $0 < H < \infty$, uniformly over $|u| \vee |v| \leq H$, as $x \rightarrow \infty$,

$$\frac{p(x+u)}{p(x+v)} = \frac{x+v}{x+u} \exp \left\{ -x(u-v) - \frac{u^2 - v^2}{2} \right\} \{1 + o(1)\}. \quad (206)$$

If $c > 0$ is fixed, then, as $q \downarrow 0$,

$$p^{-1}(cq) \sim \sqrt{2 \log(1/q)}. \quad (207)$$

In particular, for every fixed $c_1, c_2 > 0$,

$$\frac{p^{-1}(c_1 q)}{p^{-1}(c_2 q)} \longrightarrow 1 \quad (q \downarrow 0).$$

Finally, for $x \geq 0$,

$$2\bar{\Phi}(x) \leq e^{-x^2/2}. \quad (208)$$

For $x > 0$ and $\delta \geq 0$,

$$\frac{p(x+\delta)}{p(x)} \leq \left(1 + x^{-2}\right) \exp \left\{ -x\delta - \frac{\delta^2}{2} \right\}. \quad (209)$$

Consequently, if $x \rightarrow \infty$, then the following tail-ratio limits hold.

(i) For every $0 < H < \infty$, uniformly over $|u| \vee |v| \leq H$,

$$\log \frac{p(x+u)}{p(x+v)} = -x(u-v) - \frac{u^2 - v^2}{2} + o(1). \quad (210)$$

(ii) For every $0 < H < \infty$, uniformly over $|h| \leq H$,

$$\frac{p(x)}{p(x+h/x)} \longrightarrow e^h. \quad (211)$$

(iii) If $0 < \delta_0 \leq \delta \leq \delta_1 < \infty$, then

$$\frac{p(x+\delta)}{p(x)} \longrightarrow 0 \quad (212)$$

uniformly in δ .

(iv) If $\delta_x \geq 0$ and $x\delta_x \rightarrow \infty$, then

$$\frac{p(x + \delta_x)}{p(x)} \rightarrow 0. \quad (213)$$

Proof. For $x > 0$, the upper Mills bound follows from

$$\bar{\Phi}(x) < \frac{1}{x} \int_x^\infty t\phi(t) dt = \frac{\phi(x)}{x}. \quad (214)$$

Integration by parts gives

$$\bar{\Phi}(x) = \frac{\phi(x)}{x} - \int_x^\infty \frac{\phi(t)}{t^2} dt > \frac{\phi(x)}{x} - \frac{\bar{\Phi}(x)}{x^2},$$

which proves the lower Mills bound

$$\frac{x}{1+x^2}\phi(x) < \bar{\Phi}(x). \quad (215)$$

A further integration by parts gives

$$\bar{\Phi}(x) = \frac{\phi(x)}{x} - \frac{\phi(x)}{x^3} + 3 \int_x^\infty \frac{\phi(t)}{t^4} dt,$$

where

$$0 < \int_x^\infty \frac{\phi(t)}{t^4} dt < \frac{\phi(x)}{x^5}.$$

Hence

$$\bar{\Phi}(x) = \frac{\phi(x)}{x} \{1 - x^{-2} + O(x^{-4})\},$$

and inversion proves (203) in Lemma A.2. Equations (214) and (215) imply (202) in Lemma A.2 for $x > 0$. Since $\lambda(x) - x > 0$ also holds trivially for $x \leq 0$, the identity

$$\lambda'(x) = \lambda(x)\{\lambda(x) - x\}$$

shows that λ is increasing on $\mathbb{R}$.

Equations (214) and (215) give (204) in Lemma A.2. Applying this at x and σx proves (205) in Lemma A.2. Applying (204) in Lemma A.2 at $x + u$ and $x + v$ gives (206) in Lemma A.2, for every $0 < H < \infty$, uniformly over $|u| \vee |v| \leq H$. Moreover, as $x \rightarrow \infty$,

$$-\log p(x) = \frac{x^2}{2} + O(\log x),$$

which yields (207) in Lemma A.2 by inversion. To prove $2\bar{\Phi}(x) \leq e^{-x^2/2}$, set

$$d(x) = e^{-x^2/2} - 2\bar{\Phi}(x).$$

Then $d(0) = 0$, $\lim_{x \rightarrow \infty} d(x) = 0$, and

$$d'(x) = e^{-x^2/2} \left(\sqrt{\frac{2}{\pi}} - x \right).$$

Thus d first increases and then decreases to zero, so $d(x) \geq 0$ for all $x \geq 0$. Finally, (214) and (215) imply

$$\frac{p(x + \delta)}{p(x)} \leq \frac{1 + x^2}{x(x + \delta)} \exp \left\{ -x\delta - \frac{\delta^2}{2} \right\} \leq (1 + x^{-2}) \exp \left\{ -x\delta - \frac{\delta^2}{2} \right\},$$

which proves (209) in Lemma A.2.

It remains to prove the four stated tail-ratio consequences. Taking logarithms in (206) in Lemma A.2 gives (210) in Lemma A.2, because

$$\log \frac{x+v}{x+u} = o(1)$$

for every $0 < H < \infty$, uniformly over $|u| \vee |v| \leq H$. For (211) in Lemma A.2, apply (206) in Lemma A.2 with $u = 0$ and $v = h/x$. For every $0 < H < \infty$, uniformly over $|h| \leq H$,

$$\frac{p(x)}{p(x+h/x)} = \frac{x+h/x}{x} \exp \left\{ h + \frac{h^2}{2x^2} \right\} \{1 + o(1)\} \longrightarrow e^h.$$

For (212) in Lemma A.2, apply (210) in Lemma A.2 with $u = \delta$ and $v = 0$. If $0 < \delta_0 \leq \delta \leq \delta_1 < \infty$, then

$$\log \frac{p(x+\delta)}{p(x)} \leq -x\delta_0 + O(1) \longrightarrow -\infty$$

uniformly in δ . Finally, (213) in Lemma A.2 follows from (209) in Lemma A.2 with $\delta = \delta_x$, because $x\delta_x \rightarrow \infty$. $\square$